

# Database Design Tuning



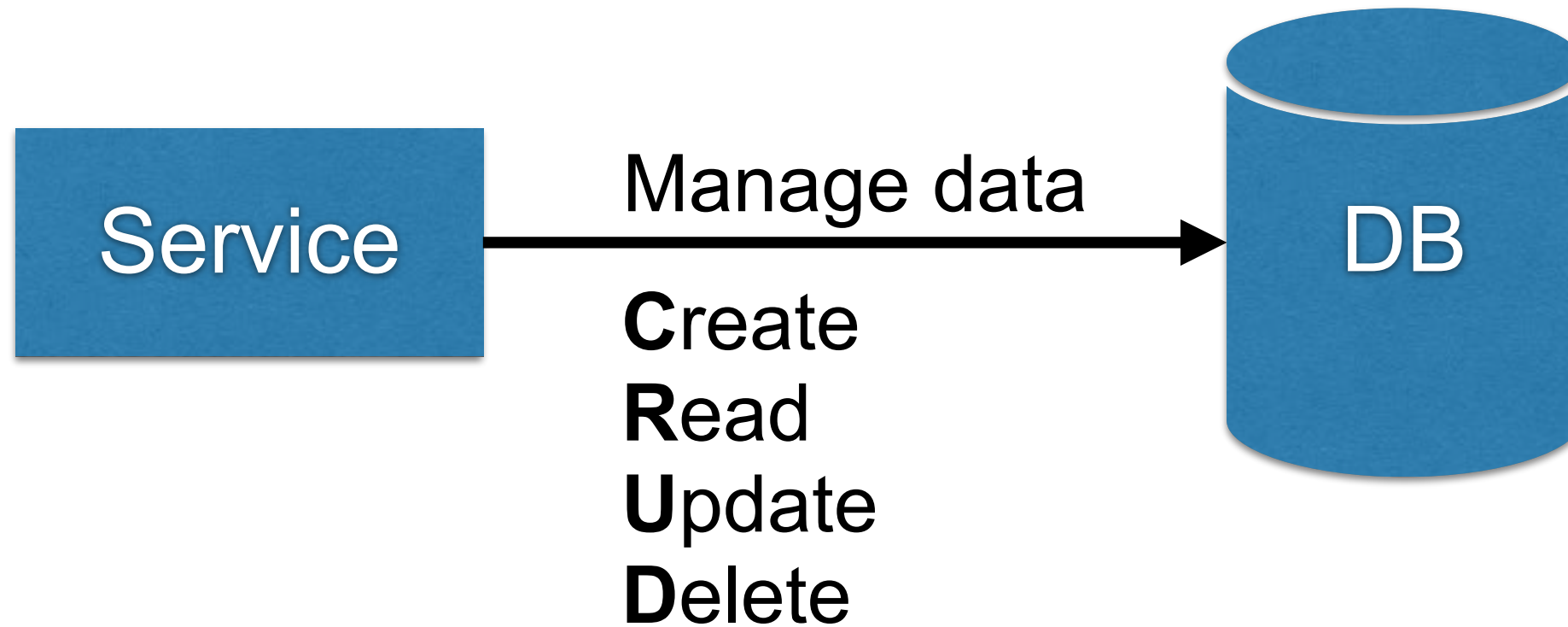
**[https://github.com/up1/  
workshop-database](https://github.com/up1/workshop-database)**



# Database

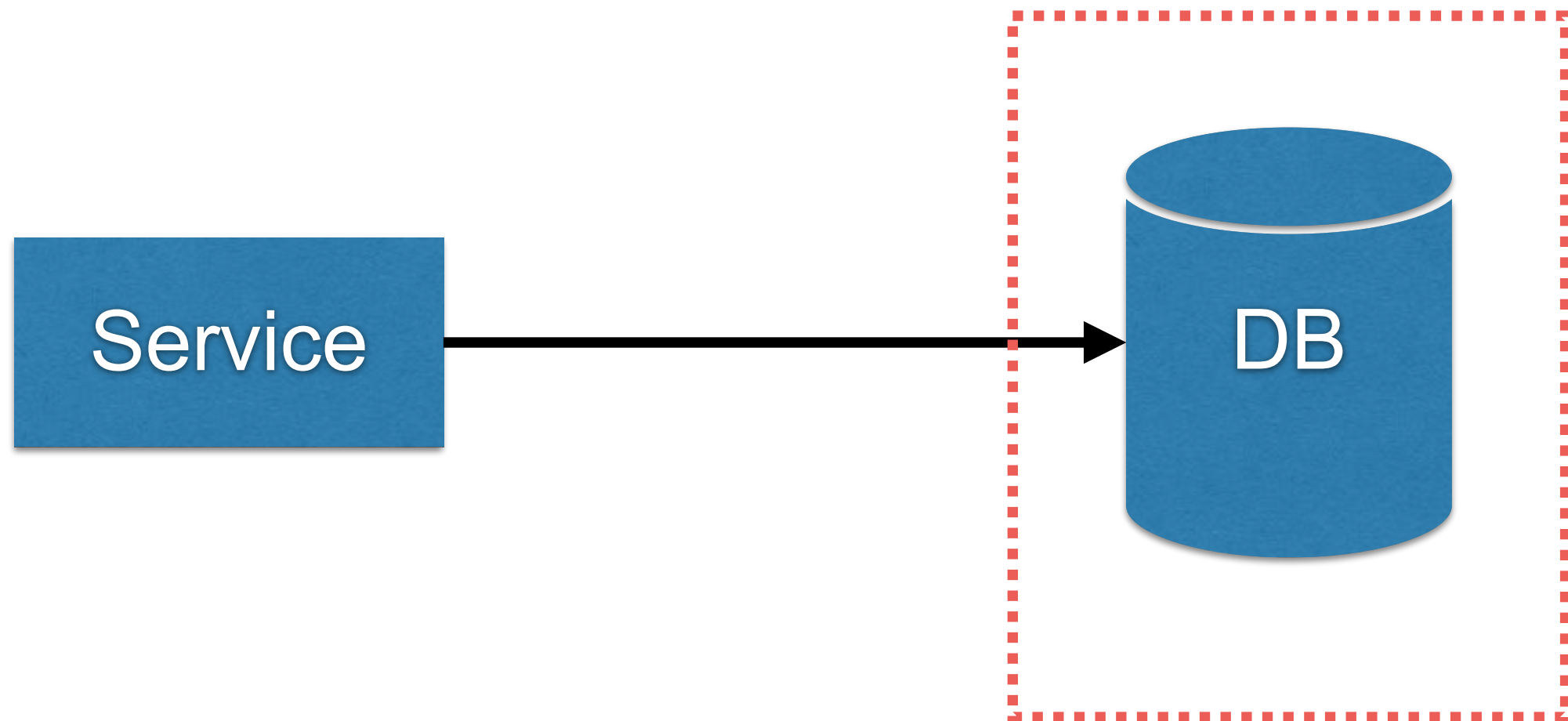


# Database



# Working with Database

Directly to database

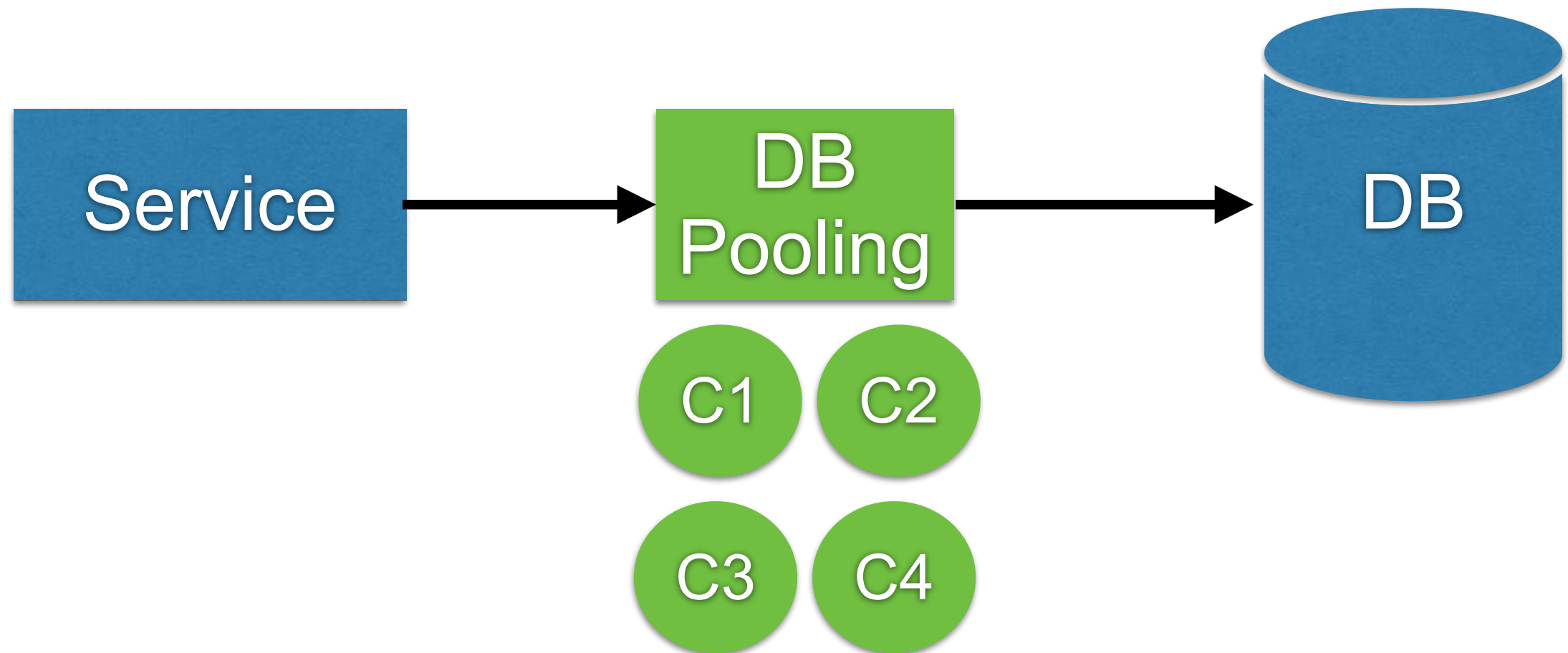


Data modeling  
Database architecture



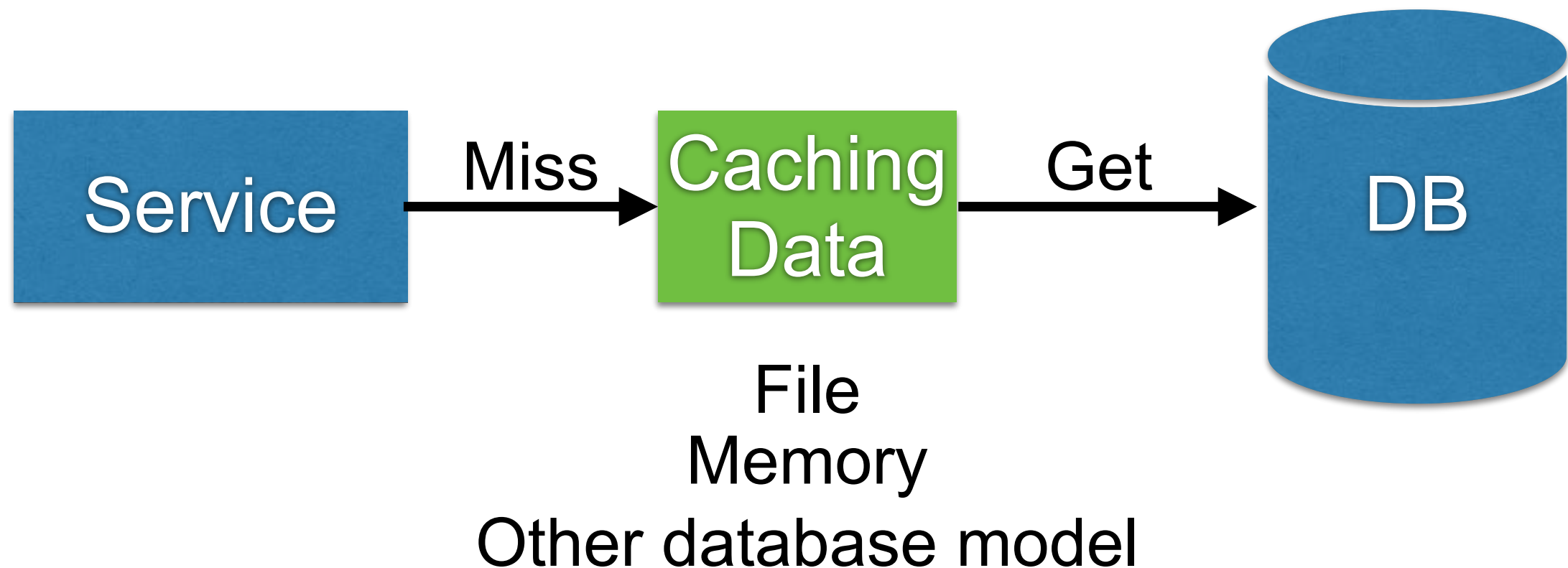
# Working with Database

## Connection pool



# Working with Database

## Caching data layer





# Scalability metrics

Throughput (transactions per second)

Latency (response time)

Availability (uptime)

Durability (data safety)





# Database ?

Structured data collection

Records

Relationships



# Database Management System ?

Software systems for storing and querying databases



# User's perspective

Data modeling

Querying data

Store data

Read

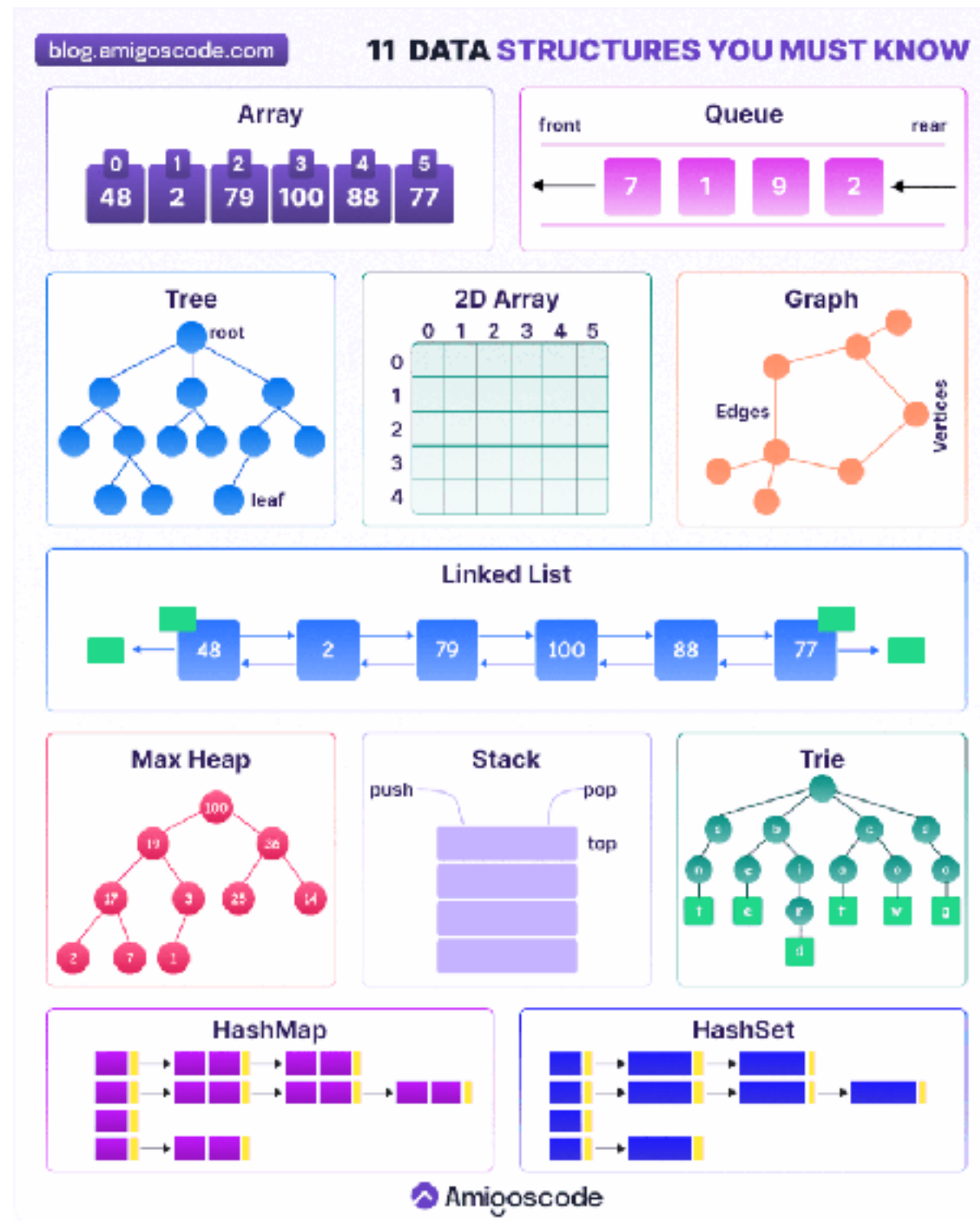
Write



# Database Models ?



# Data Structure



<https://blog.amigoscode.com/p/11-data-structures-every-developer>



# Database Models ?

424 systems in ranking, July 2025

| Rank     |          |                                                                                         | DBMS                                                                                        | Database Model                                                                                                 | Score    |          |          |
|----------|----------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------|----------|----------|
| Jul 2025 | Jun 2025 | Jul 2024                                                                                |                                                                                             |                                                                                                                | Jul 2025 | Jun 2025 | Jul 2024 |
| 1.       | 1.       | 1.                                                                                      | Oracle                                                                                      | Relational, Multi-model     | 1217.05  | -13.33   | -23.31   |
| 2.       | 2.       | 2.                                                                                      | MySQL                                                                                       | Relational, Multi-model     | 940.73   | -12.85   | -98.73   |
| 3.       | 3.       | 3.                                                                                      | Microsoft SQL Server                                                                        | Relational, Multi-model     | 771.14   | -5.61    | -36.51   |
| 4.       | 4.       | 4.                                                                                      | PostgreSQL                                                                                  | Relational, Multi-model     | 680.89   | +0.23    | +41.98   |
| 5.       | 5.       | 5.                                                                                      | MongoDB    | Document, Multi-model       | 403.83   | +0.99    | -25.99   |
| 6.       | 6.       |  7.    | Snowflake                                                                                   | Relational                                                                                                     | 176.17   | +1.68    | +39.64   |
| 7.       | 7.       |  6.  | Redis                                                                                       | Key-value, Multi-model    | 149.72   | -2.01    | -7.05    |
| 8.       | 8.       |  9.  | IBM Db2                                                                                     | Relational, Multi-model   | 127.51   | +2.38    | +3.11    |
| 9.       | 9.       |  8.  | Elasticsearch                                                                               | Multi-model               | 118.83   | -2.45    | -12.00   |
| 10.      | 10.      | 10.                                                                                     | SQLite                                                                                      | Relational                                                                                                     | 115.44   | -1.60    | +5.49    |
| 11.      | 11.      |  12. | Apache Cassandra                                                                            | Wide column, Multi-model  | 108.76   | +0.49    | +9.63    |
| 12.      | 12.      |  15. | Databricks                                                                                  | Multi-model               | 108.04   | +3.36    | +24.74   |
| 13.      | 13.      |  14. | MariaDB  | Relational, Multi-model   | 95.44    | +0.91    | +4.85    |
| 14.      | 14.      |  11. | Microsoft Access                                                                            | Relational                                                                                                     | 90.47    | +2.19    | -10.17   |
| 15.      | 15.      |  17. | Amazon DynamoDB                                                                             | Multi-model               | 84.15    | +0.81    | +13.20   |

<https://db-engines.com/en/ranking>



# Database Model

**Relational**

Document

Key-value

Search engine

Wide column

Graph

Time series

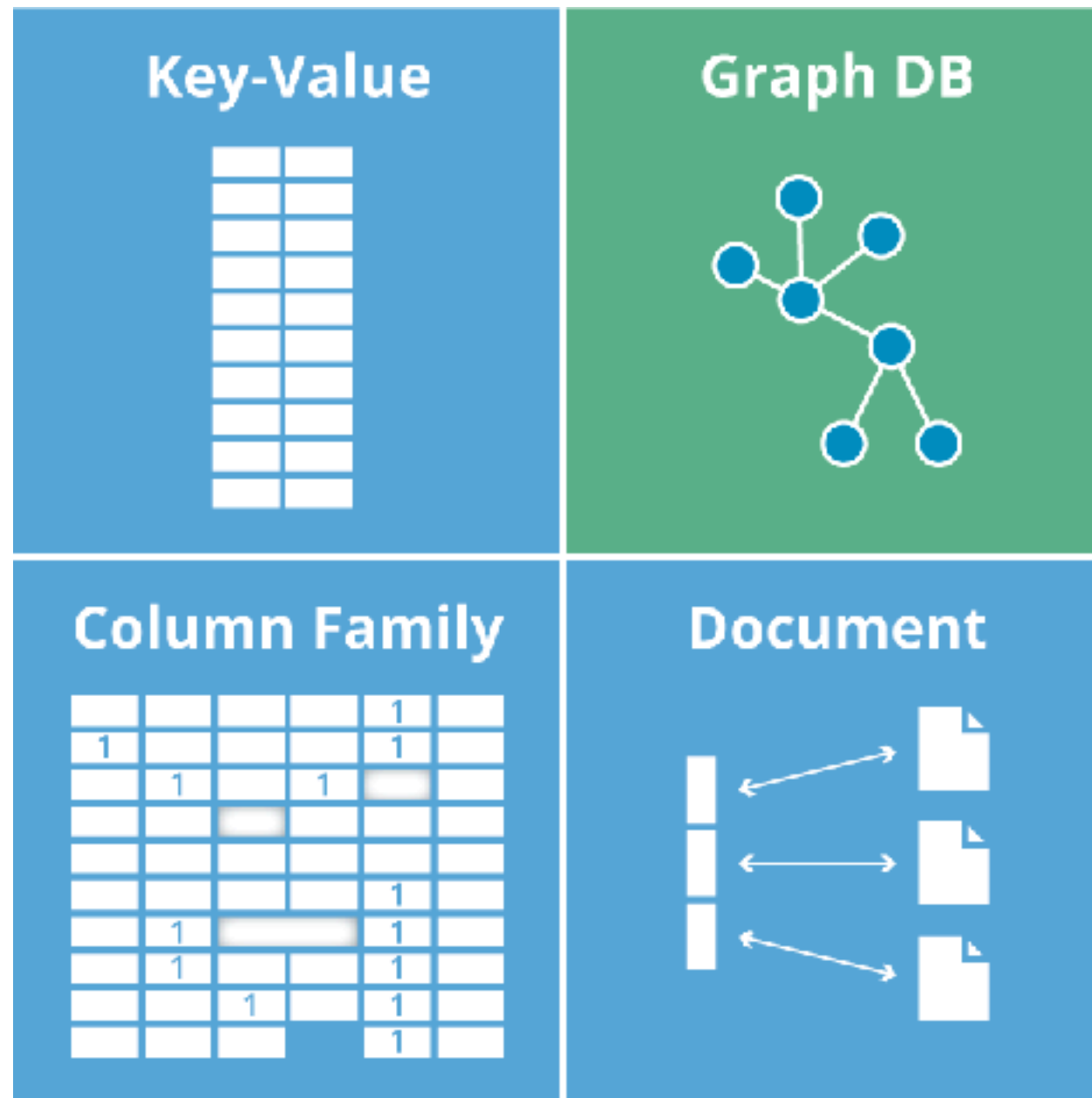
Vector

**Non-Relational**  
**NoSQL**





# NoSQL





# Relational Database



PostgreSQL

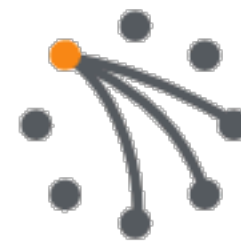


**ORACLE®**  
DATABASE

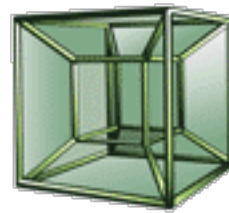


# NoSQL

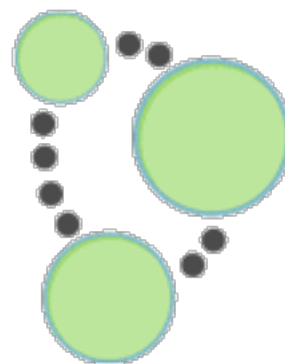
APACHE  
**HBASE**



**riak**



**HYPERTABLE** INC

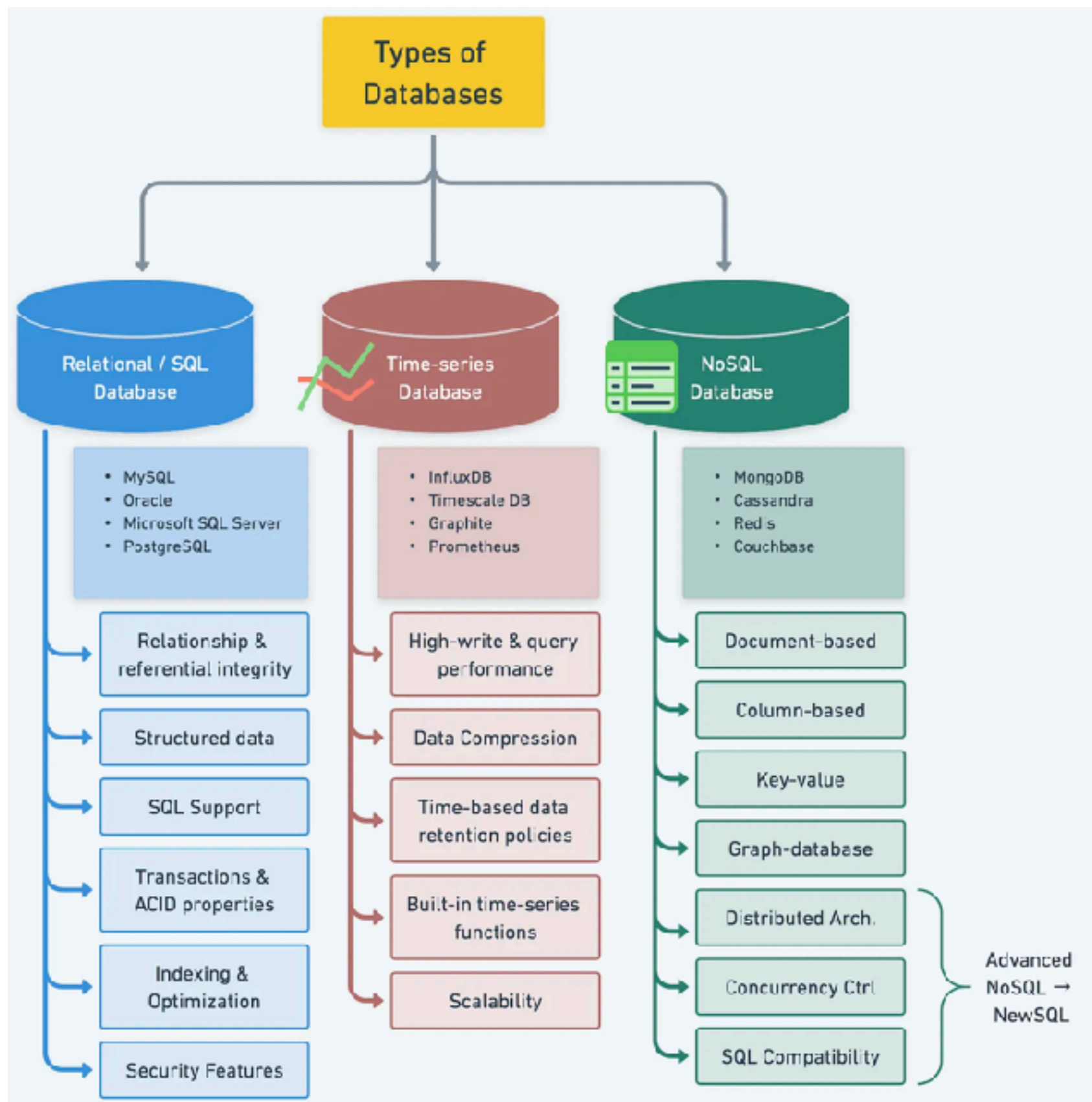


**Neo4j**



**redis**





<https://blog.bytebytego.com/p/understanding-database-types>



# Relational Database ?



# Relational Database ?

**Relational DataBase Managment System**

Maintain data in **tables**

**Pre-define** structure

Each table has **Primary key**

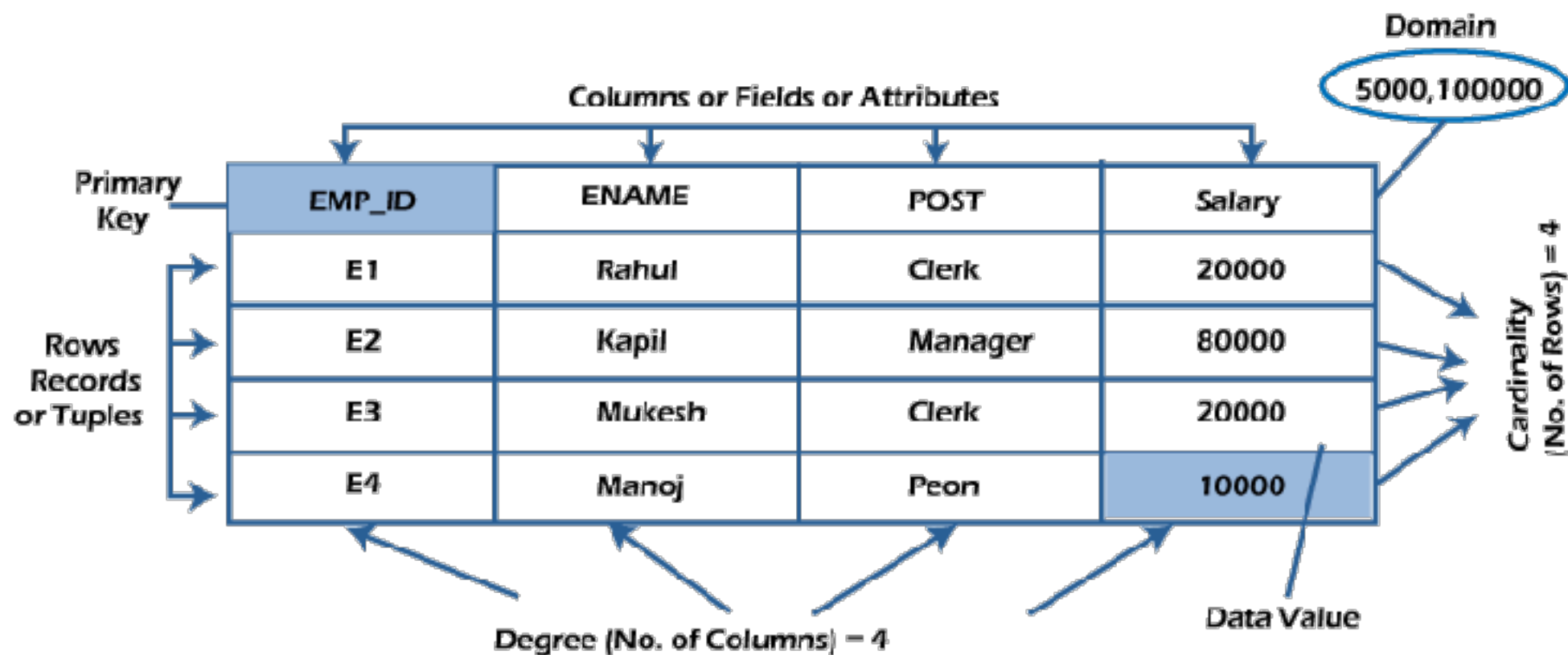
Many **columns**

Data is represented in term of **row**



# Relational Database ?

Table = Employee



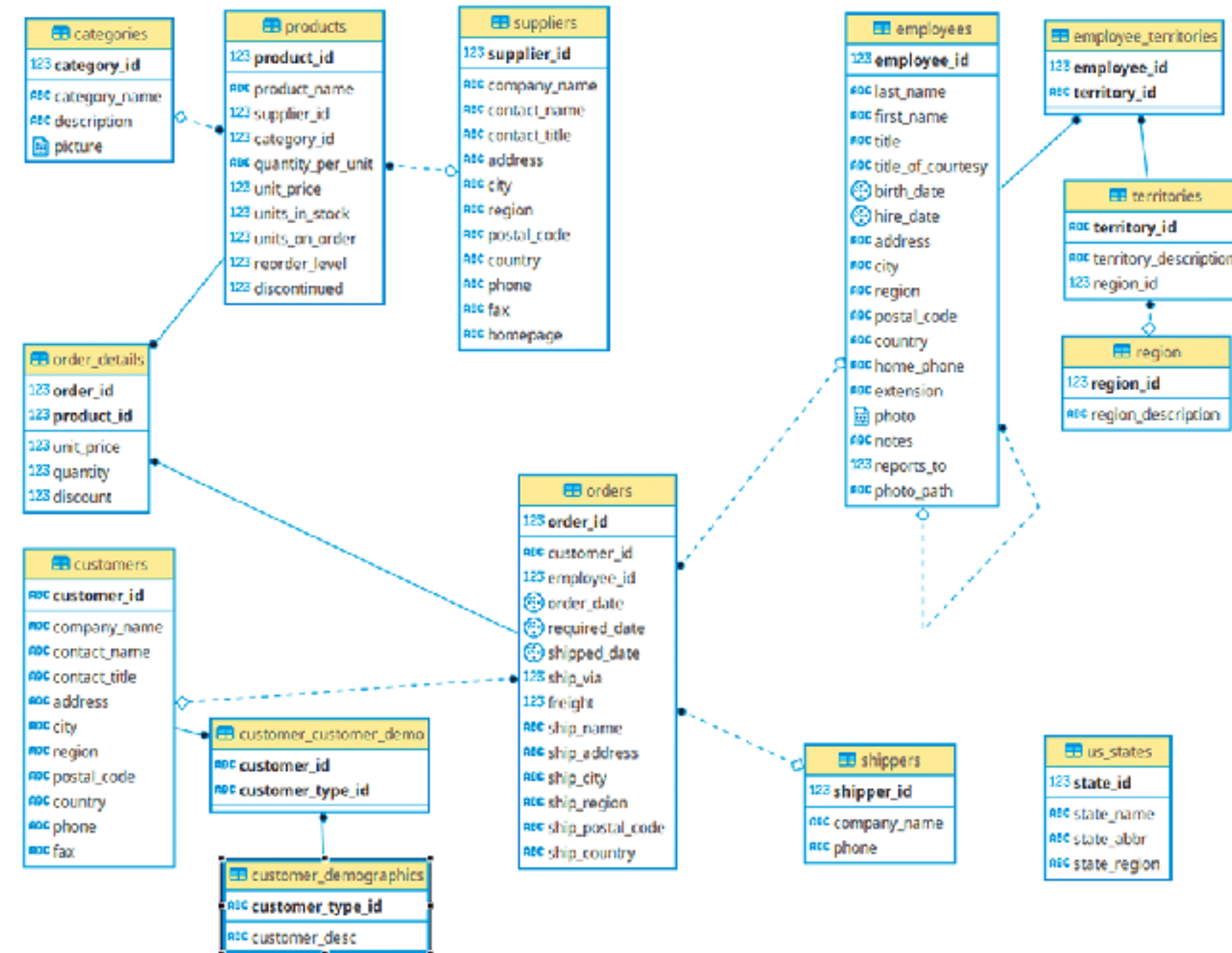
# Relational Database

| Database     | Ease of use | Support complex query | Community support | Licence     |
|--------------|-------------|-----------------------|-------------------|-------------|
| MySQL        | High        | High                  | Strong            | Open source |
| PostgreSQL   | Medium      | Very high             | Strong            | Open source |
| MSSQL Server | Medium      | Very high             | Strong            | Proprietary |
| Oracle       | Low         | Very high             | Moderate          | Proprietary |





# Real Design of RDBMS ?





# Questions ?

How to write data ?

How to read data ?

Aggregate data ?

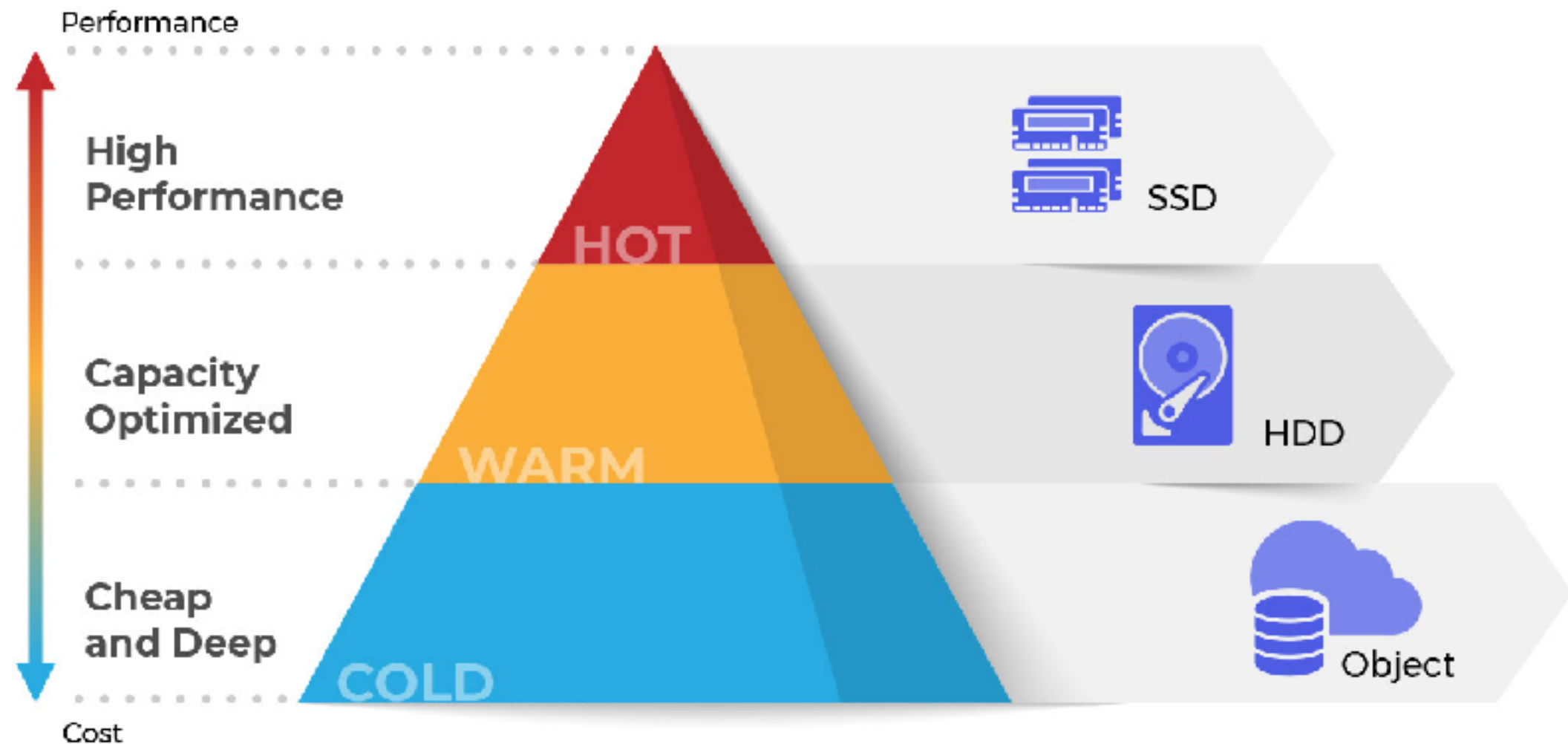
Reporting

Data retention ?



# Questions ?

Hot vs Warm vs Cold data ?

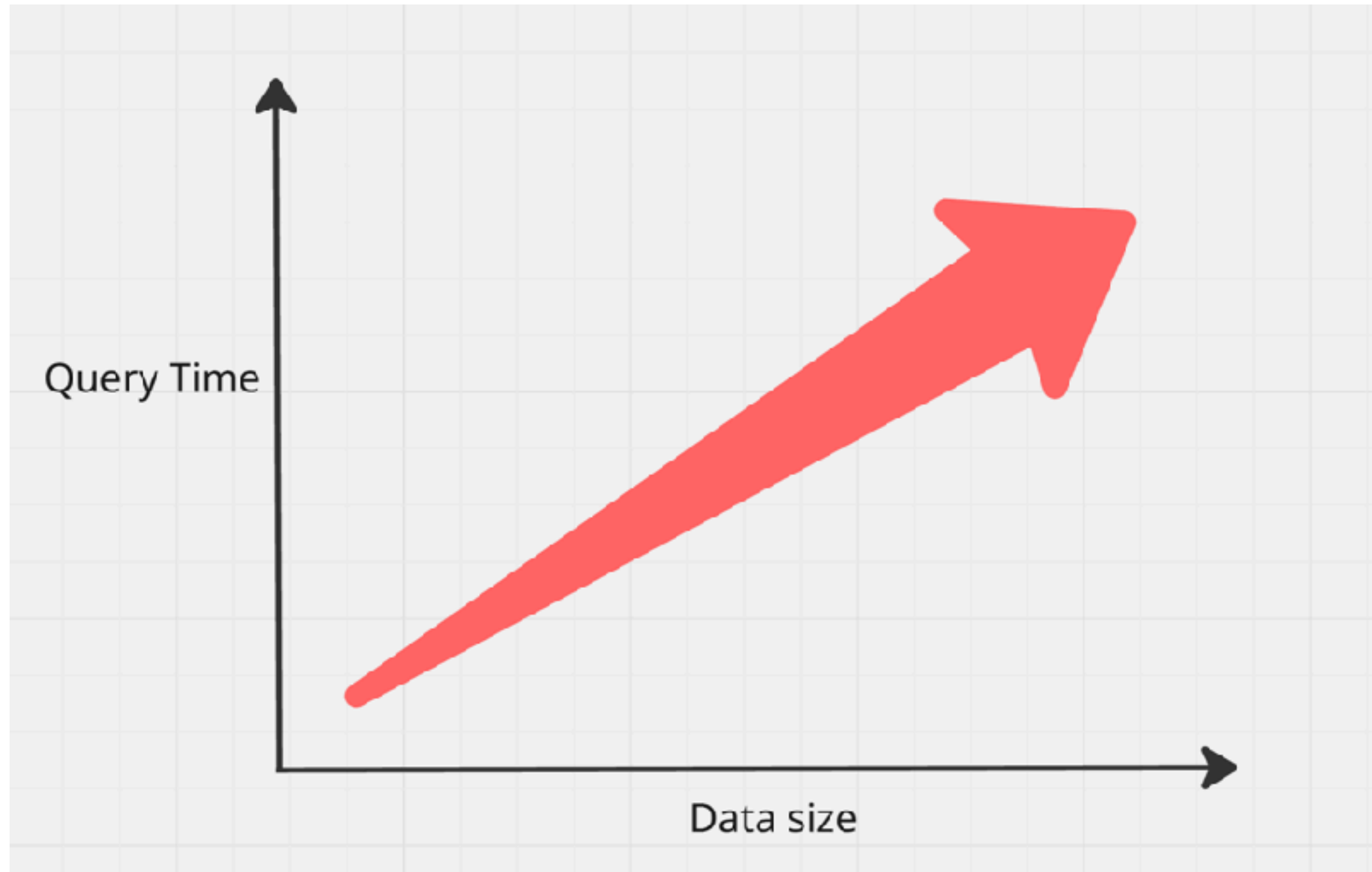


# Complex Query ?

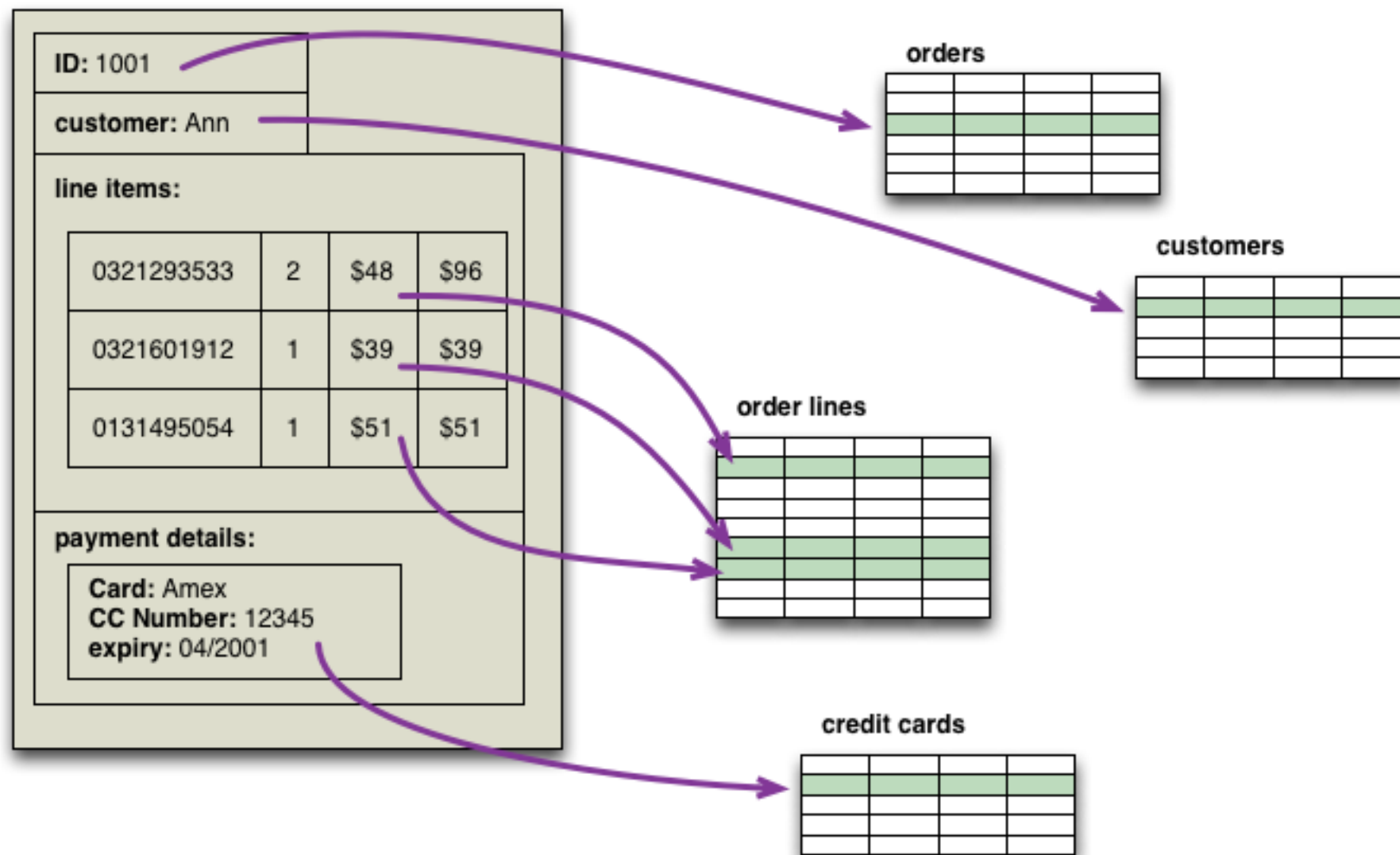
```
1  WITH tmp_1 AS
2  (
3  SELECT Calc1 =
4  ( (SELECT TOP 1 DataValue
5  FROM (
6  SELECT TOP 50 PERCENT DataValue
7  FROM SOMEDATA
8  WHERE DataValue IS NOT NULL
9  ORDER BY DataValue
10 ) AS A
11 ORDER BY DataValue DESC
12 ) + (SELECT TOP 1 DataValue
13 FROM (
14 SELECT TOP 50 PERCENT DataValue
15 FROM SOMEDATA
16 WHERE DataValue Is NOT NULL
17 ORDER BY DataValue DESC
18 )AS A
19 ORDER BY DataValue ASC))/2
20 ),tmp_2 AS
21 (SELECT AVG(DataValue) Mean, MAX(DataValue) - MIN(DataValue) AS MaxMinRange
22 FROM SOMEDATA
23 ),tmp_3 AS
24 (
25 SELECT TOP 1 DataValue AS Calc2, COUNT(*) AS Calc2Count
26 FROM SOMEDATA
27 GROUP BY DataValue
28 ORDER BY Calc2Count DESC
29 )
30 SELECT Mean, Calc1, Calc2 , MaxMinRange AS [Range]
31 FROM tmp_1 CROSS JOIN tmp_2 CROSS JOIN tmp_3;
```



# Problems ?



# NoSQL

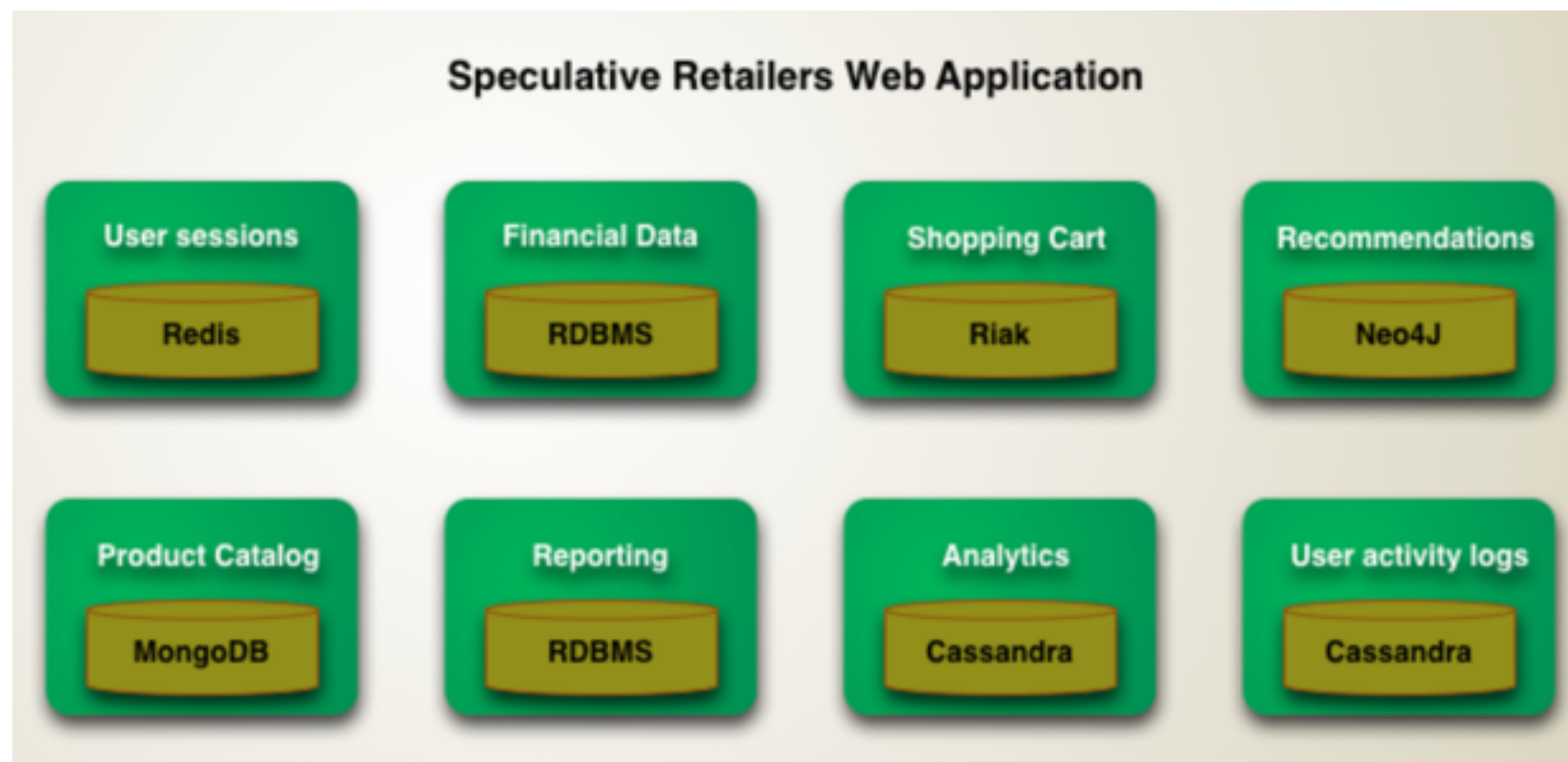


<https://martinfowler.com/bliki/AggregateOrientedDatabase.html>



# Polyglot Persistence

Right tool for the right job



<https://martinfowler.com/bliki/PolyglotPersistence.html>



# Data Modeling

Read

Write





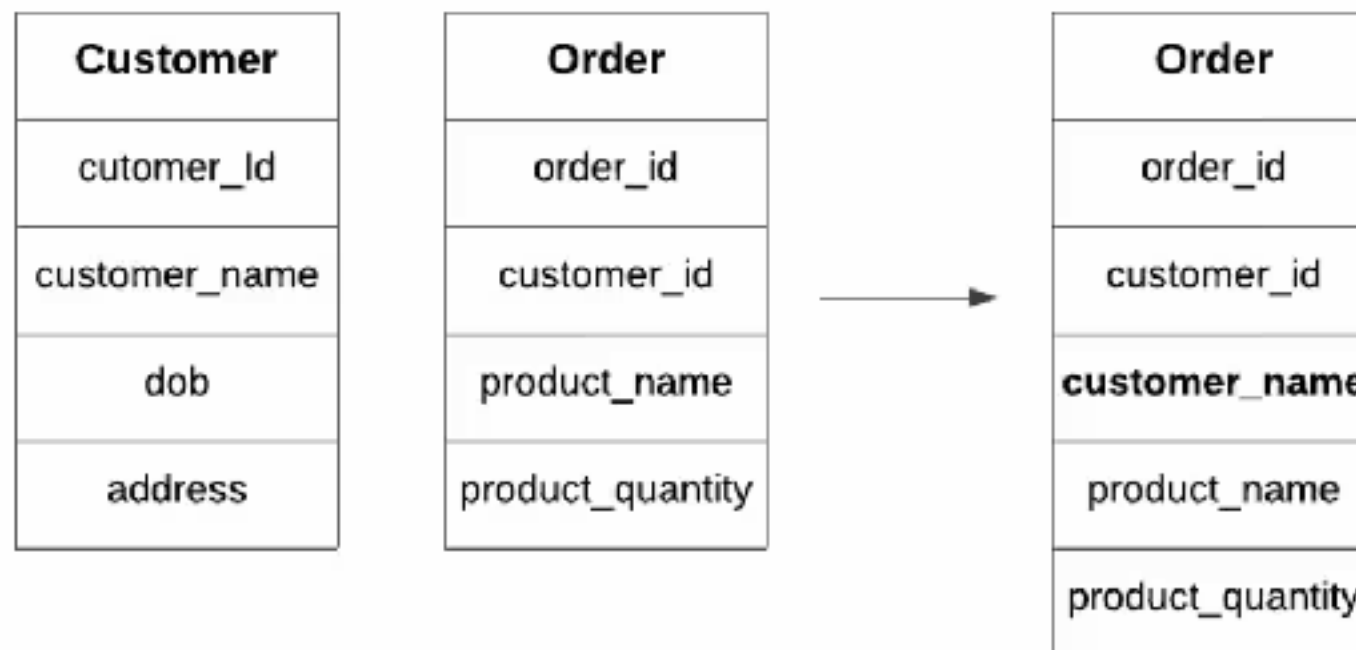
# Data Modeling

Normalization

Denormalization

Pre-joins, pre-aggregation

Caching strategies





# Normalization ?

Cleansing data

Reorganized data

Remove redundant data

Improve efficient for store data

***Organize data in consistent way***



# Normalization

Divide in to several tables  
Link tables with **relationships !!**

Design for write or read ?

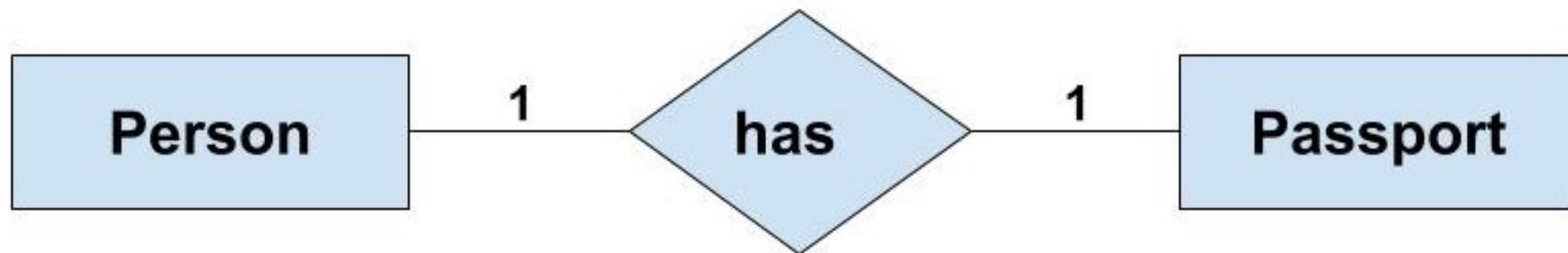


# Relationships



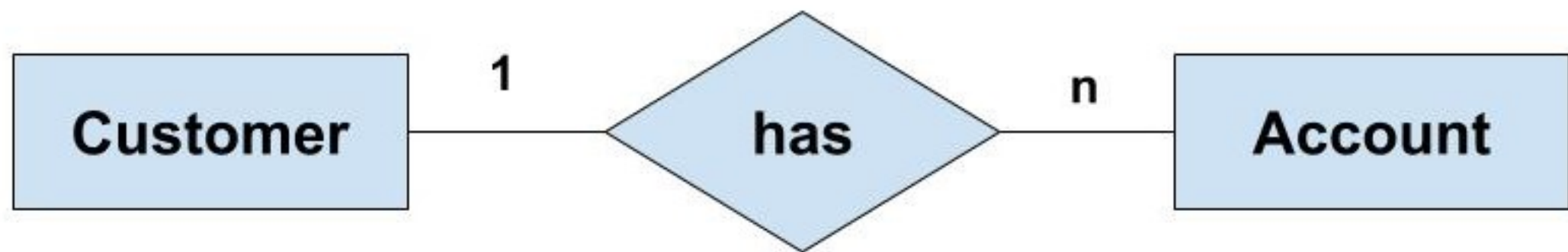
# Relationship between table

One-to-one  
One-to-many  
Many-to-many



# Relationship between table

One-to-one  
**One-to-many**  
Many-to-many



# Relationship between table

One-to-one  
One-to-many  
**Many-to-many**

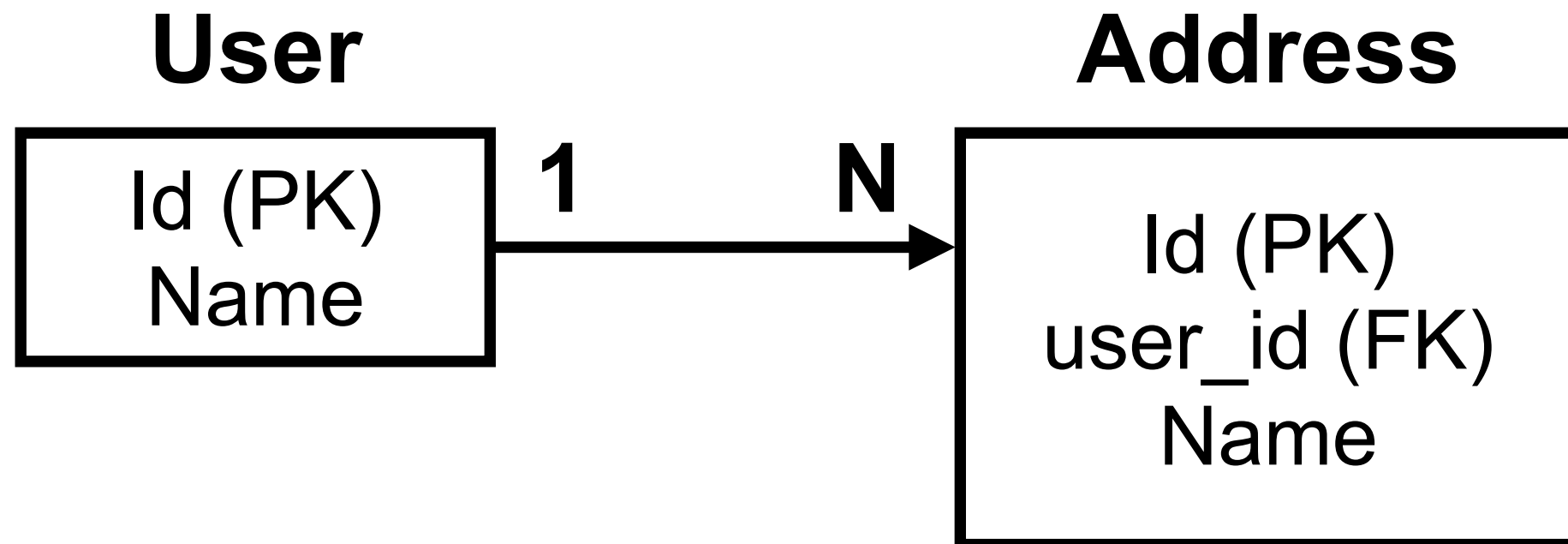


# Relationships

Primary key (PK)

Composite key

Foreign key (FK)





# Normalization forms

**First Normal form (1NF)**

**2NF**

**3NF**

**4NF**

**5NF**

**6NF**





# First Normal Form (1NF)

No repeat data in table

Every column should only have single value

Every row should be unique



# Sample data in 1NF

| Name   | Address   | Gender | T-shirt Size |
|--------|-----------|--------|--------------|
| User 1 | Address 1 | Male   | Large        |
| User 2 | Address 2 | Female | Small        |
| User 2 | Address 2 | Female | Medium       |
| User 3 | Address 3 | Male   | Medium       |



# Second Normal Form (2NF)

Apply 1NF

Have one primary key



# Problem in 1NF

| Name   | Address   | Gender | T-shirt Size |
|--------|-----------|--------|--------------|
| User 1 | Address 1 | Male   | Large        |
| User 2 | Address 2 | Female | Small        |
| User 2 | Address 2 | Female | Medium       |
| User 3 | Address 3 | Male   | Medium       |



# Sample data in 2NF

| User ID | Name   | Address   | Gender |
|---------|--------|-----------|--------|
| 1       | User 1 | Address 1 | Male   |
| 2       | User 2 | Address 2 | Female |
| 3       | User 3 | Address 3 | Male   |



# Sample data in 2NF

| User ID | T-shirt Size |
|---------|--------------|
| 1       | Large        |
| 2       | Small        |
| 2       | Medium       |
| 3       | Medium       |



# Third Normal Form (3NF)

Apply 2NF

It should only be dependent on the primary key

if any changes to the primary key are made,  
all data that is impacted must be put into a  
new table





# Problem in 2NF

| User ID | Name   | Address   | Gender |
|---------|--------|-----------|--------|
| 1       | User 1 | Address 1 | Male   |
| 2       | User 2 | Address 2 | Female |
| 3       | User 3 | Address 3 | Male   |





# Sample data in 3NF

| User ID | Name   | Address   | Gender |
|---------|--------|-----------|--------|
| 1       | User 1 | Address 1 | 1      |
| 2       | User 2 | Address 2 | 2      |
| 3       | User 3 | Address 3 | 1      |



# Sample data in 3NF

| Gender ID | Gender name |
|-----------|-------------|
| 1         | Male        |
| 2         | Female      |



# Sample data in 3NF

| User ID | T-shirt Size |
|---------|--------------|
| 1       | Large        |
| 2       | Small        |
| 2       | Medium       |
| 3       | Medium       |



# 3NF is normal form ...



# Benefits of data normalization

Freeing up space  
Improve query response time  
Reduce data inconsistency  
Easier to use and analyze



# Challenges of data normalization

Slower query response rates

Accurate knowledge is required

Add complexity for teams

Denormalization as an alternative !!



# Primary Key (PK)





# Foreign Key (FK)





# Other Constraints

Ensure data quality

Prevent inconsistency at the database level

Not Null

Unique

Check

Default



# Workshop with PostgreSQL



PostgreSQL



# PostgreSQL Database

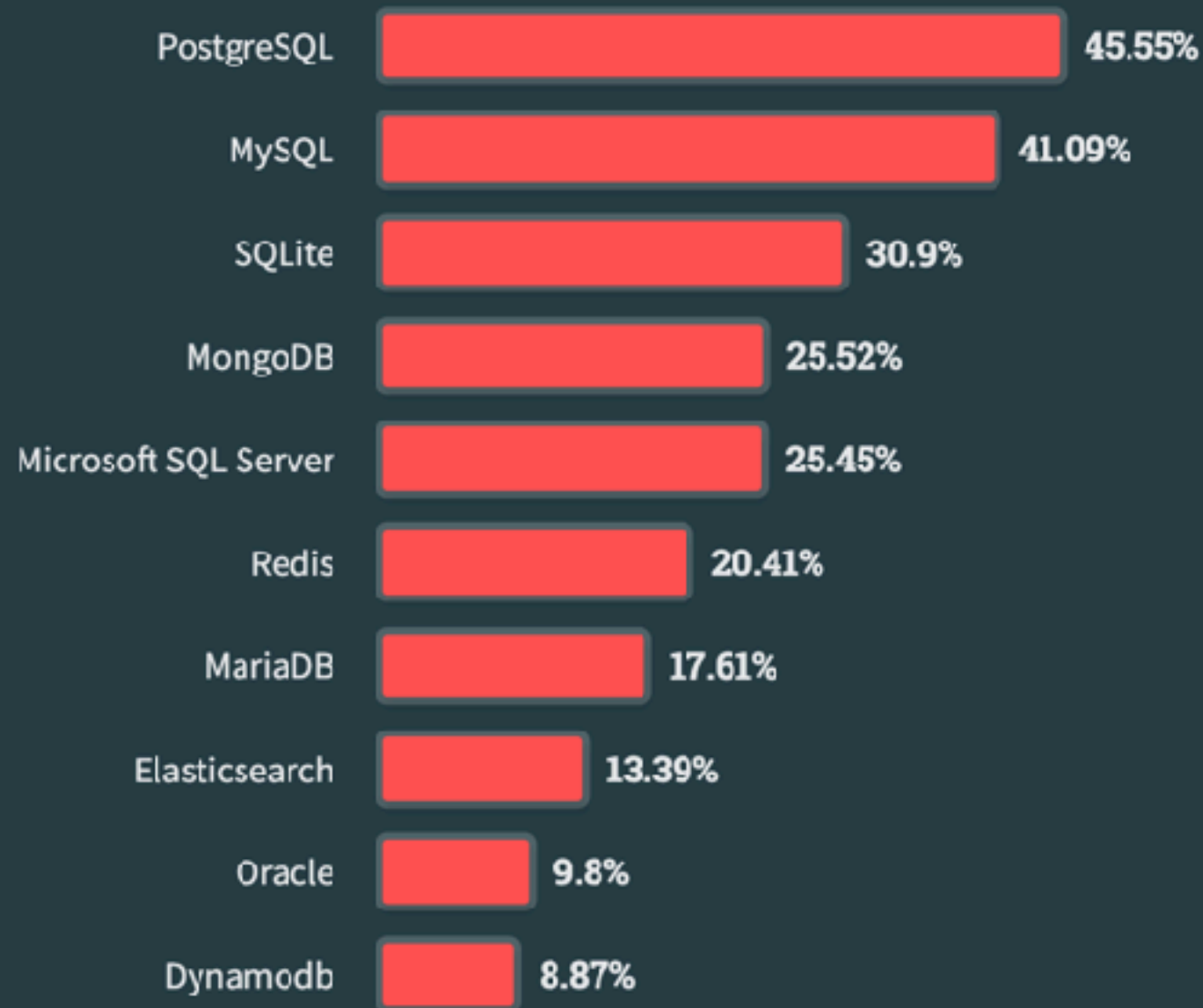
All Respondents

Professional Developers

Learning to Code

Other Coders

76,634 responses



<https://survey.stackoverflow.co/2023/#most-popular-technologies-database>



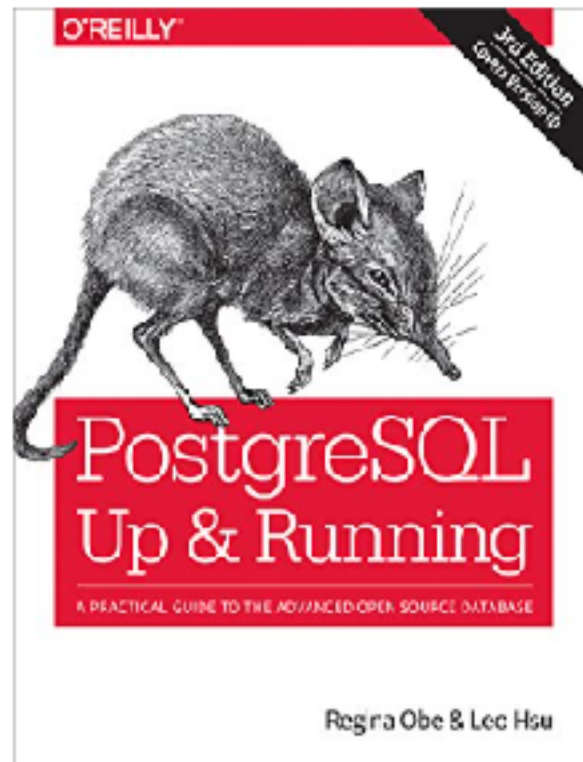
# Book Store

QuickStart  
Guides

Master SQL and Manage, Analyze and Manipulate Data

\$25<sup>19</sup>

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Read sample

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## PostgreSQL: Up and Running: A Practical Guide to the Advanced Open Source Database 3rd Edition, Kindle Edition

by Regina O. Obe (Author), Leo S. Hsu (Author) | Format: Kindle Edition

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With examples throughout, this book shows you how to achieve tasks that are difficult or impossible in other databases. This third edition covers new features, such as ANSI-SQL constructs found only in proprietary databases until now: foreign data wrapper (FDW) enhancements, new full-text search and partitioning introduced in version 9.5, and more. [Read more](#)

ISBN-13



978-1491963418

Edition



3rd

Sticky notes



[On Kindle Scribe](#)

Publisher



O'Reilly Media

Publication date



October 10, 2017

Language



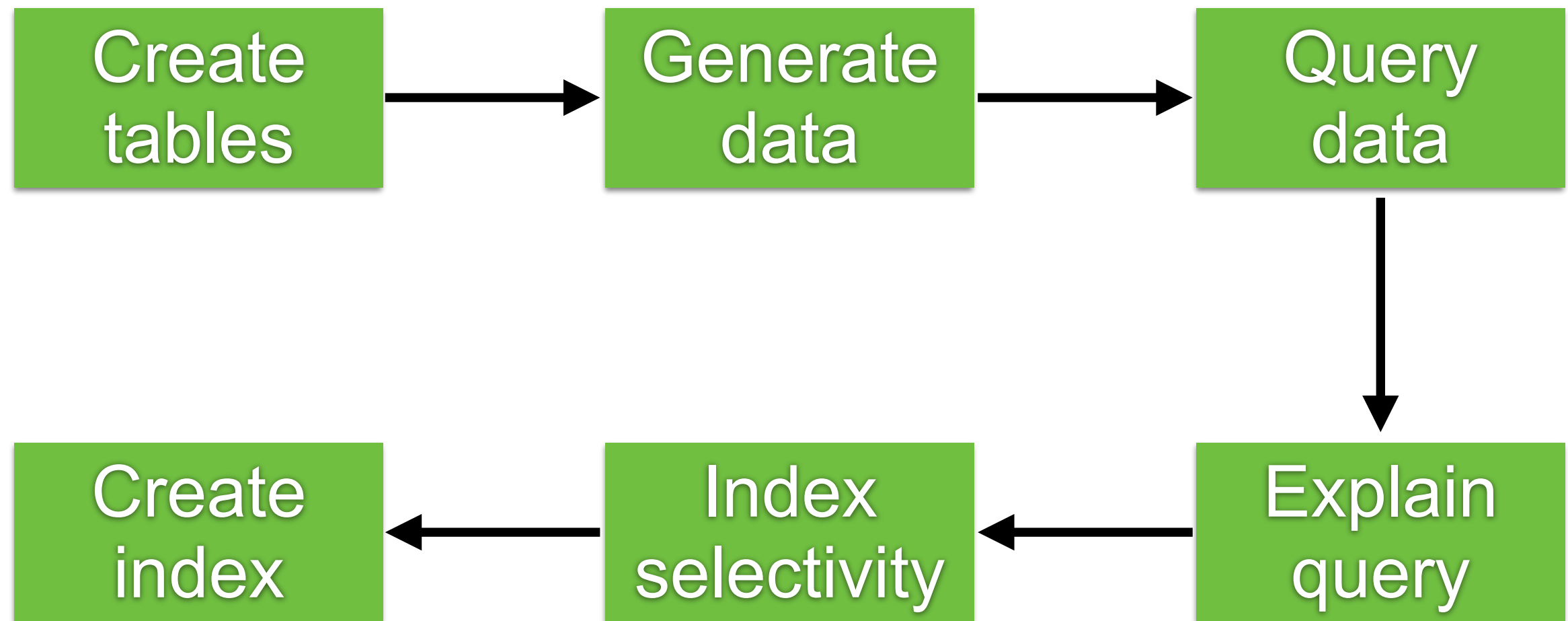
English



# Book Store



# Step in workshop



<https://github.com/up1/workshop-database/tree/main/workshop/basic-workshop>



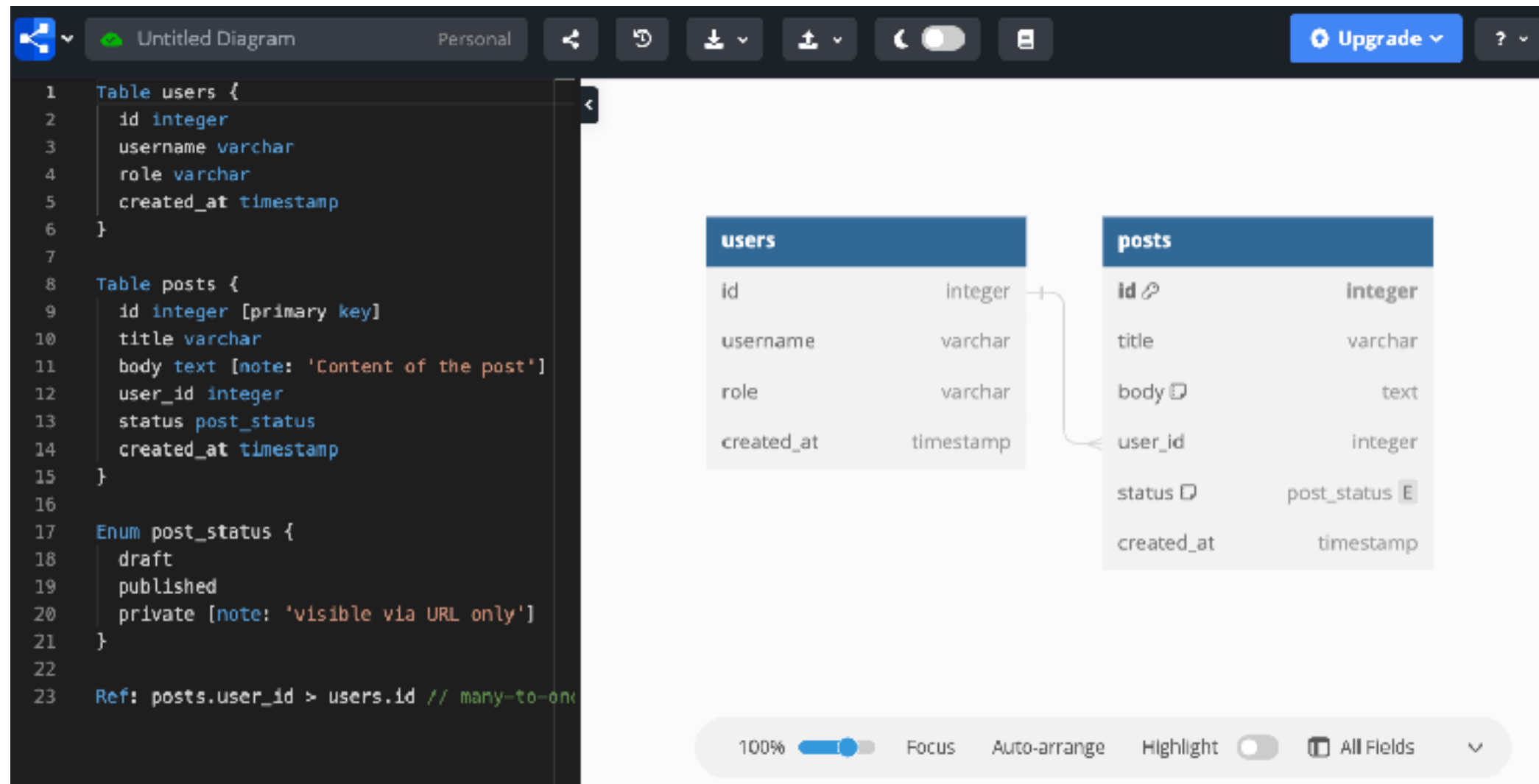
# Design





# Design table with DBML

## Database Markup Language



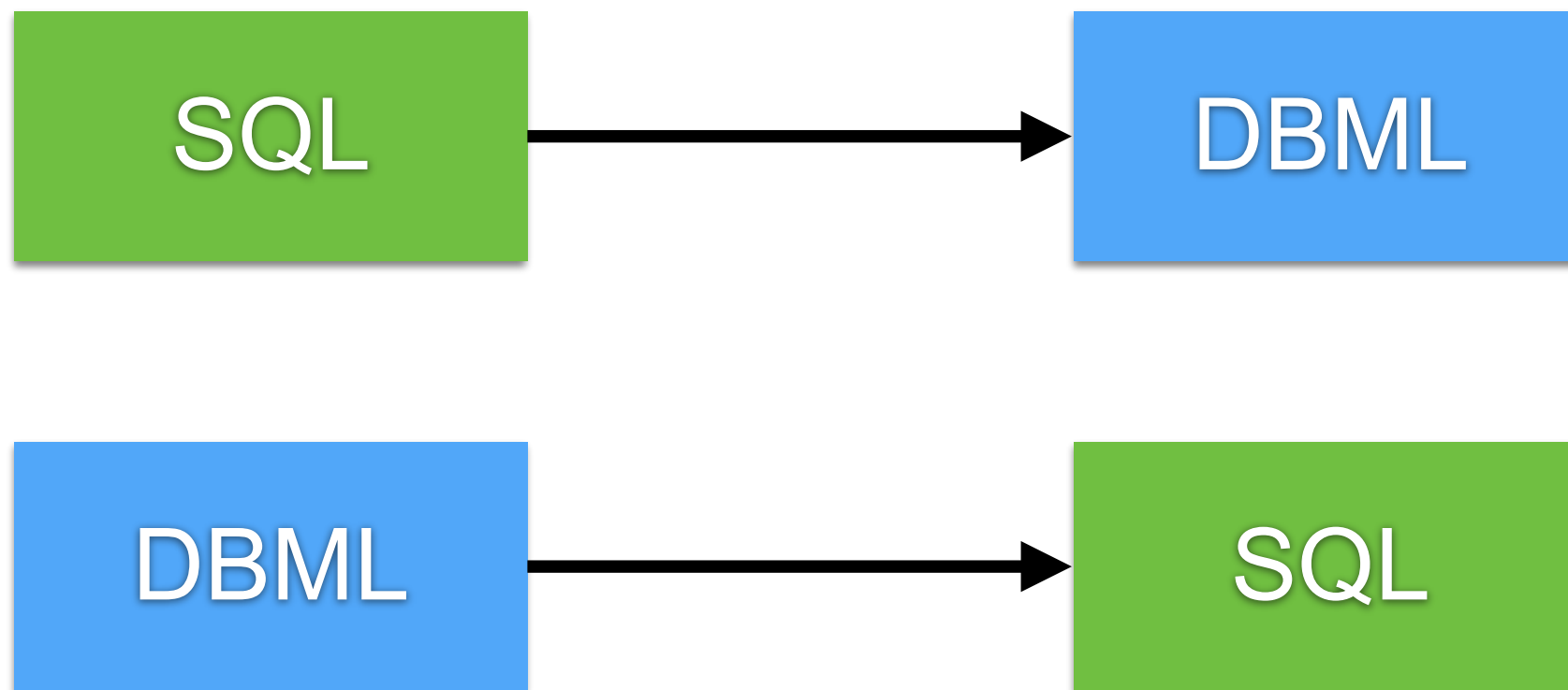
<https://dbml.dbdiagram.io/home/>





# Convert SQL to DBML

```
$npm install -g @dbml/cli
```

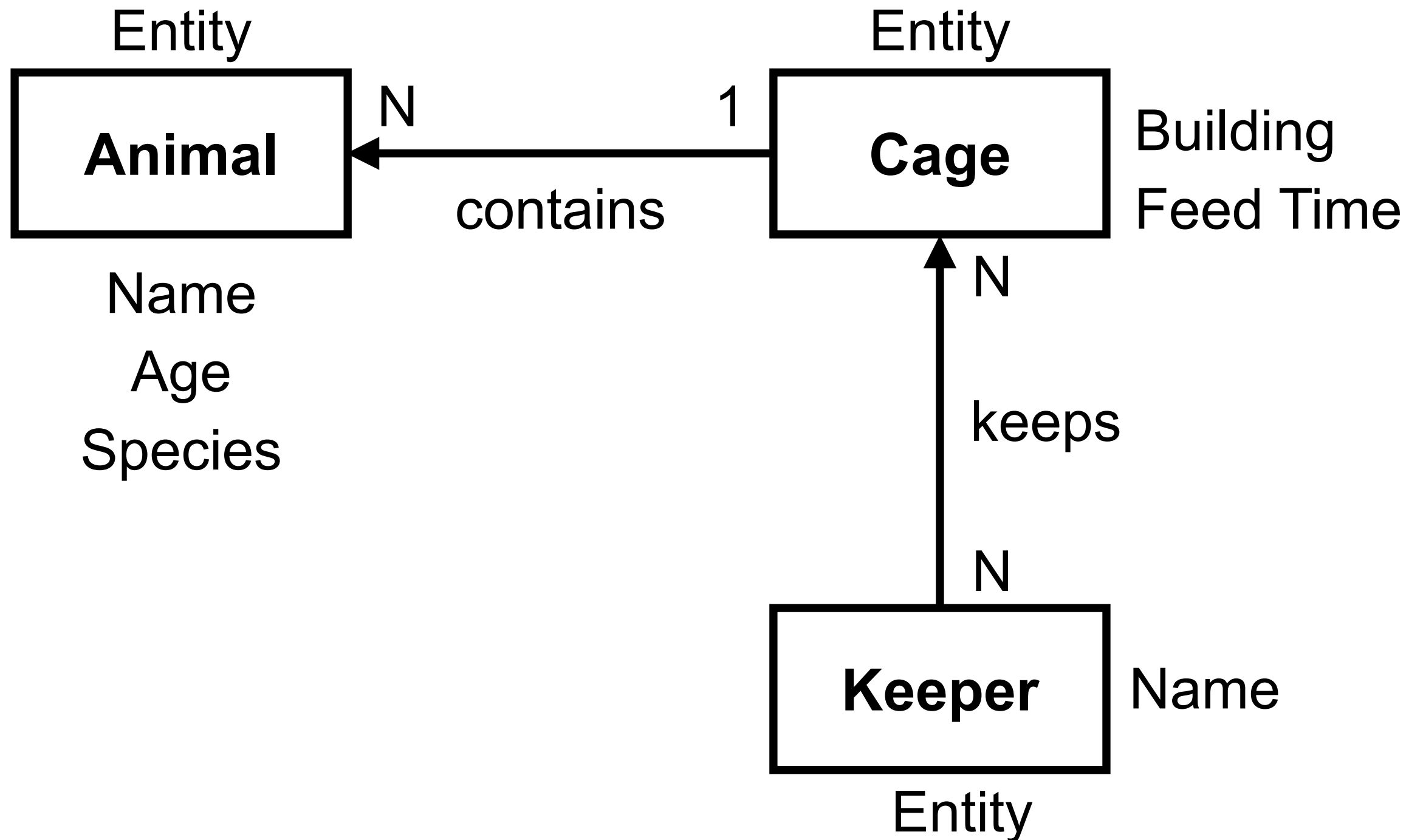


<https://dbml.dbdiagram.io/cli/#convert-a-sql-file-to-dbml>



# Design with entity relation diagram





# How to manage RDBMS ?



# Database Language

**Data Definition Language (DDL)**  
**Data Manipulation Language (DML)**  
Data Control Language (DCL)  
Transaction Control Language (TCL)



# Data Definition Language (DDL)

Create  
Alter  
Drop  
Truncate



# Data Manipulation Language (DML)

Select  
Insert  
Update  
Delete





# Delete vs Truncate ?





# Delete vs Truncate

## Delete

Row-level operation

Mark every row matching the WHERE-clause as deleted

Slow operation

## Truncate

Table operation

Empty the table

Start a new data file



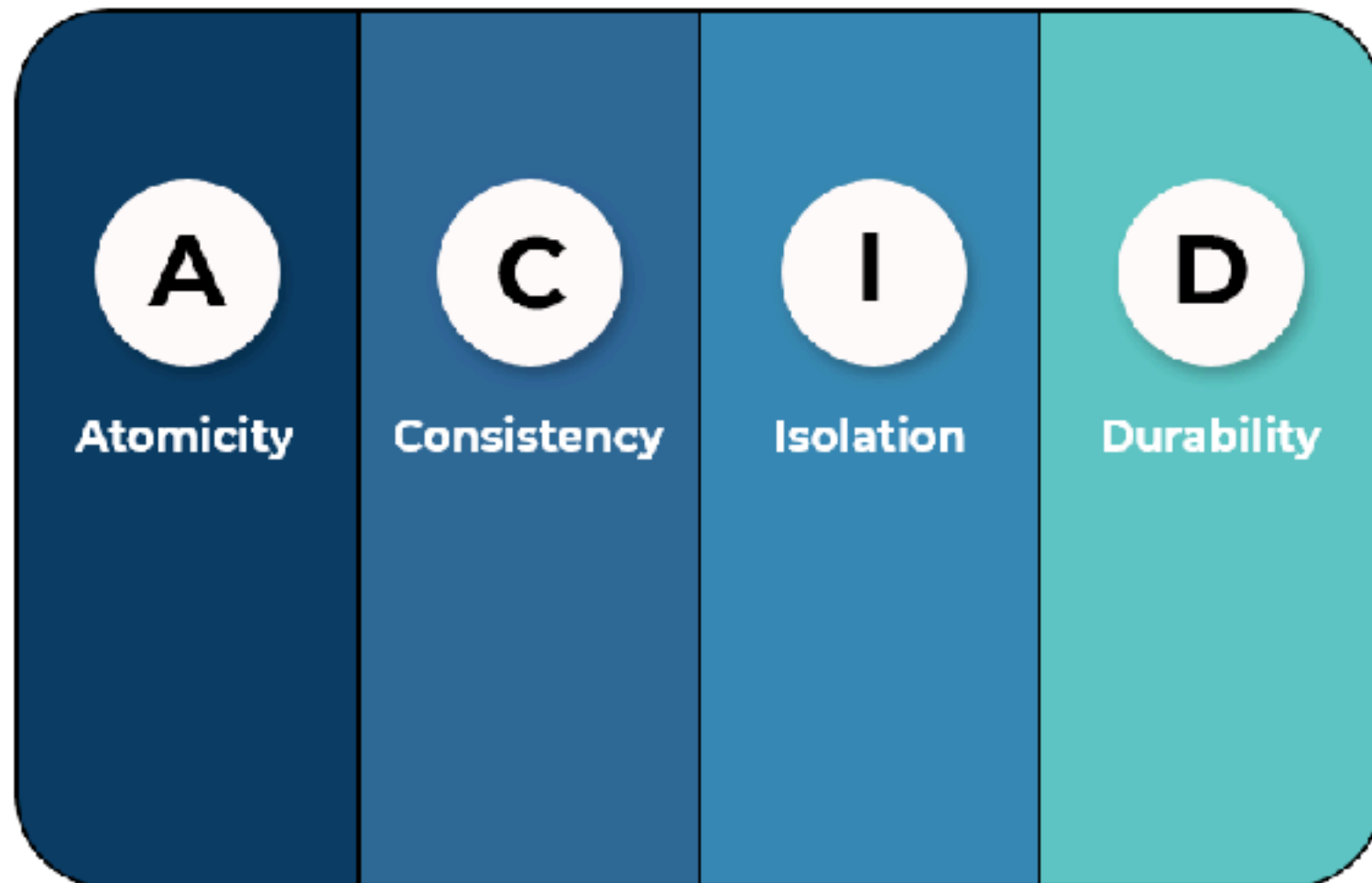
# ACID in RDBMS

<https://en.wikipedia.org/wiki/ACID>

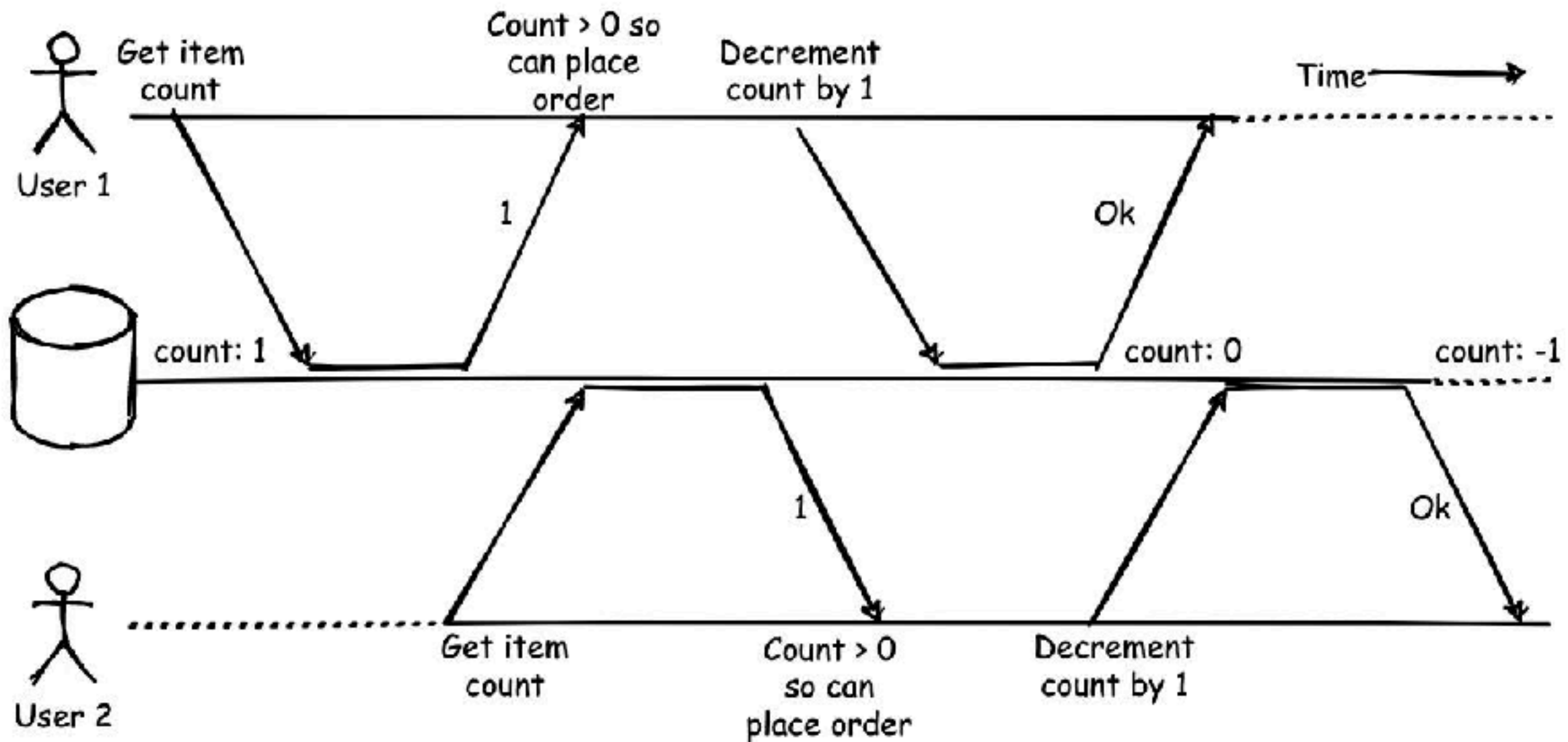


# ACID in RDBMS

Ensure database in valid state even in the event of unexpected errors



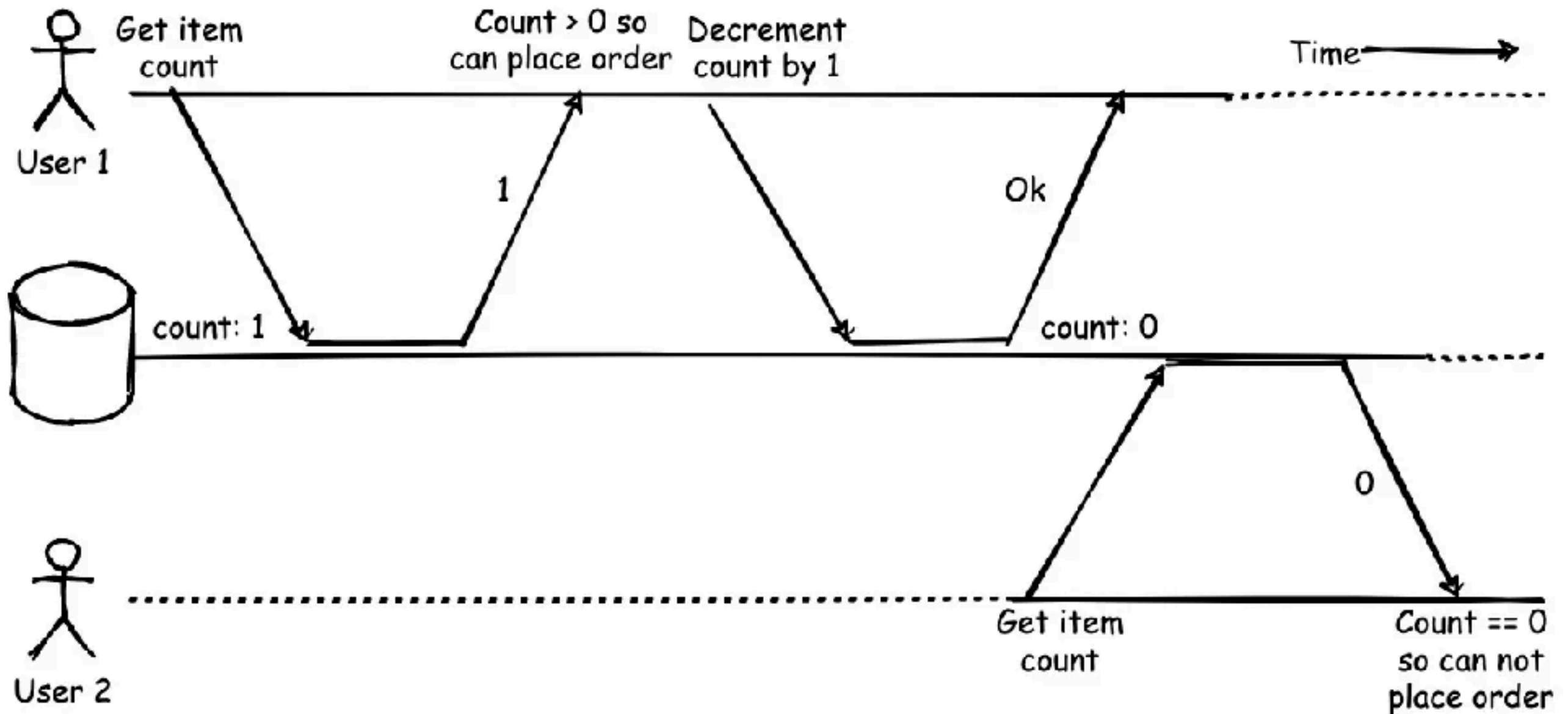
# Race condition



[https://en.wikipedia.org/wiki/Race\\_condition#Example](https://en.wikipedia.org/wiki/Race_condition#Example)



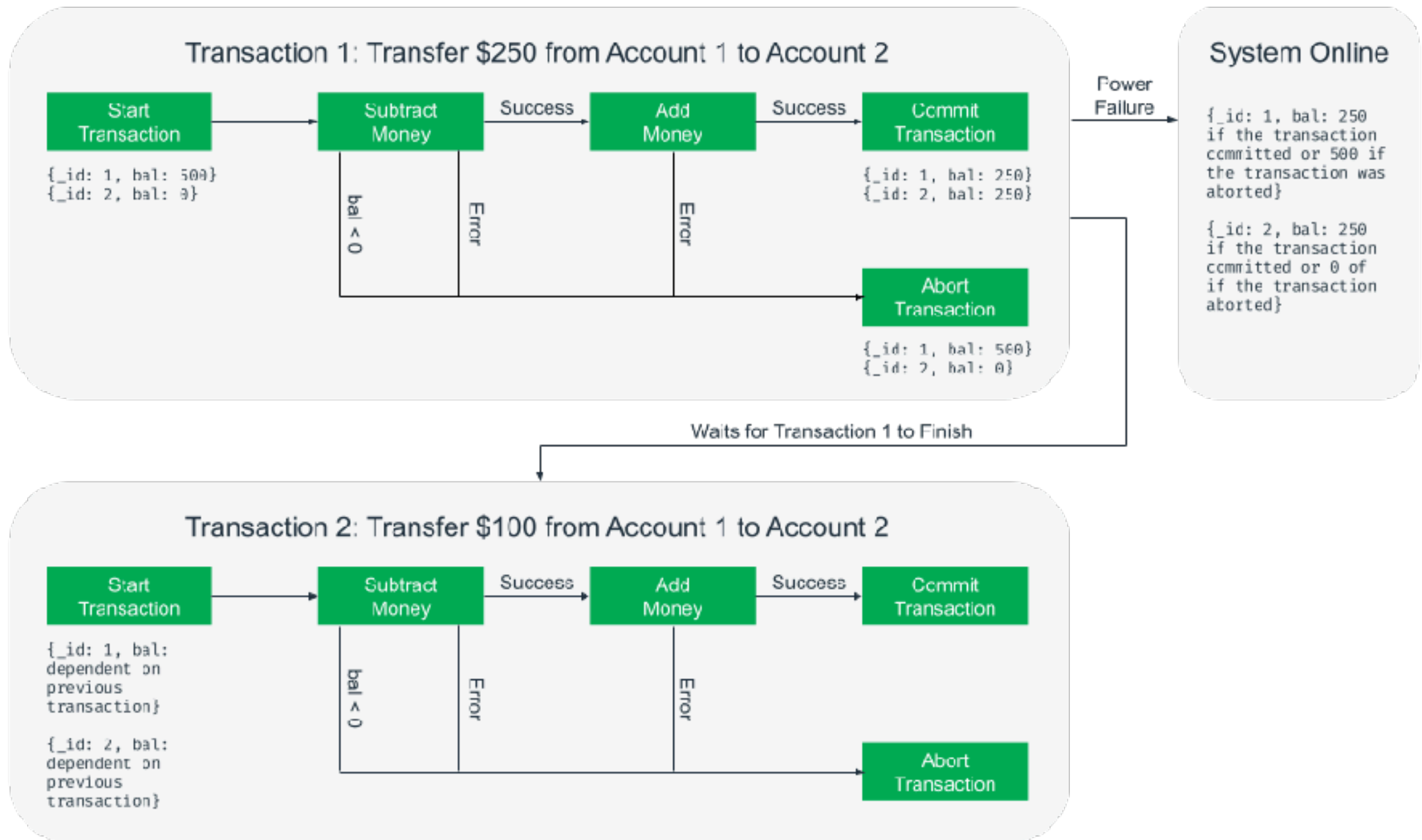
# Sequential



<https://www.mongodb.com/basics/acid-transactions>



# ACID in RDBMS



<https://www.mongodb.com/basics/acid-transactions>



# Structure Query Language





# SQL

Maintain data in **tables**

**Pre-define** structure

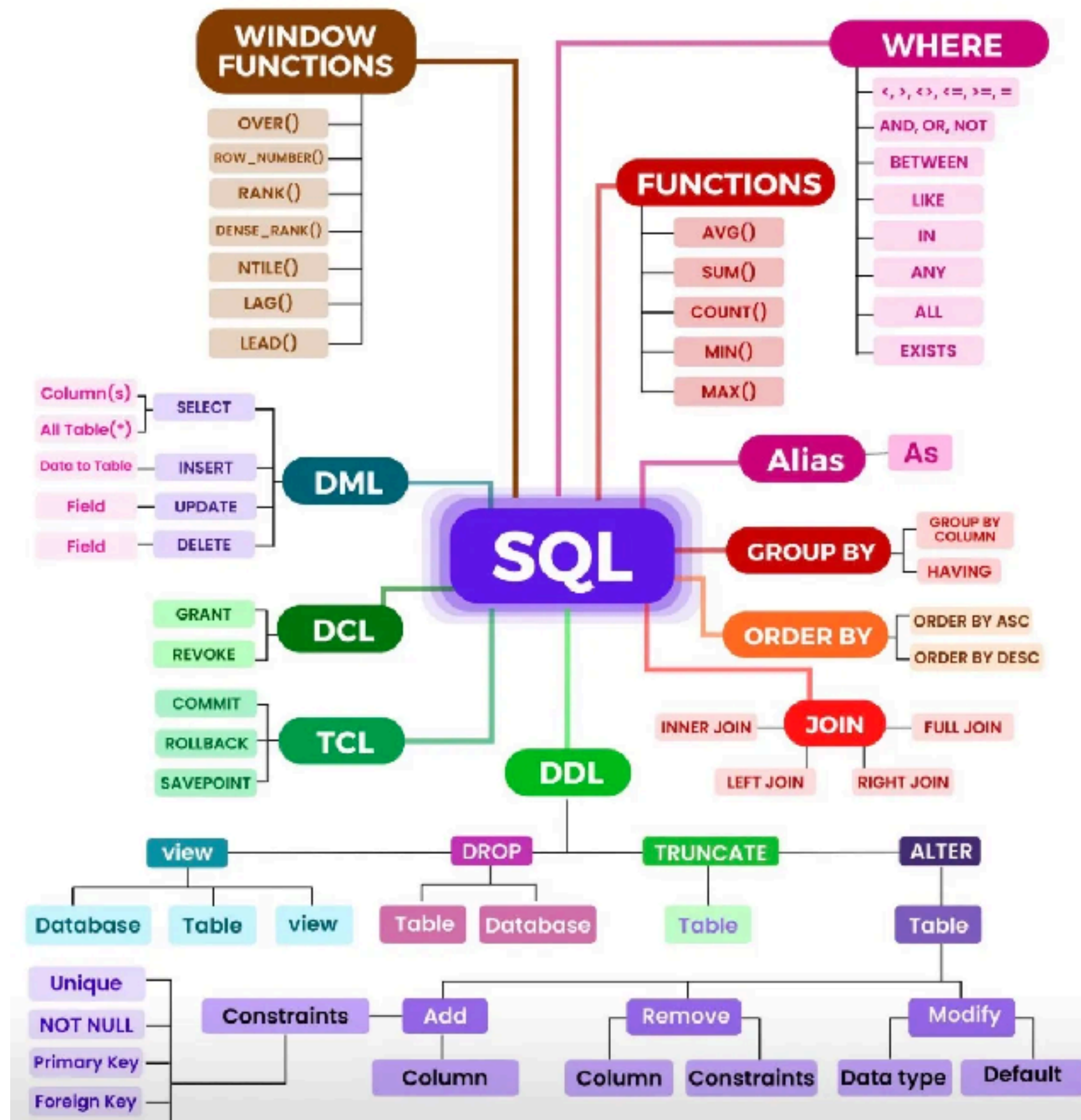
Each table has **Primary key**

Many **columns**

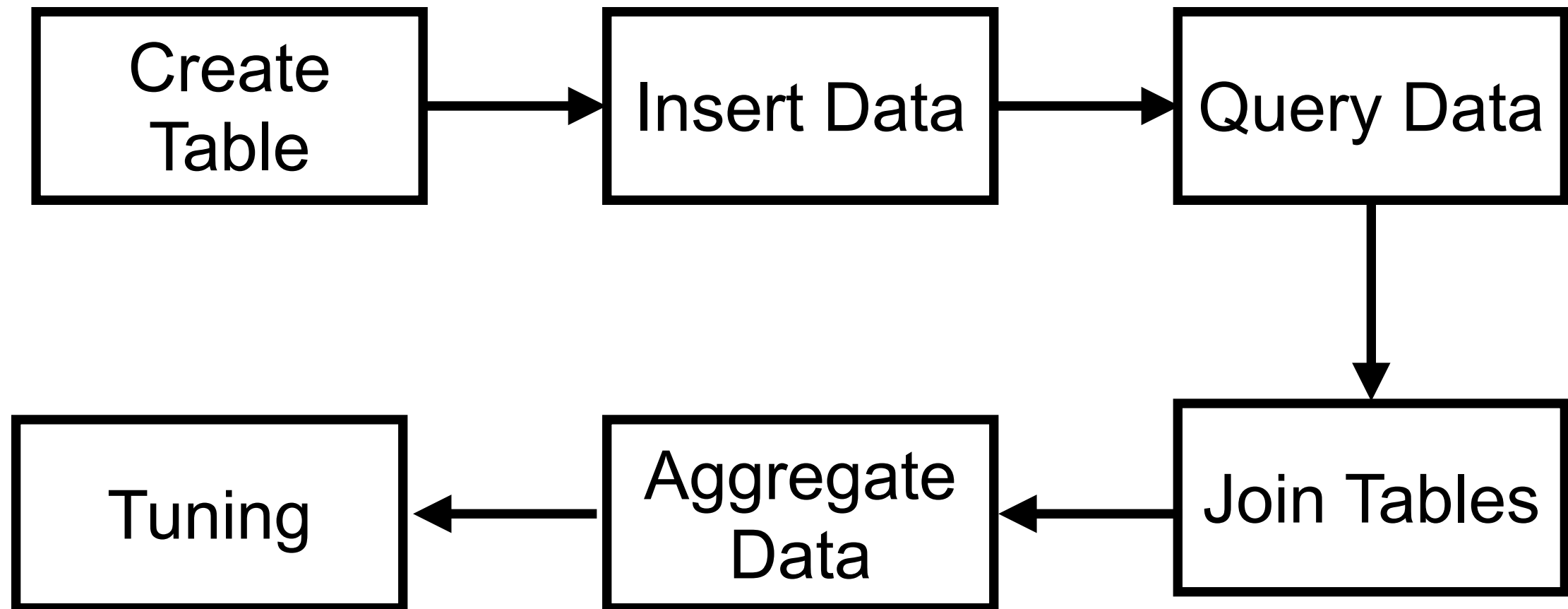
Data is represented in term of **row**







# Learning paths



# Query Data



**SELECT** columns  
**FROM** table  
**WHERE** condition A  
**AND** condition B  
**GROUP BY** columns  
**ORDER BY** columns **ASC|DESC**



# Select Data process

Table A

Table B

Table C



# Select Data process

Table A

Table B

Table C

FROM + JOIN  
(merge)

Table A, B, C



# Select Data process

Table A

Table B

Table C

FROM + JOIN  
(merge)

Table A, B, C

WHERE  
(filter)

Table A, B, C





# Select Data process

Table A

Table B

Table C

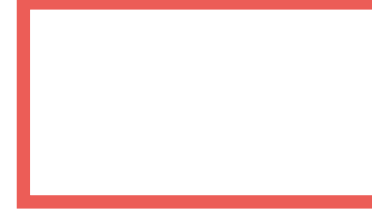
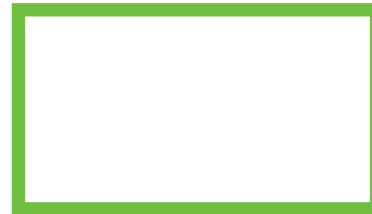
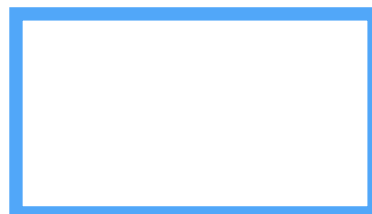
FROM + JOIN  
(merge)

Table A, B, C

WHERE  
(filter)

Table A, B, C

Group By  
(Group)





# Select Data process

Table A

Table B

Table C

FROM + JOIN  
(merge)

Table A, B, C

WHERE  
(filter)

Table A, B, C

Group By  
(Group)

Having  
(filter)

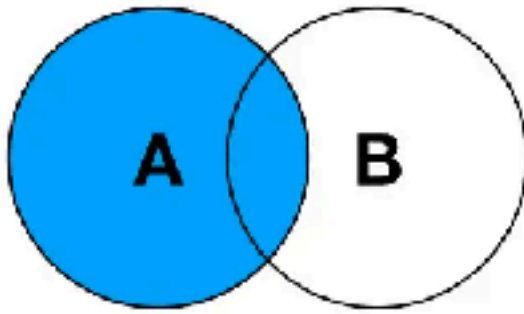


# Joining

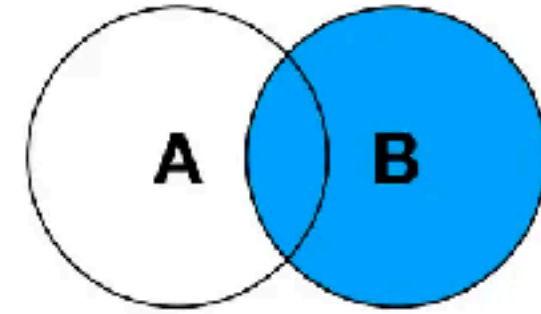
<https://www.postgresql.org/docs/current/tutorial-join.html>



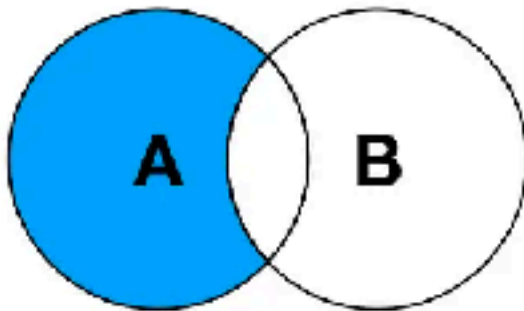
# SQL JOINS



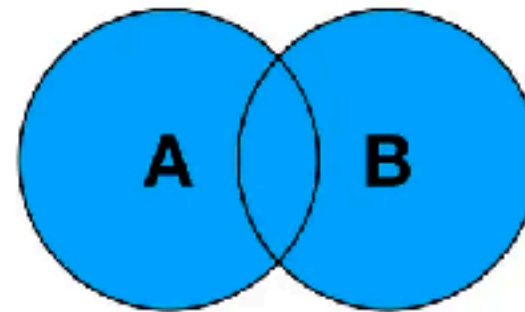
**LEFT JOIN**



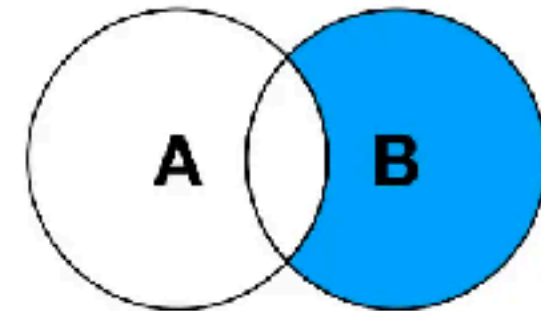
**RIGHT JOIN**



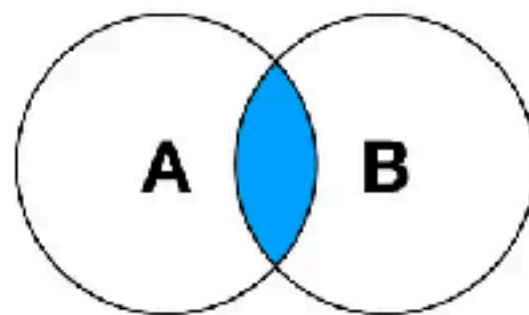
**LEFT JOIN EXCLUDING  
INNER JOIN**



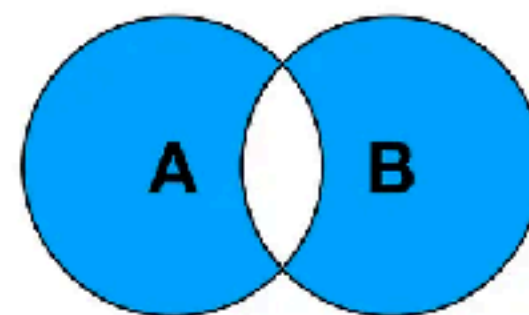
**FULL OUTER JOIN**



**RIGHT JOIN EXCLUDING  
INNER JOIN**



**INNER JOIN**



**FULL OUTER JOIN EXCLUDING  
INNER JOIN**



Self-join

# Join in PostgreSQL

Left join

Inner join

Right join

Full outer  
join



# Workshop

Basket A

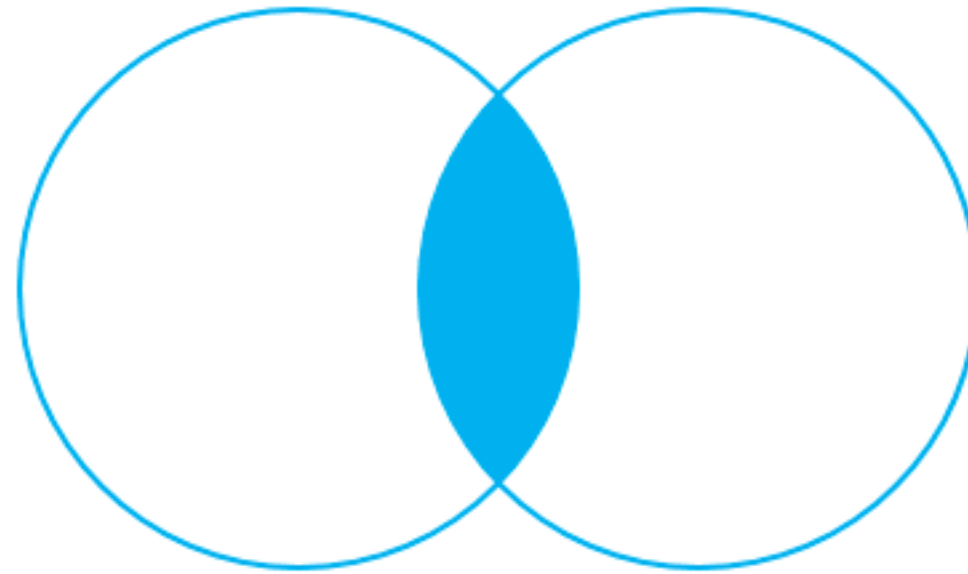
Apple  
Orange  
Banana  
Cucumber

Basket B

Orange  
Apple  
Watermelon  
Pear



# Inner join



INNER JOIN



# Inner join

Basket A

**Apple**  
**Orange**  
Banana  
Cucumber

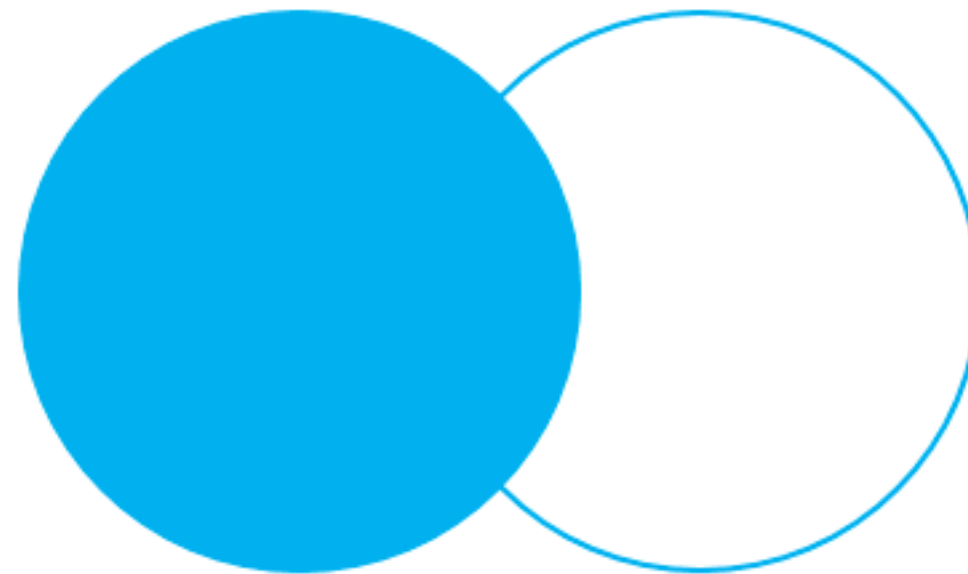
Basket B

**Orange**  
**Apple**  
Watermelon  
Pear





# Left join (1)



LEFT OUTER JOIN





# Left join (1)

**Basket A**

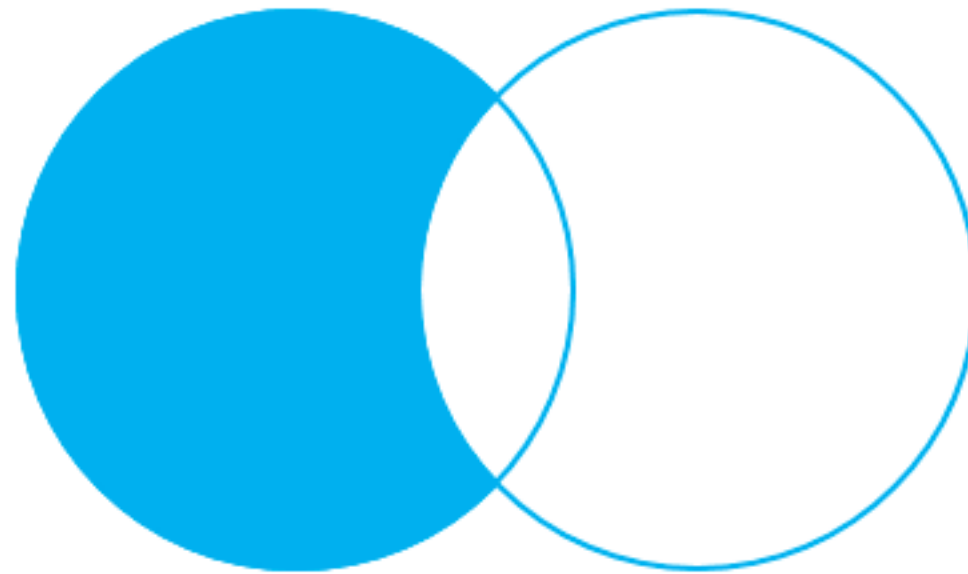
**Apple**  
**Orange**  
**Banana**  
**Cucumber**

**Basket B**

**Orange**  
**Apple**  
**Watermelon**  
**Pear**



# Left join (2)



LEFT OUTER JOIN – only  
rows from the left table



# Left join (2)

Basket A

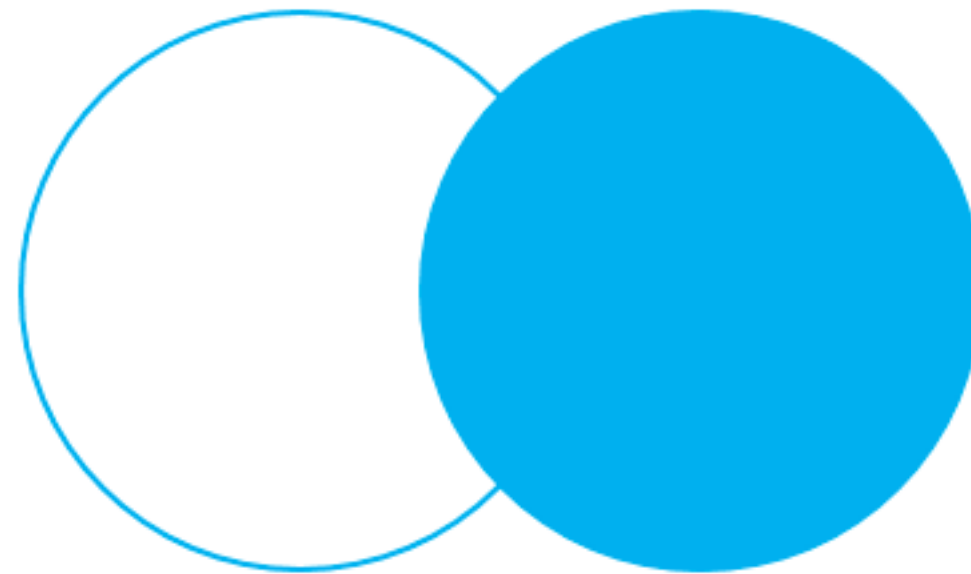
Apple  
Orange  
**Banana**  
**Cucumber**

Basket B

Orange  
Apple  
Watermelon  
Pear



# Right join (1)



RIGHT OUTER JOIN



# Right join (1)

Basket A

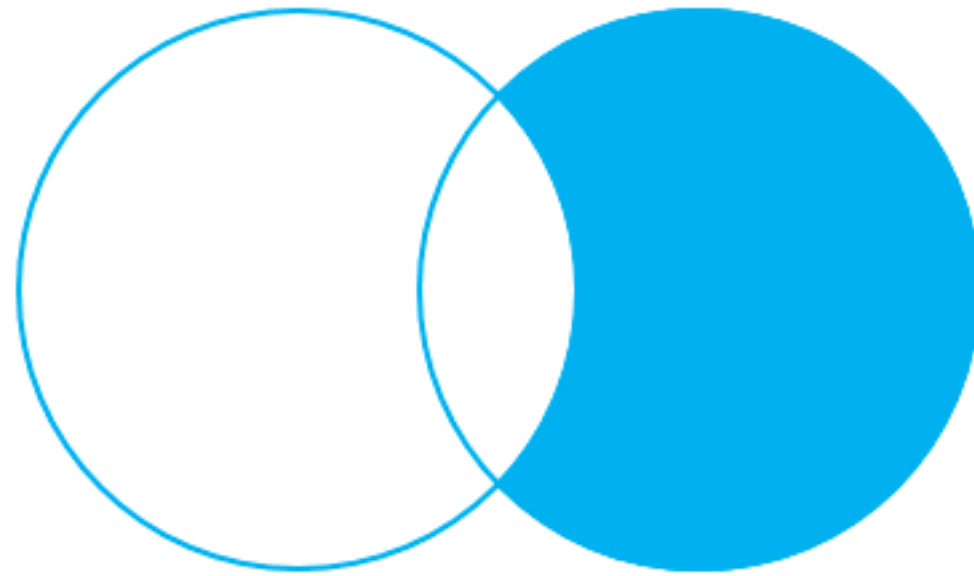
Apple  
Orange  
**Banana**  
**Cucumber**

Basket B

**Orange**  
**Apple**  
**Watermelon**  
**Pear**



# Right join (2)



RIGHT OUTER JOIN – only  
rows from the right table



# Right join (2)

Basket A

Apple  
Orange  
Banana  
Cucumber

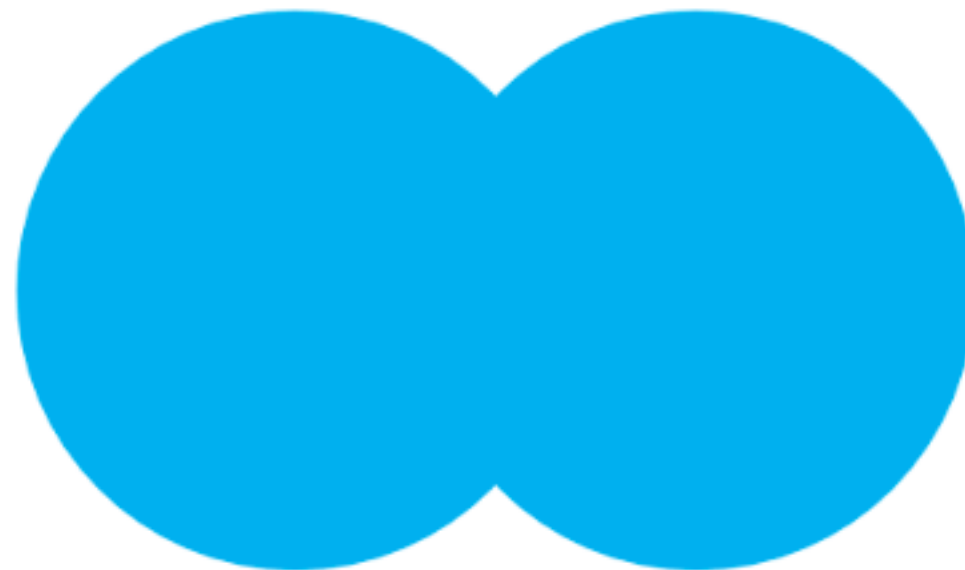
Basket B

Orange  
Apple  
**Watermelon**  
**Pear**





# Full outer join (1)

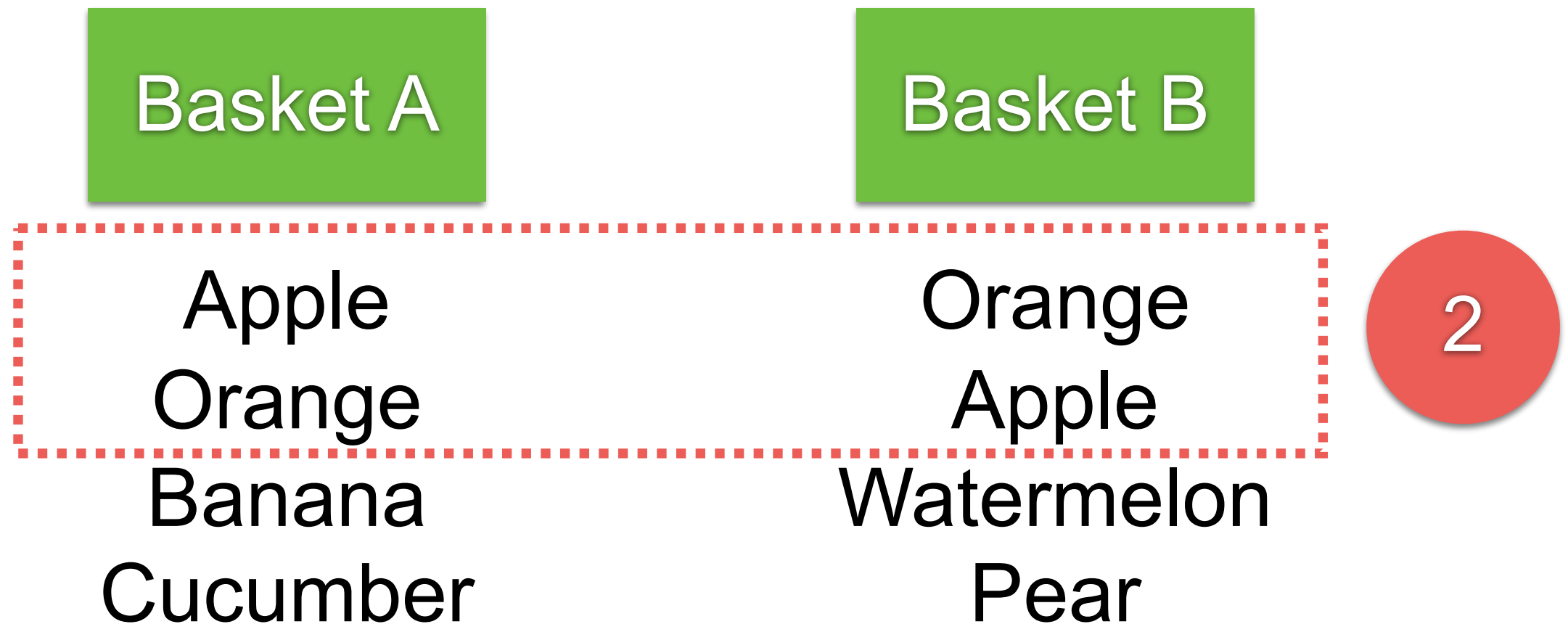


FULL OUTER JOIN





# Full outer join (1)



# Full outer join (1)

Basket A

Apple  
Orange  
Banana  
Cucumber

Basket B

Orange  
Apple  
Watermelon  
Pear

2



# Full outer join (1)

Basket A

Apple  
Orange  
Banana  
Cucumber

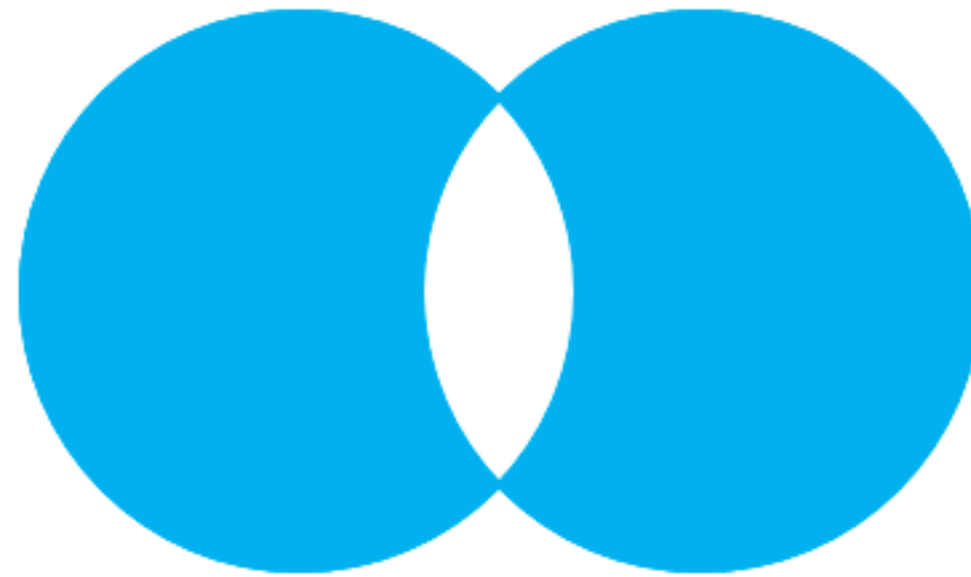
Basket B

Orange  
Apple  
Watermelon  
Pear

2



# Full outer join (2)



FULL OUTER JOIN – only  
rows unique to both tables



# Full outer join (2)

Basket A

Apple  
Orange  
Banana  
Cucumber

Basket B

Orange  
Apple  
Watermelon  
Pear

2



# Full outer join (2)

Basket A

Apple  
Orange  
Banana  
Cucumber

Basket B

Orange  
Apple  
Watermelon  
Pear

2



# Workshop with Joins

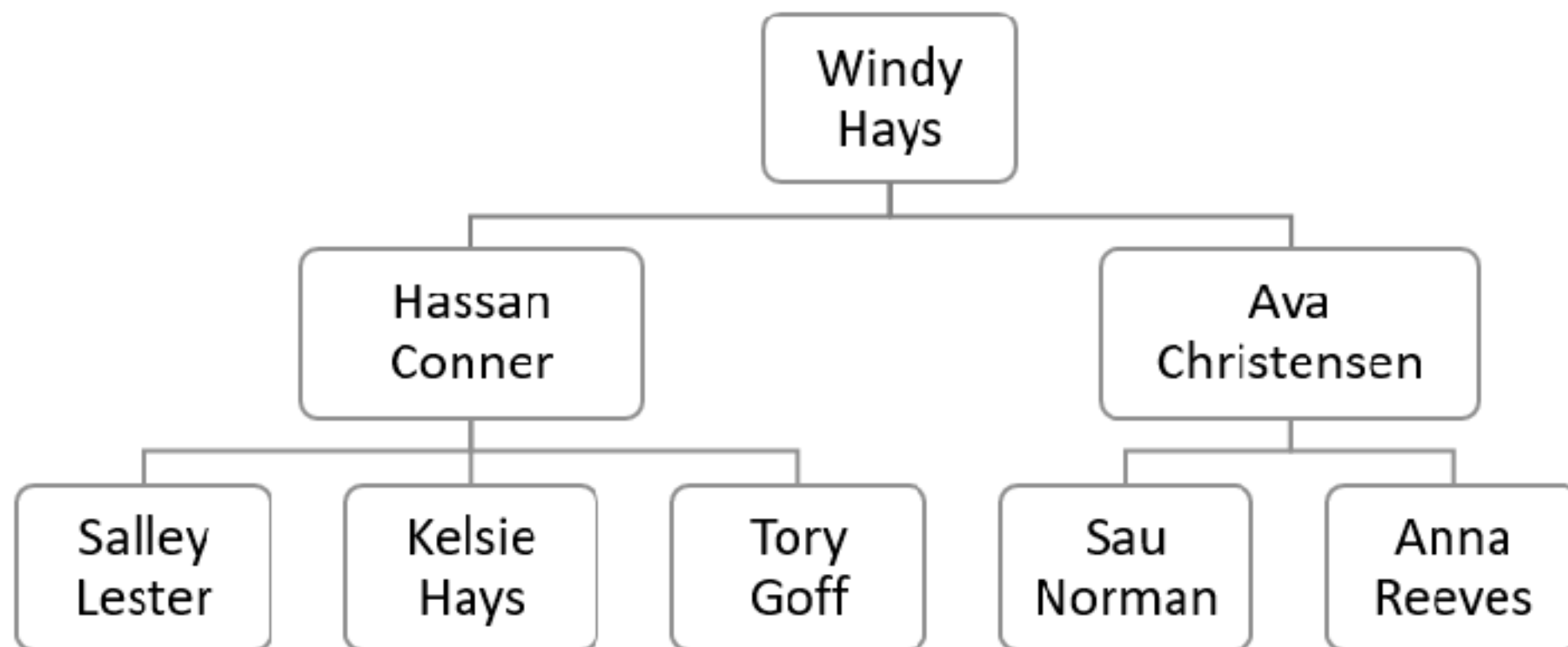
<https://github.com/up1/workshop-database/blob/main/workshop/joins.md>





# Self-join !!!

Regular join that joins a table to itself  
Query hierarchical data





# Workshop with Self-Join

<https://github.com/up1/workshop-database/blob/main/workshop/self-join.md>



# Problems ?



# Problems

Slow query

Accurate knowledge is required

Add complexity

Need denormalization ?

Rewrite query ?

Database vs application caching ?

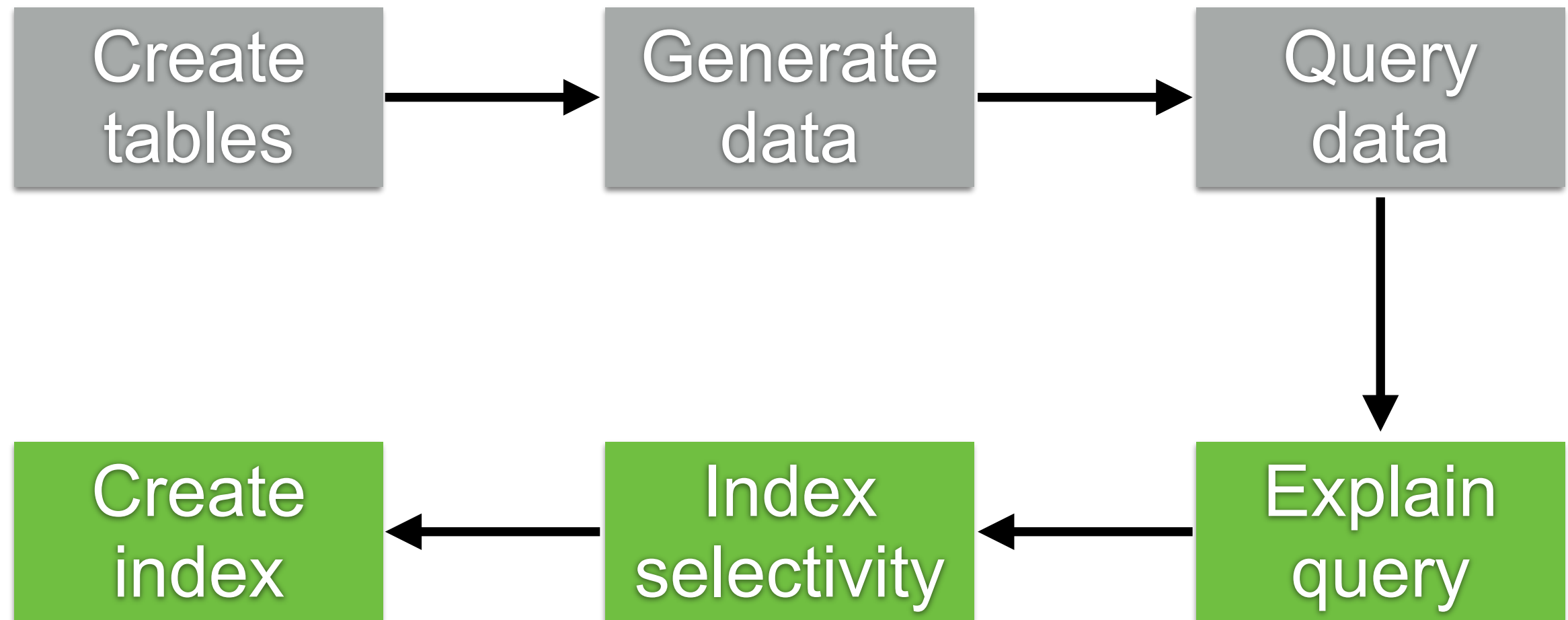


# Improvement for query data

**When ? Why ? How ? Who ?**



# Step in workshop



# Tips for SQL Optimization

| Bad                                   | Good                                     |
|---------------------------------------|------------------------------------------|
| SELECT *                              | SELECT column1, column2                  |
| SELECT ALL                            | SELECT with Limit, Offset, Top           |
| Many joins !!                         | Use appropriate joins                    |
| SELECT distinct !! For large datasets | Use group by, union , exists, row number |
| %column%                              | Use full text search feature             |
| count(*) > 0 or IN                    | exists (select 1 from table WHERE ...)   |
| Don't use subquery                    | Use derived table or pre-joined tables   |



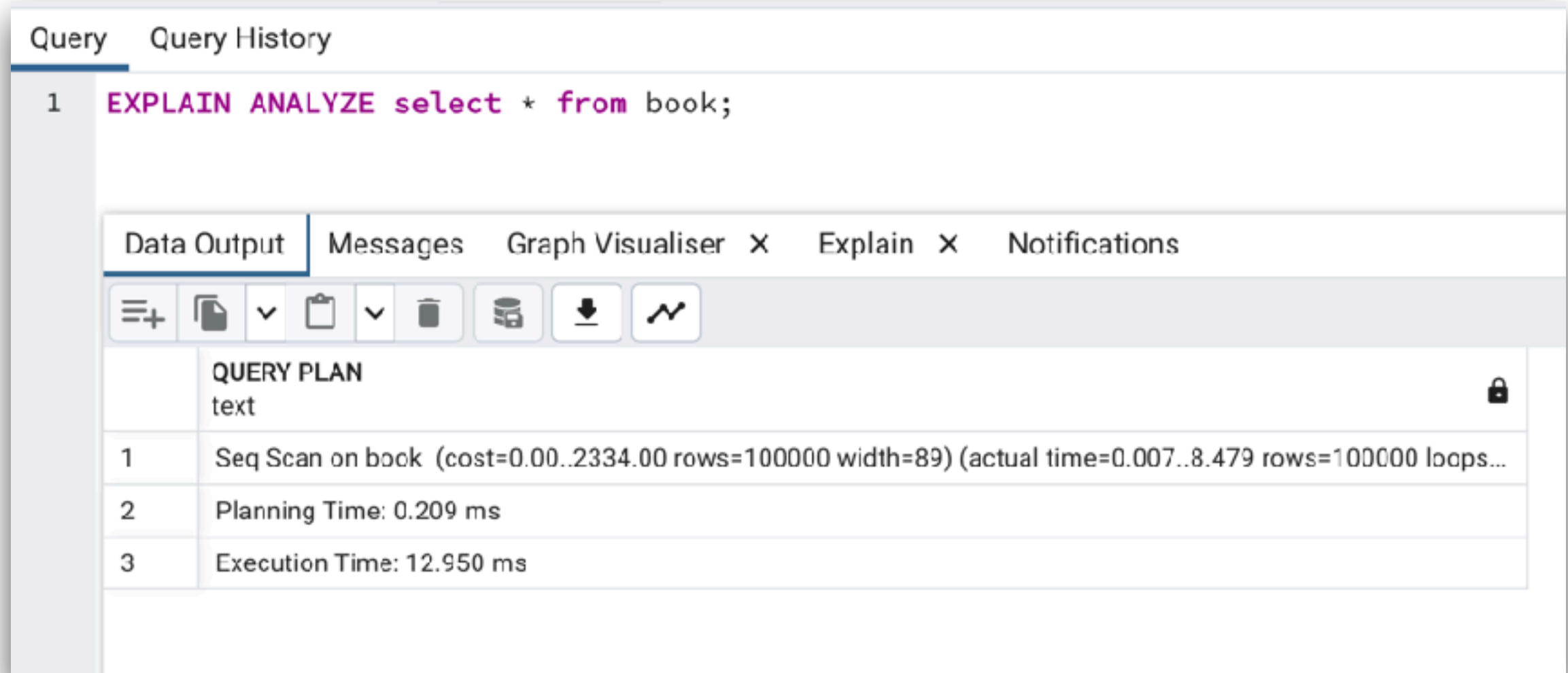
**Always  
Explain Analyze Query !!**





# Query plan in PostgreSQL

## EXPLAIN ANALYZE <SQL>



The screenshot shows a PostgreSQL query editor interface. The top bar has tabs for 'Query' and 'Query History'. The main area displays the query: `1 EXPLAIN ANALYZE select * from book;`. Below the query, there are tabs for 'Data Output', 'Messages', 'Graph Visualiser', 'X', 'Explain', 'X', and 'Notifications'. The 'Data Output' tab is active, showing a table with the query plan. The table has three rows: 1. 'Seq Scan on book (cost=0.00..2334.00 rows=100000 width=89) (actual time=0.007..8.479 rows=100000 loops...)', 2. 'Planning Time: 0.209 ms', and 3. 'Execution Time: 12.950 ms'.

|   | QUERY PLAN                                                                                                 |
|---|------------------------------------------------------------------------------------------------------------|
| 1 | Seq Scan on book (cost=0.00..2334.00 rows=100000 width=89) (actual time=0.007..8.479 rows=100000 loops...) |
| 2 | Planning Time: 0.209 ms                                                                                    |
| 3 | Execution Time: 12.950 ms                                                                                  |

<https://github.com/up1/workshop-database/tree/main/workshop#query-plan>





# Indexing Strategies

<https://www.postgresql.org/docs/current/indexes-types.html>



# Index

## Symbols

2PC, 219, 346, 347, 349–351

## A

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Amazon, 137, 199, 317

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Amazon Web Services, 251, 302

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Automatic Market Making, 409

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118, 120, 122

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2

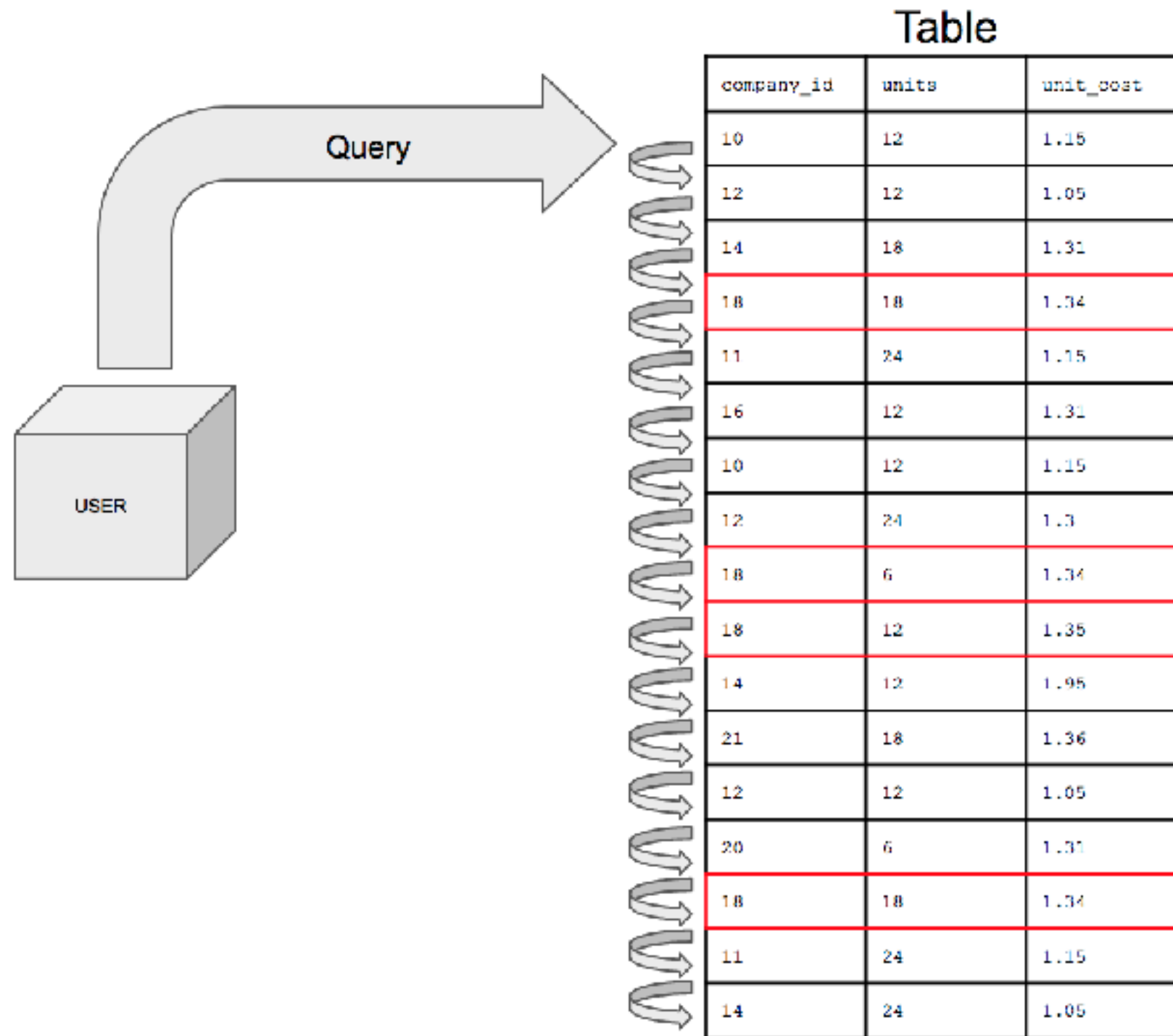
candlestick chart, 381

candlestick charts, 384, 387, 396,  
407

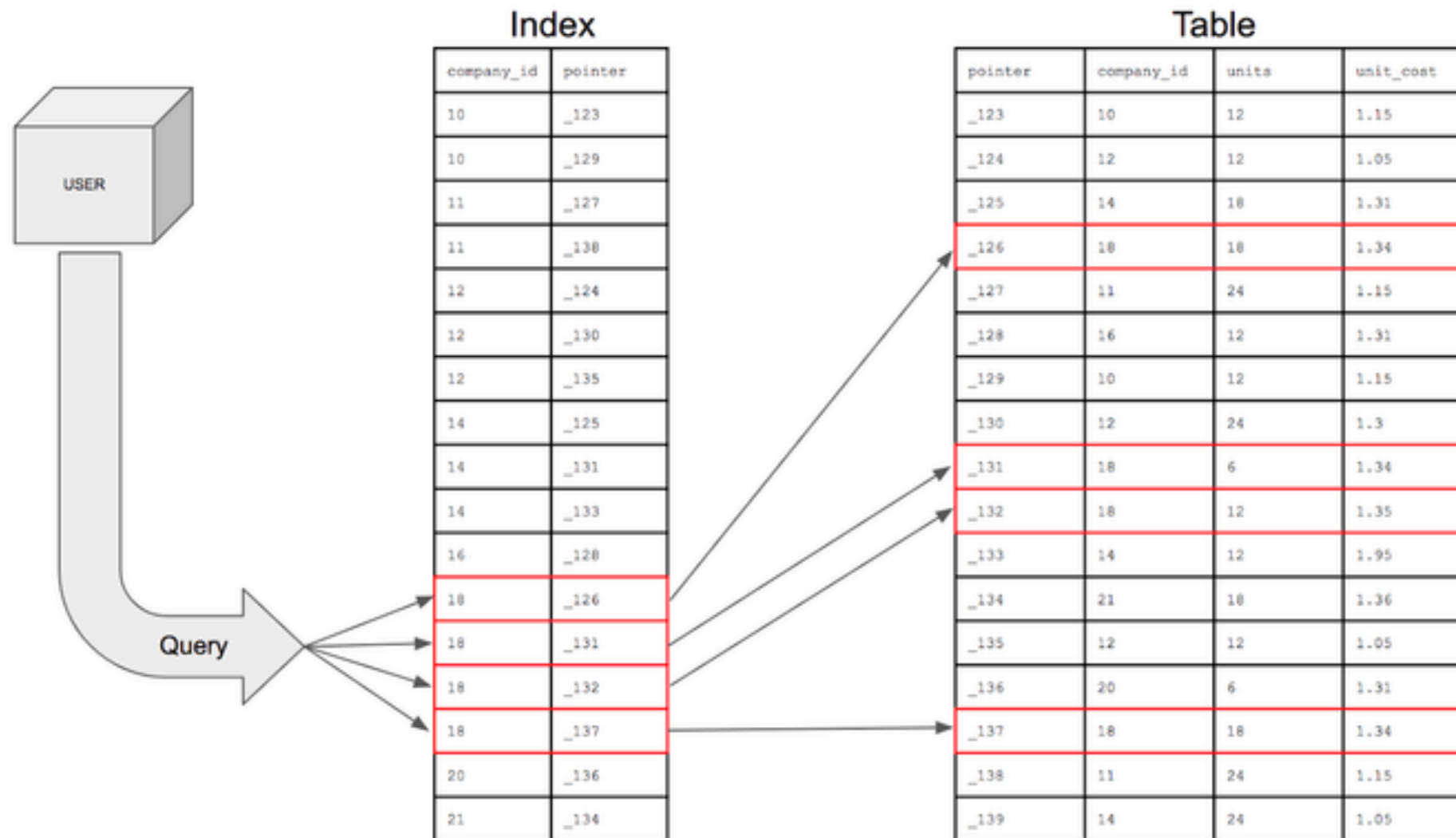
CAP theorem, 79



# Without Index (full table scan)



# With Index



# With Index

Table

|         | Column A     | Column B             |
|---------|--------------|----------------------|
| RowId 1 | Some value 1 | Some another value 1 |
| RowId 2 | Some value 2 | Some another value 2 |
| RowId 3 | Some value 3 | Some another value 3 |
| RowId 4 | Some value 1 | Some another value 4 |
| RowId 5 | Some value 5 | Some another value 5 |
| RowId 6 | Some value 6 | Some another value 6 |
| RowId 7 | Some value 1 | Some another value 7 |
| RowId 8 | Some value 8 | Some another value 8 |

Index

| Some value 1 | "RowId 1" "RowId 4" "RowId 7" |
|--------------|-------------------------------|
| Some value 2 | RowId 2                       |
| Some value 3 | RowId 3                       |
| Some value 5 | RowId 5                       |
| Some value 6 | RowId 6                       |
| Some value 8 | RowId 8                       |
|              |                               |
|              |                               |

Index by column "Column A"



# Main factors for good index ?

How much space does the index use ?

How fast does it read ?

How much overhead does it add to writes ?

How well does it handle concurrency ?

Quick lookup, fast insert, reliable delete



# Index selectivity

Factor to select a column to create an index

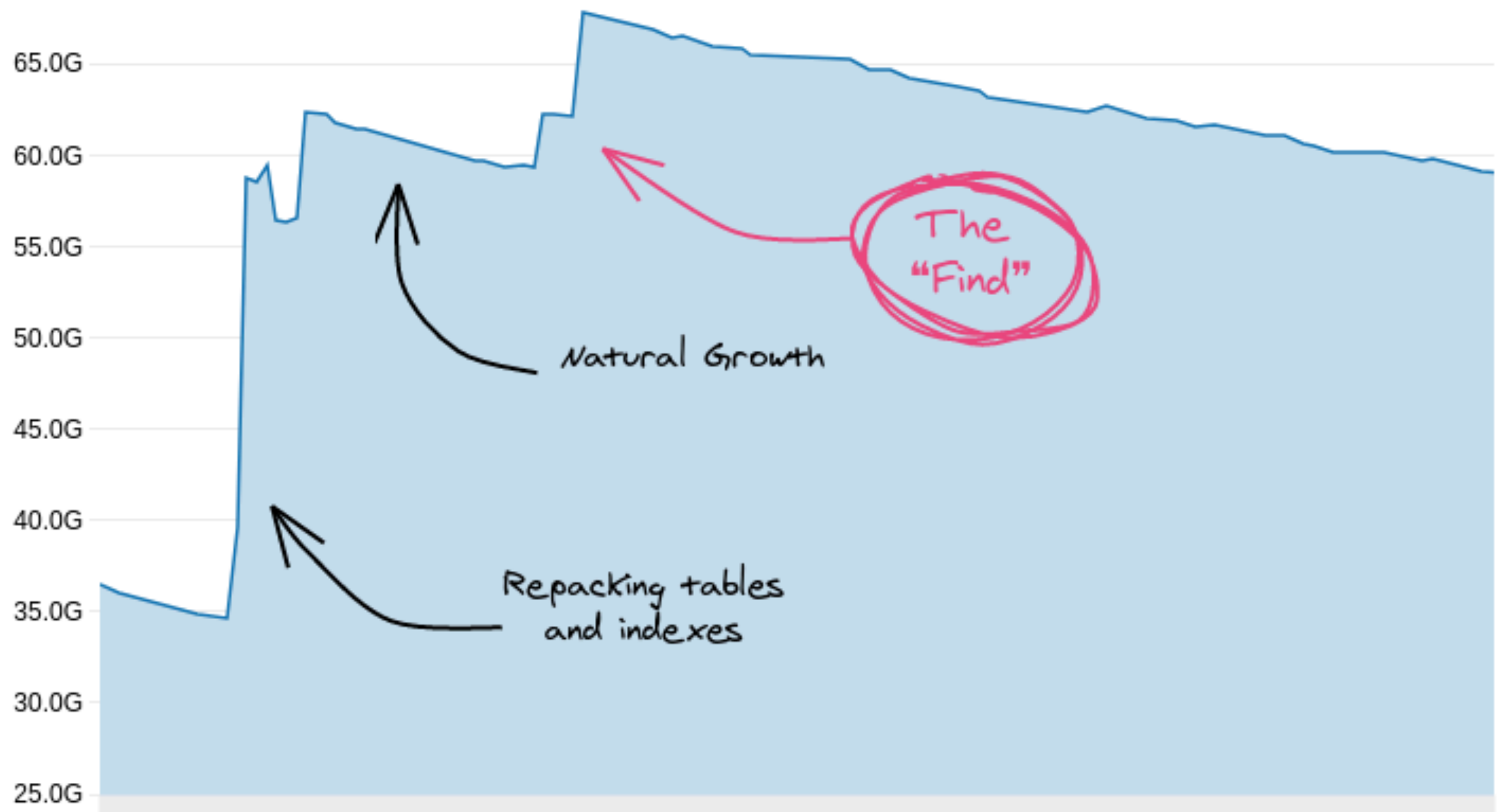
The number of **distinct values** in the indexed column divided by the number of records in the table is called a selectivity of an index

***Prefer indexing columns with selectivity greater than  $> 0.85$  (use Index scan)***





# Unused Index ?



<https://hakibenita.com/postgresql-unused-index-size>





# Types Index in PostgreSQL

B-Tree

Hash

Block range indexes (BRIN)

Generalized inverted index (GIN)

Generalized inverted search tree index (GiST)

Space partitioned GiST (SP-GiST)

<https://www.postgresql.org/docs/current/indexes-types.html>

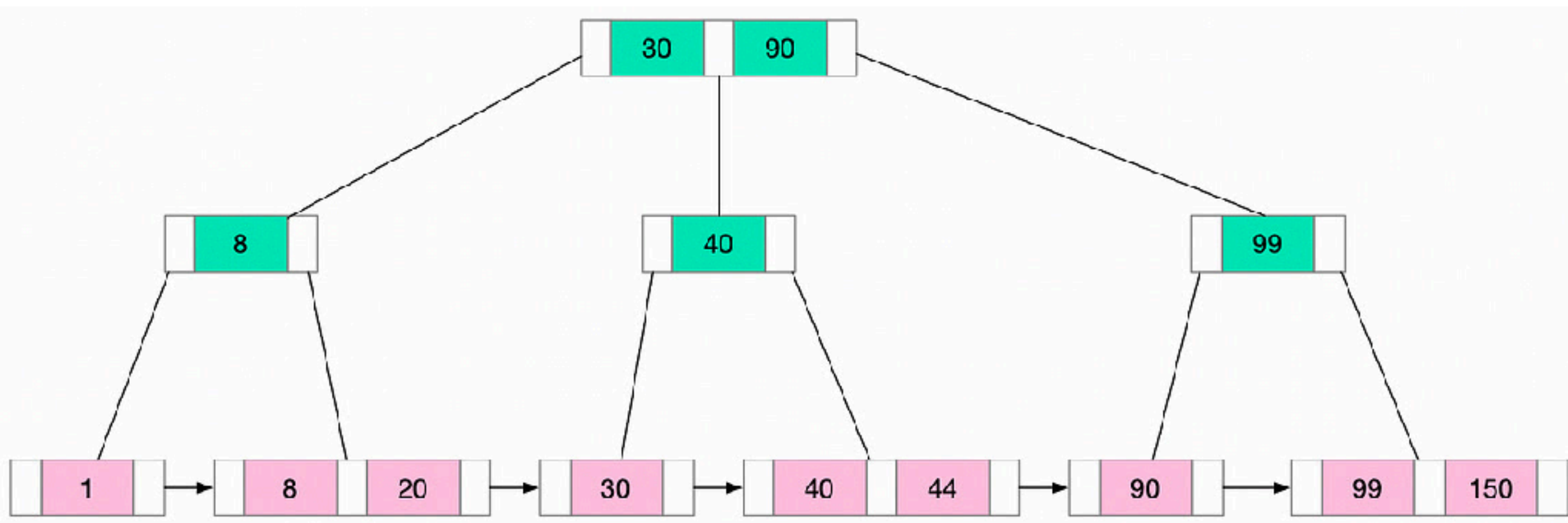


# PostgreSQL Index Types

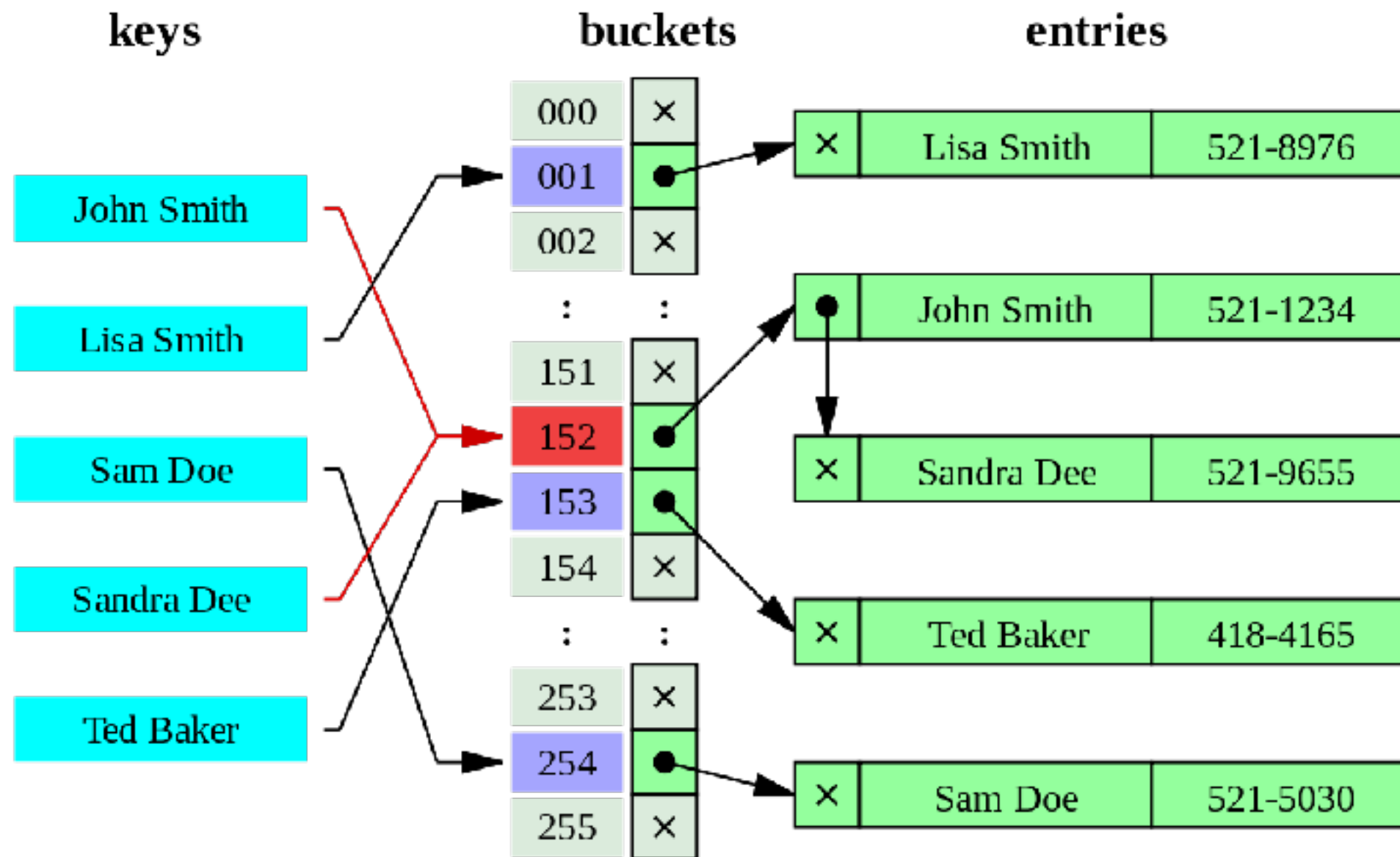
| Index type | Data                                       | Target                                           | Use case                                                 |
|------------|--------------------------------------------|--------------------------------------------------|----------------------------------------------------------|
| B-Tree     | High-cardinality scalar columns            | Fast equality, range, between operation          | PK, FK, timestamp/range scan, sorting                    |
| GIN        | Composite values (arrays, JSONB, tsvector) | Efficient contains, full-text, array membership  | Full text search, JSONB containment, array lookup        |
| GiST       | Complex or non-scalar types                | Flexible tree, Range and text                    | PostGIS spatial queries, trigram similarity, range types |
| BRIN       | Very large data, ordered table             | Lightweight index for append-only and range scan | Time-series data<br>Log data                             |
| Bloom      | Many low-cardinality columns               | Compact multi-column equality filters            | Filter data, check data existing                         |



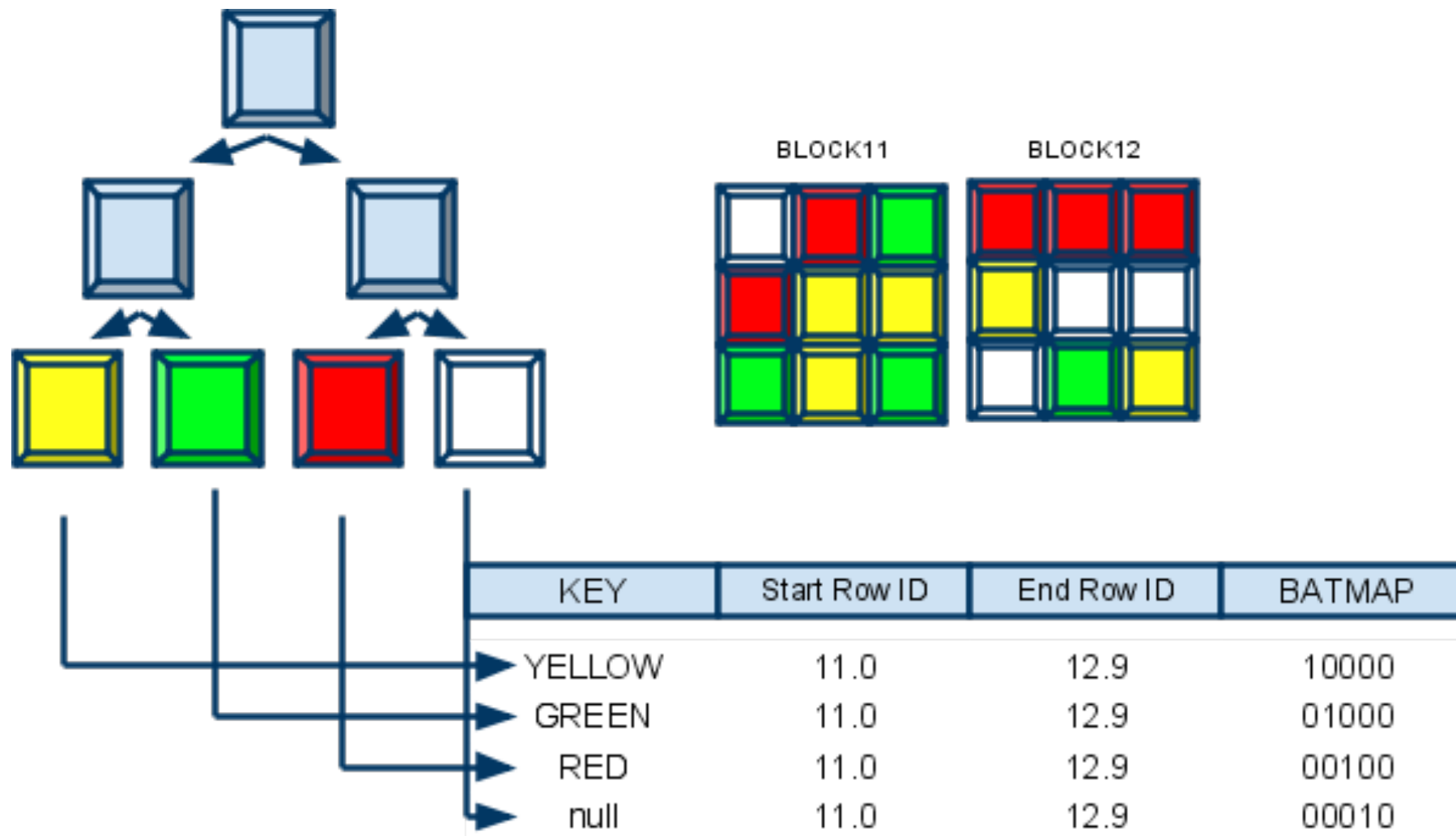
# B-Tree +



# Hash



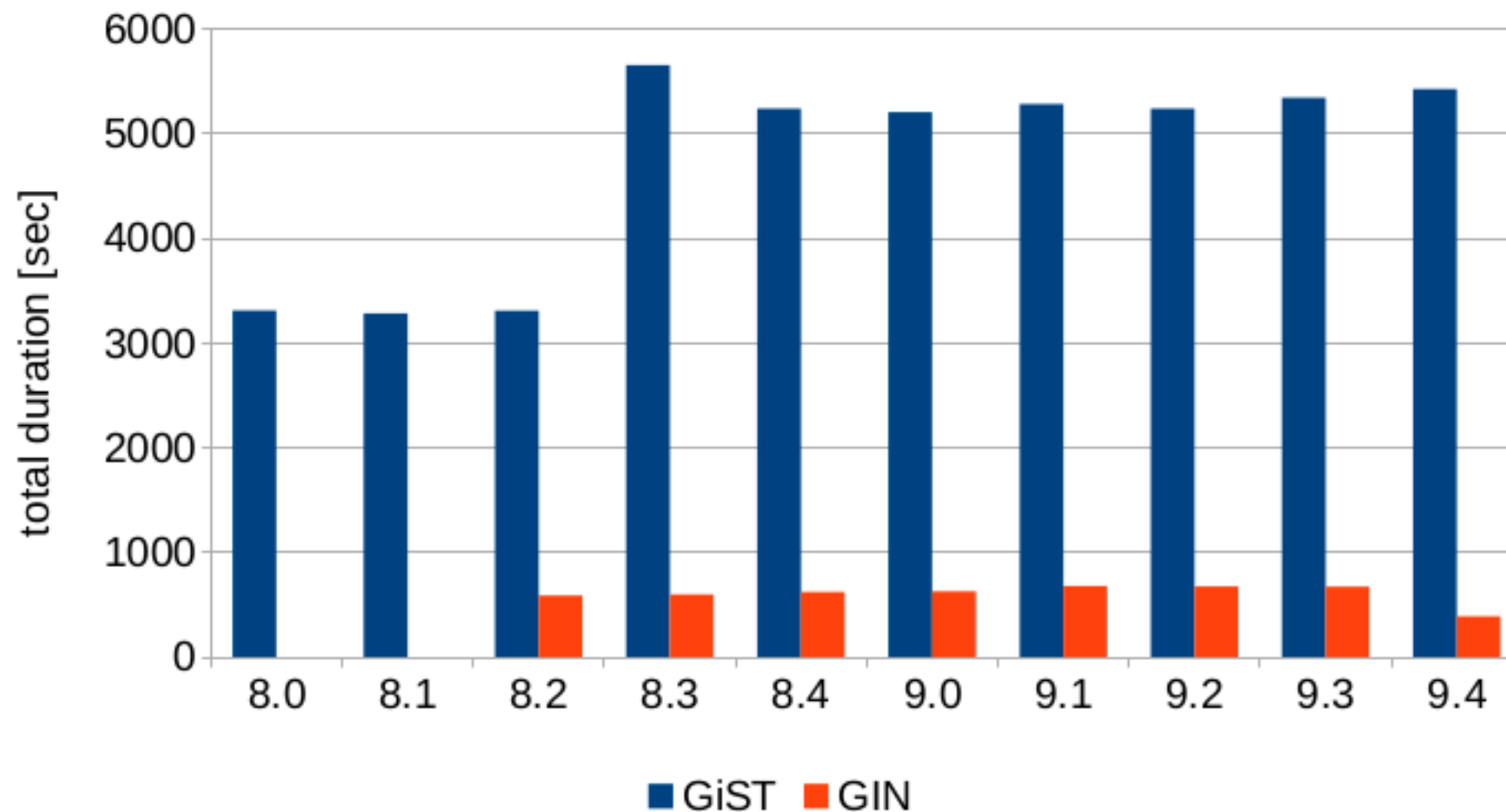
# Bitmap



# GIN vs GiST

## Fulltext benchmark - GiST vs. GIN

33k queries from postgresql.org [TOP 100]



# Tuning and Use Index ?

Size vs Speed

Extension setup

Combine indexes

Measure impact with **EXPLAIN** (before and after)





# Don't over-index Unused index

Size of index ?





# Workshop with Indexing

<https://github.com/up1/workshop-database/tree/main/workshop/basic-workshop>



# Tuning Performance



# Tuning performance

Normalization vs Denormalization

**Partition**

**Sharding**

Database architecture

Database parameteres



# Partition

<https://hakibenita.com/postgresql-unused-index-size>



# Partition

Table partitioning refers to **splitting** what is logically one large table into **smaller physical pieces**

Reduce index size



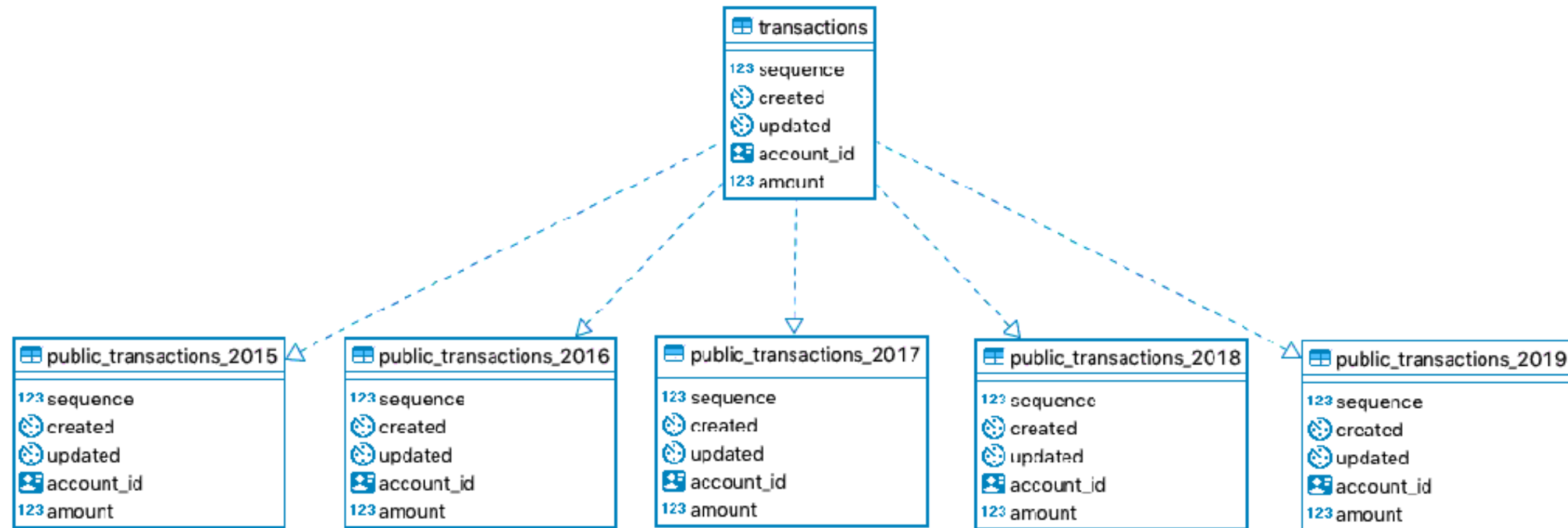
# Partition

Horizontal partitioning  
Vertical partitioning



# Horizontal partitioning

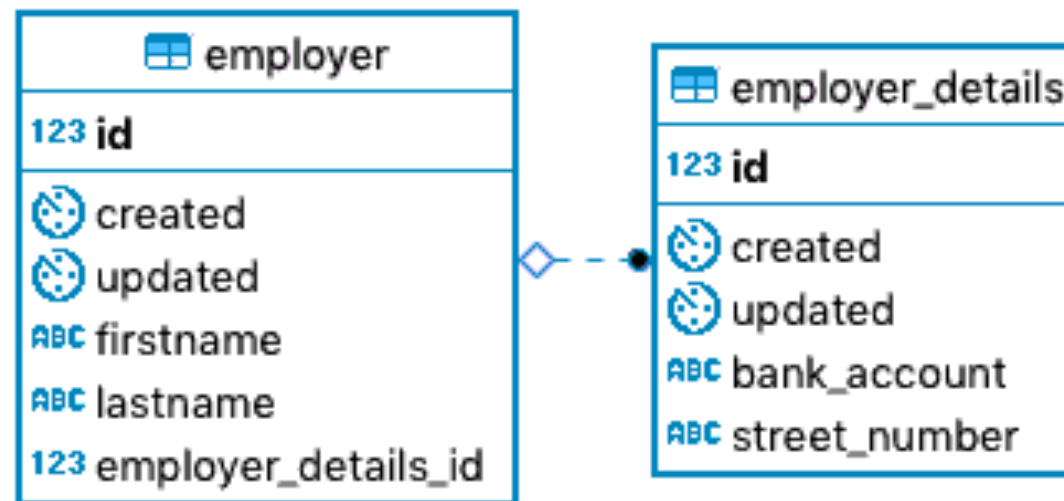
Putting different rows into different table





# Vertical partitioning

## Horizontal partitioning





# Types Horizontal partitioning

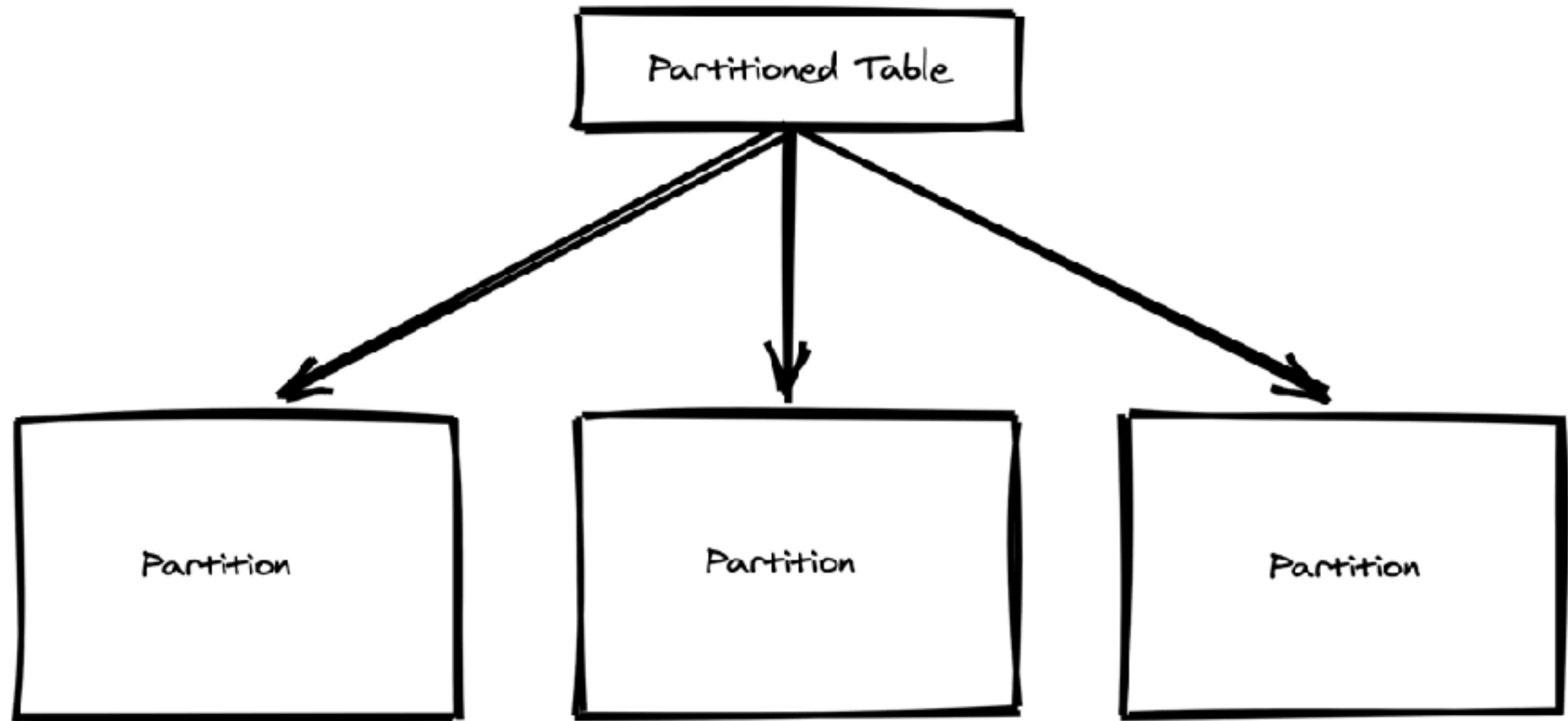
Range  
List



# Example



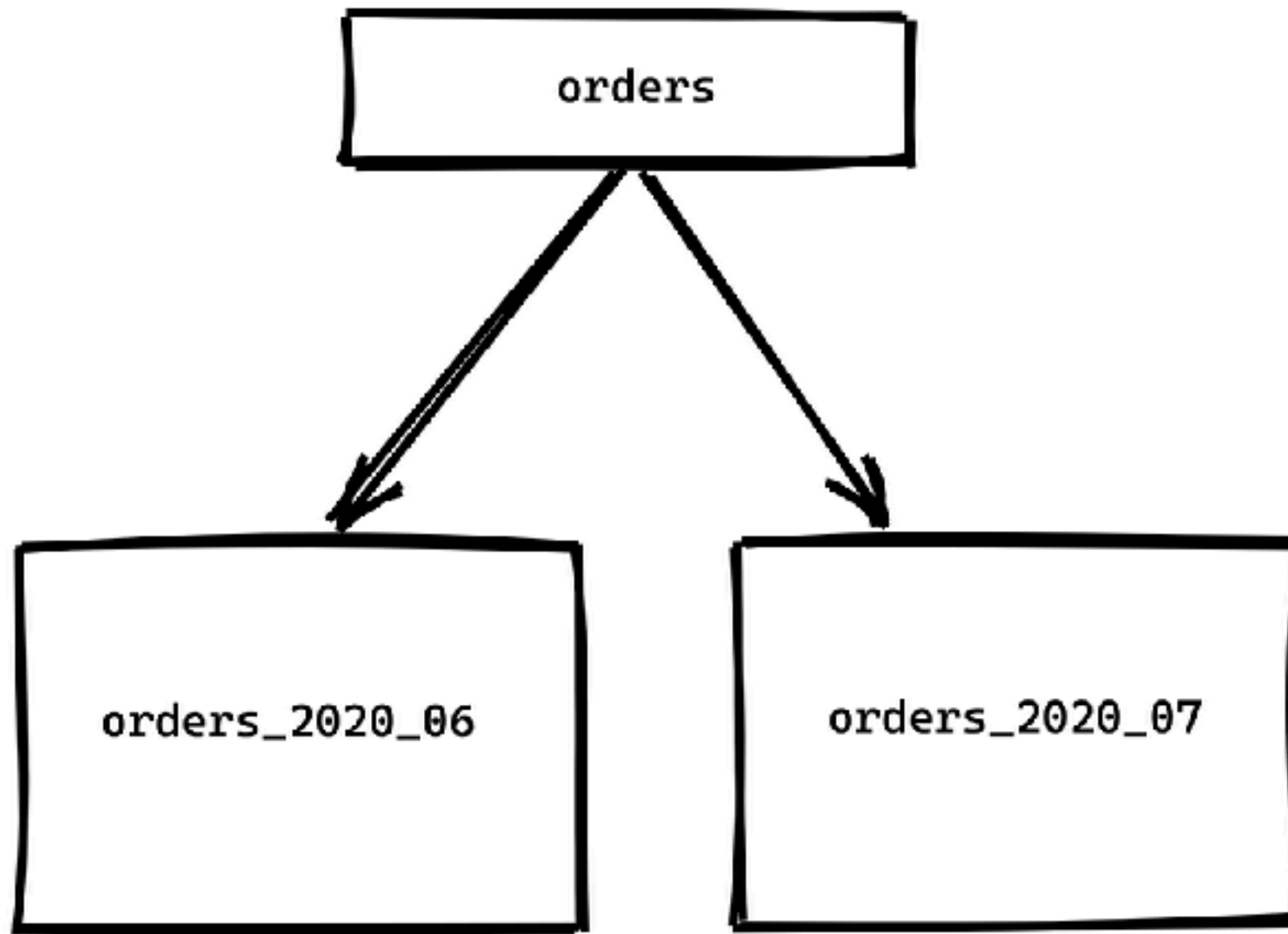
# Partition tables



<https://chrisherwin.com/table-partitioning/>



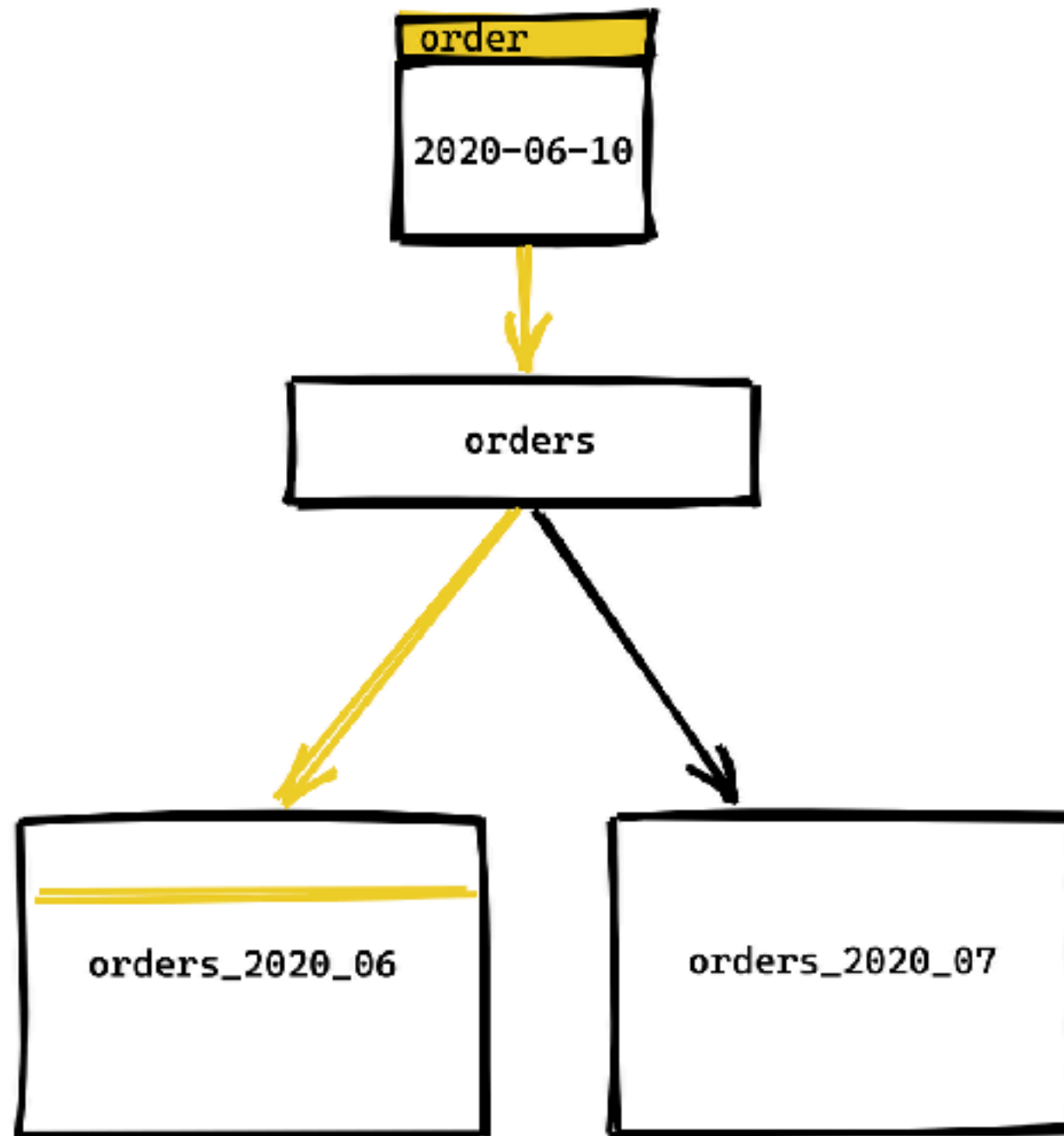
# Orders table



<https://chriserwin.com/table-partitioning/>



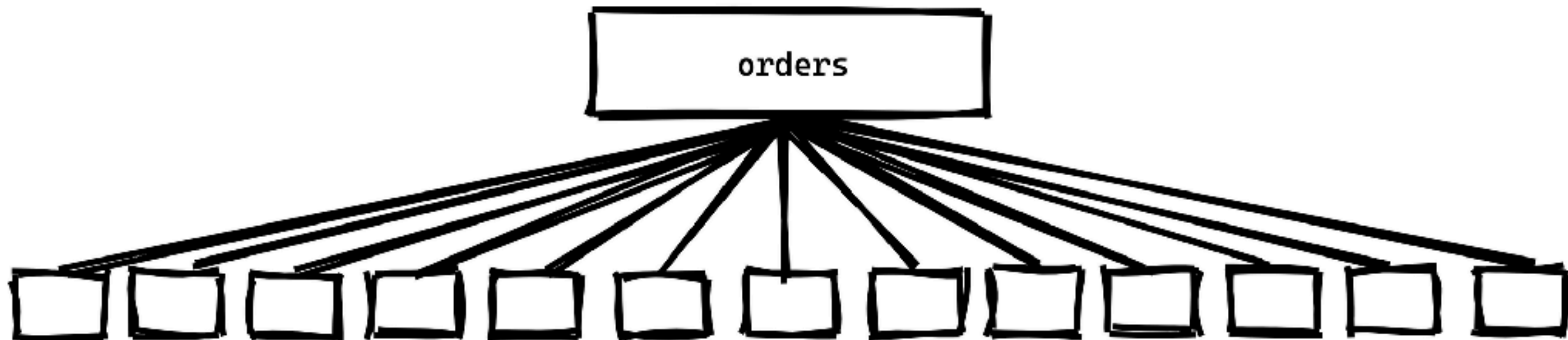
# Insert data into orders



<https://chriserwin.com/table-partitioning/>



...



<https://chrisherwin.com/table-partitioning/>



# Workshop with Partition

<https://github.com/up1/workshop-database/blob/main/workshop/basic-workshop/partition.md>



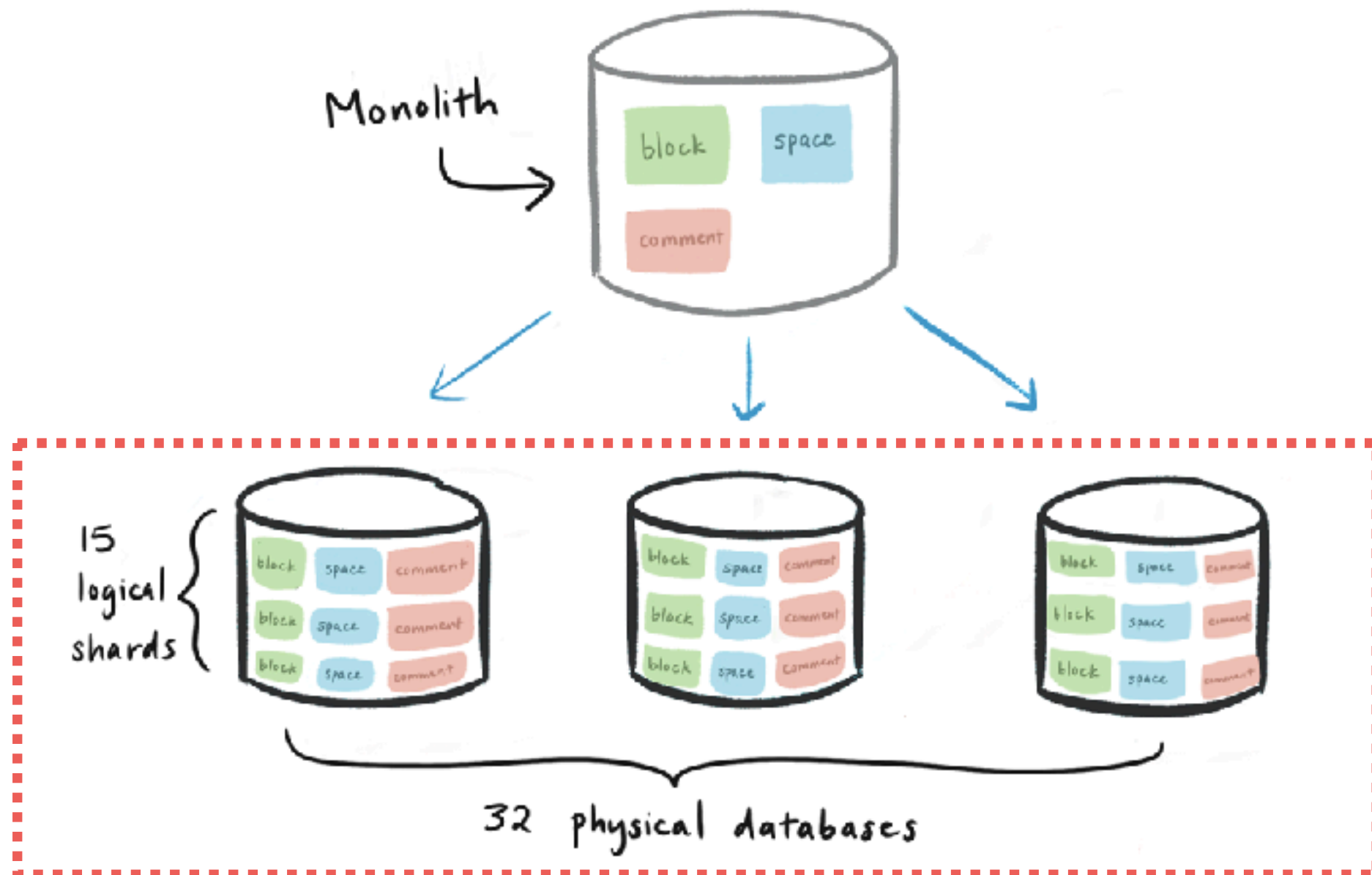


# Sharding





# Sharding (Physical server)



<https://www.notion.so/blog/sharding-postgres-at-notion>



# Sharding techniques

Hash

Range

Location

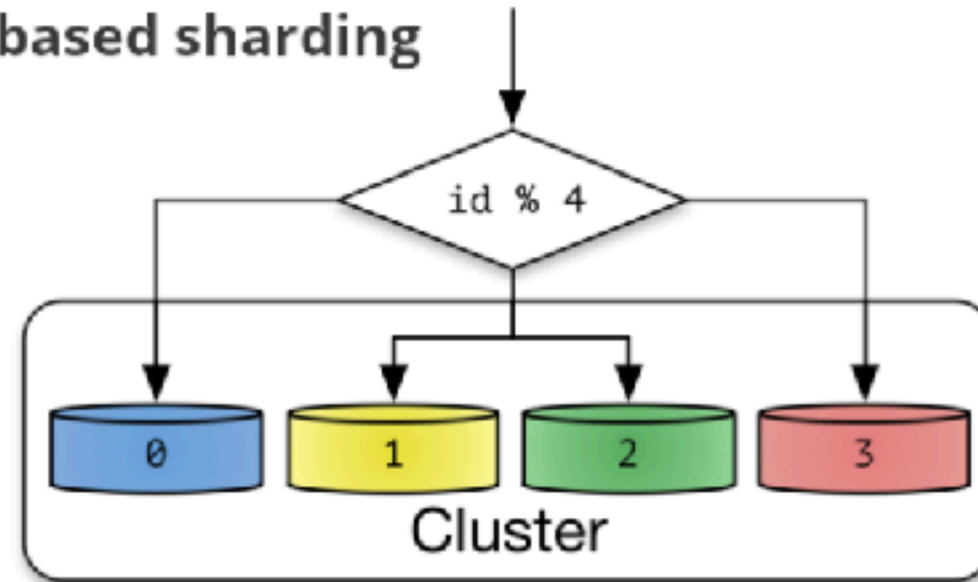
Key/Directory

<https://www.somkiat.cc/introduction-database-sharding/>

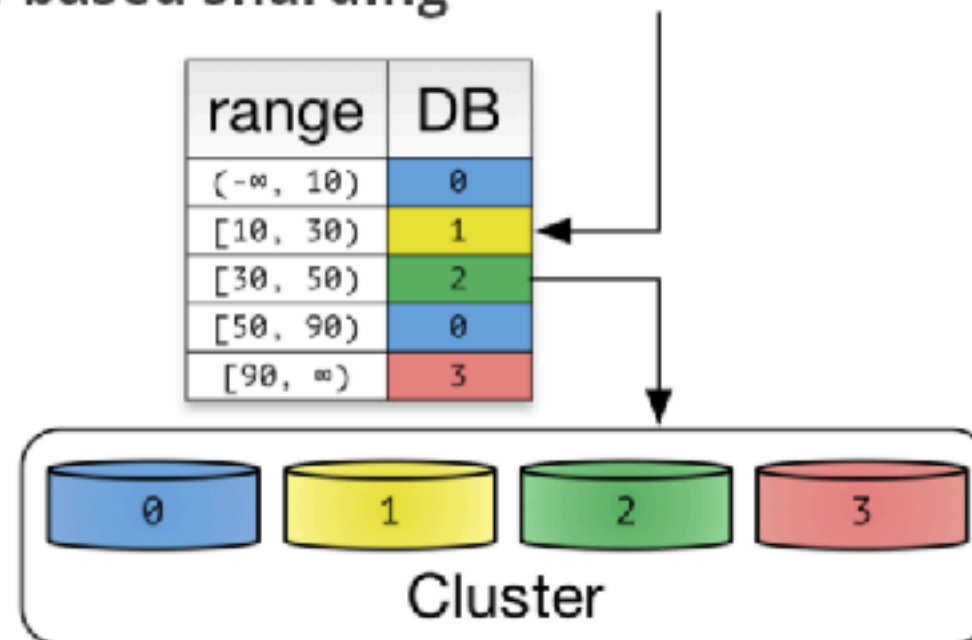


# Sharding

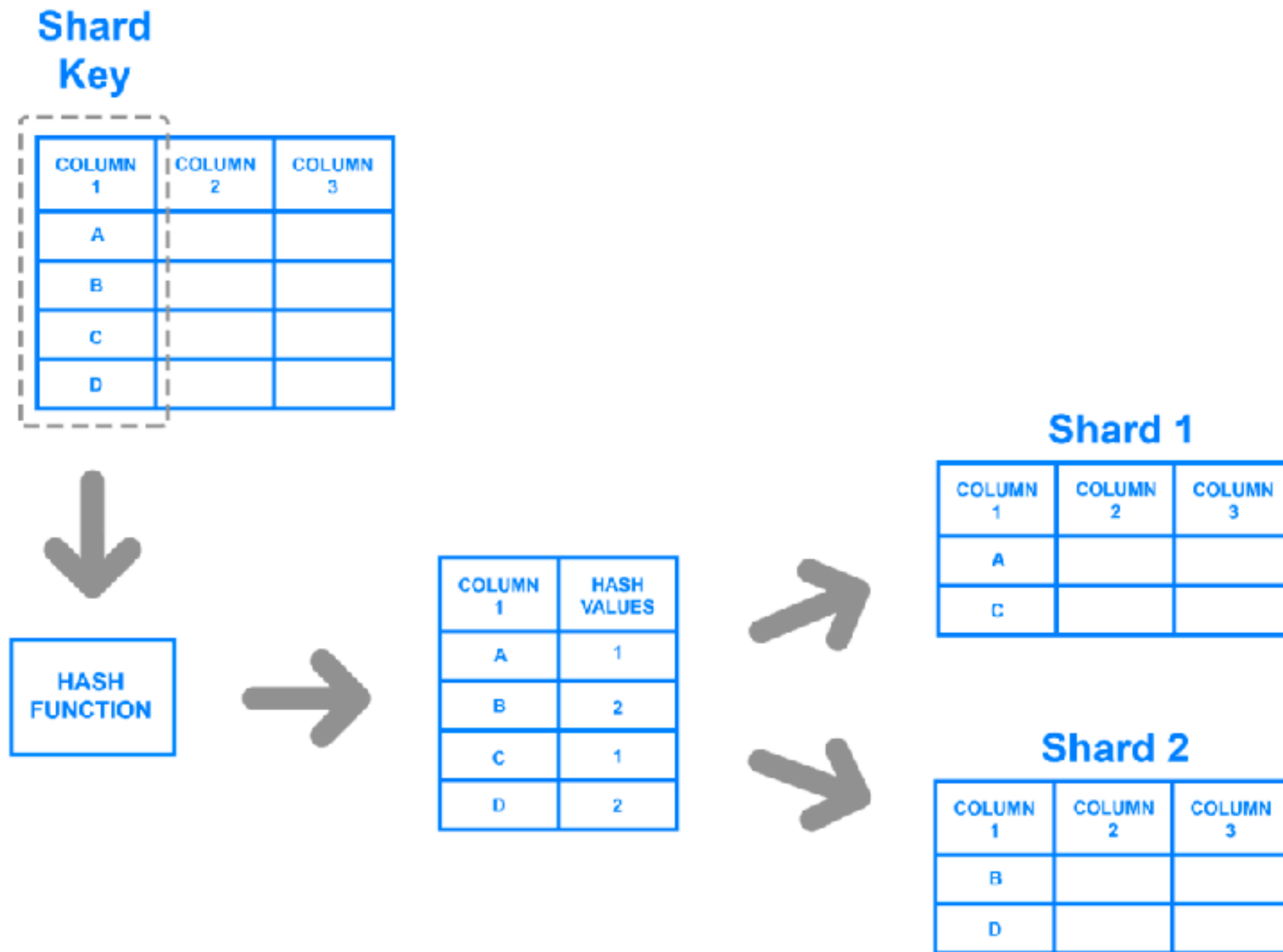
(a) Hash-based sharding



(b) Range-based sharding



# Shard Key



# Range

| PRODUCT     | PRICE |
|-------------|-------|
| WIDGET      | \$118 |
| GIZMO       | \$88  |
| TRINKET     | \$37  |
| THINGAMAJIG | \$18  |
| DOODAD      | \$60  |
| TCHOTCHKE   | \$999 |



(\$0-\$49.99)

| PRODUCT     | PRICE |
|-------------|-------|
| TRINKET     | \$37  |
| THINGAMAJIG | \$18  |

(\$50-\$99.99)

| PRODUCT | PRICE |
|---------|-------|
| GIZMO   | \$88  |
| DOODAD  | \$60  |

(\$100+)

| PRODUCT   | PRICE |
|-----------|-------|
| WIDGET    | \$118 |
| TCHOTCHKE | \$999 |



# Directory

Pre-Sharded Table

Shard Key

| DELIVERY ZONE | FIRST NAME | LAST NAME |
|---------------|------------|-----------|
| 3             | DARCY      | CLAY      |
| 1             | DENISE     | LASALLE   |
| 2             | HIROSHI    | YOSHIMURA |
| 4             | KIRSTY     | MACCOLL   |



Shard Key

| DELIVERY ZONE | SHARD ID |
|---------------|----------|
| 1             | S1       |
| 2             | S2       |
| 3             | S3       |
| 4             | S4       |



S1

|   |        |         |
|---|--------|---------|
| 1 | DENISE | LASALLE |
|---|--------|---------|

S2

|   |         |           |
|---|---------|-----------|
| 2 | HIROSHI | YOSHIMURA |
|---|---------|-----------|

S3

|   |       |      |
|---|-------|------|
| 3 | DARCY | CLAY |
|---|-------|------|

S4

|   |        |         |
|---|--------|---------|
| 4 | KIRSTY | MACCOLL |
|---|--------|---------|



# Sharding

Shard 1

| Stores |                |
|--------|----------------|
| id     | name           |
| 1      | my book store  |
| 5      | my other store |

| Products |      |          |
|----------|------|----------|
| id       | name | store_id |
| 1        | foo  | 1        |
| 2        | bar  | 1        |
| 3        | baz  | 1        |

| Purchases |            |          |       |
|-----------|------------|----------|-------|
| id        | product_id | store_id | price |
| 1         | 2          | 1        | 1000  |
| 2         | 1          | 1        | 1200  |
| 3         | 3          | 1        | 1199  |

Shard 2

| Stores |               |
|--------|---------------|
| id     | name          |
| 2      | my sock store |
| 6      | old things    |

| Products |           |          |
|----------|-----------|----------|
| id       | name      | store_id |
| 33       | new socks | 2        |
| 34       | old socks | 6        |
| 35       | old tie   | 6        |

| Purchases |            |          |       |
|-----------|------------|----------|-------|
| id        | product_id | store_id | price |
| 102       | 35         | 6        | 600   |
| 43        | 33         | 2        | 800   |

<https://www.citusdata.com/blog/2016/08/10/sharding-for-a-multi-tenant-app-with-postgres/>





# Challenge of Large datasets





# Challenge of Large datasets

Disk I/O, CPU, Memory pressure

Index size and maintenance

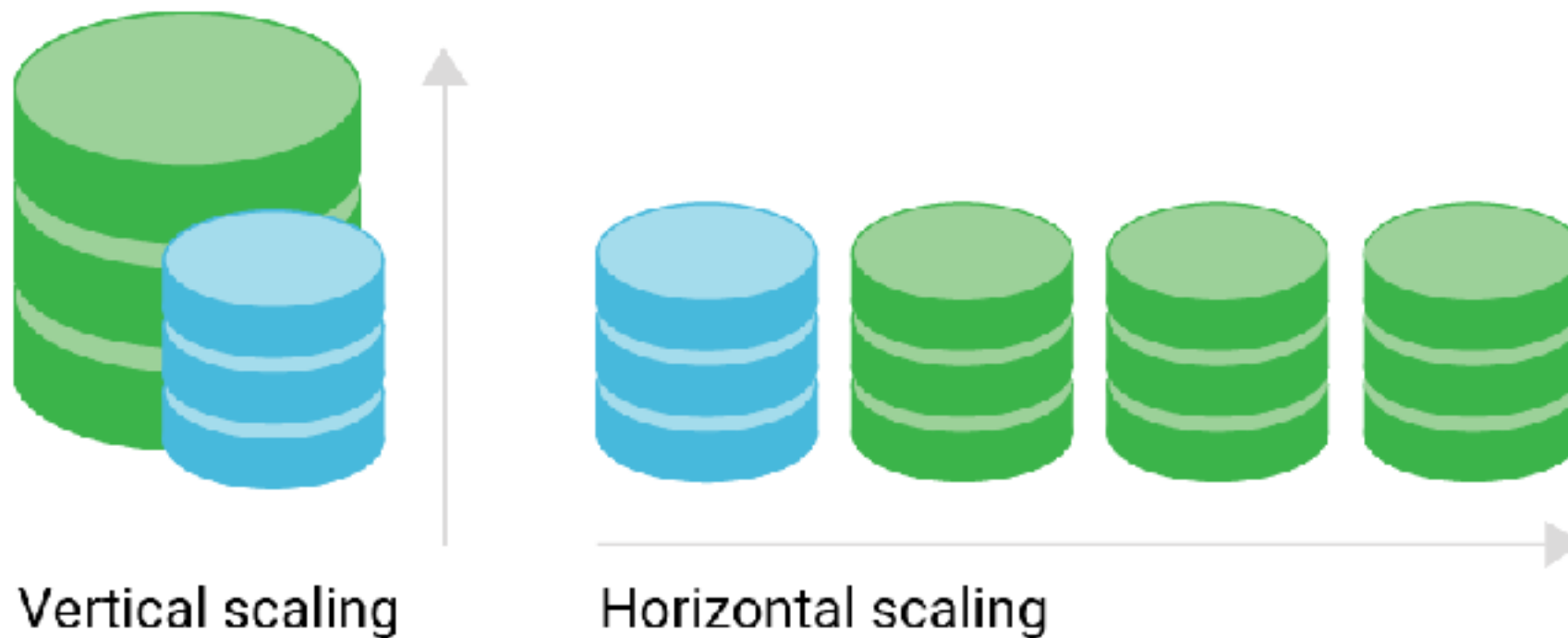
Locking and concurrency

Why database slow down with more data ?



# Scaling Database ?

## Vertical vs Horizontal



# Scalability metrics

Throughput (transactions per second)

Latency (response time)

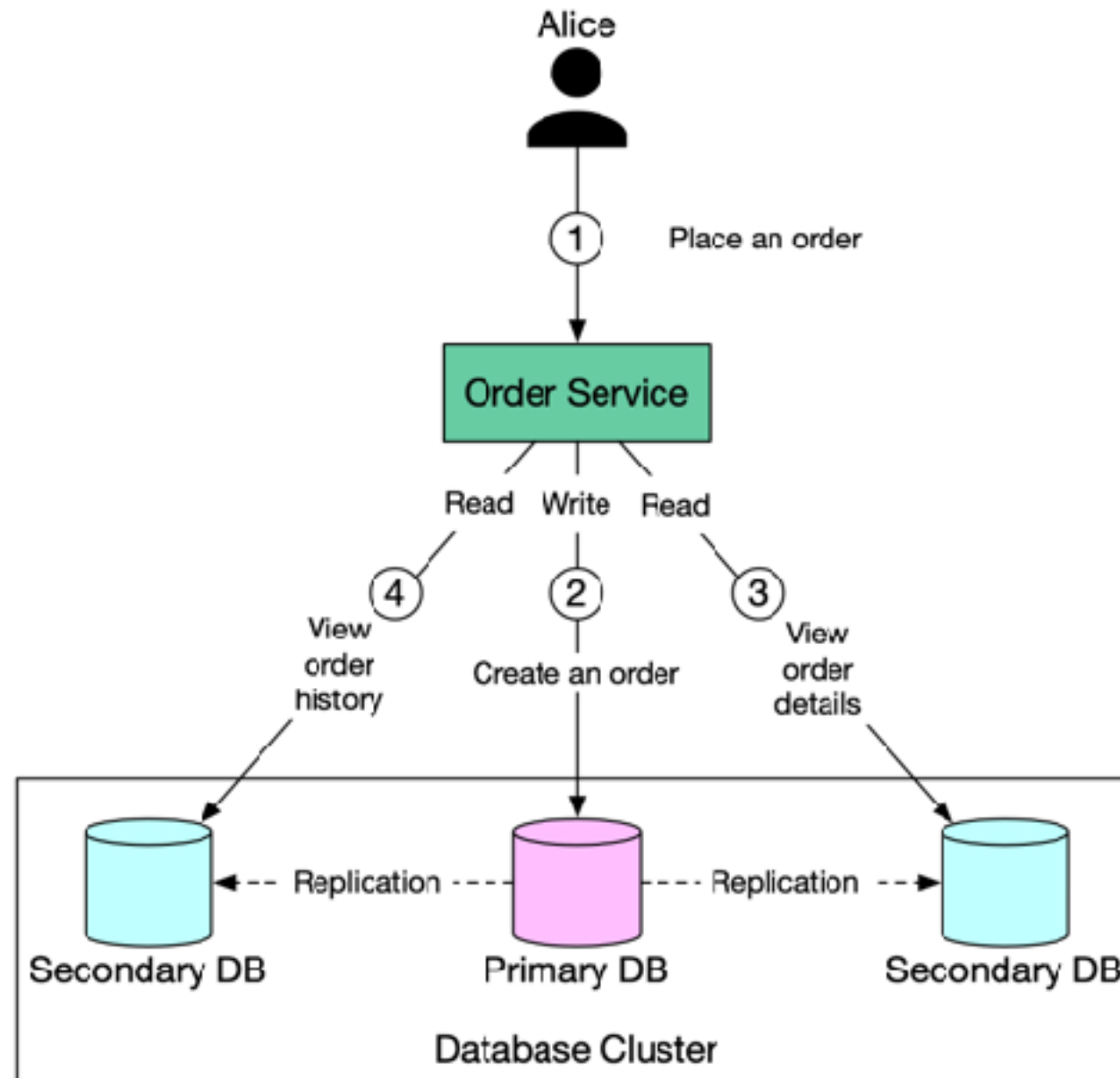
Availability (uptime)

Durability (data safety)



# Read replicas

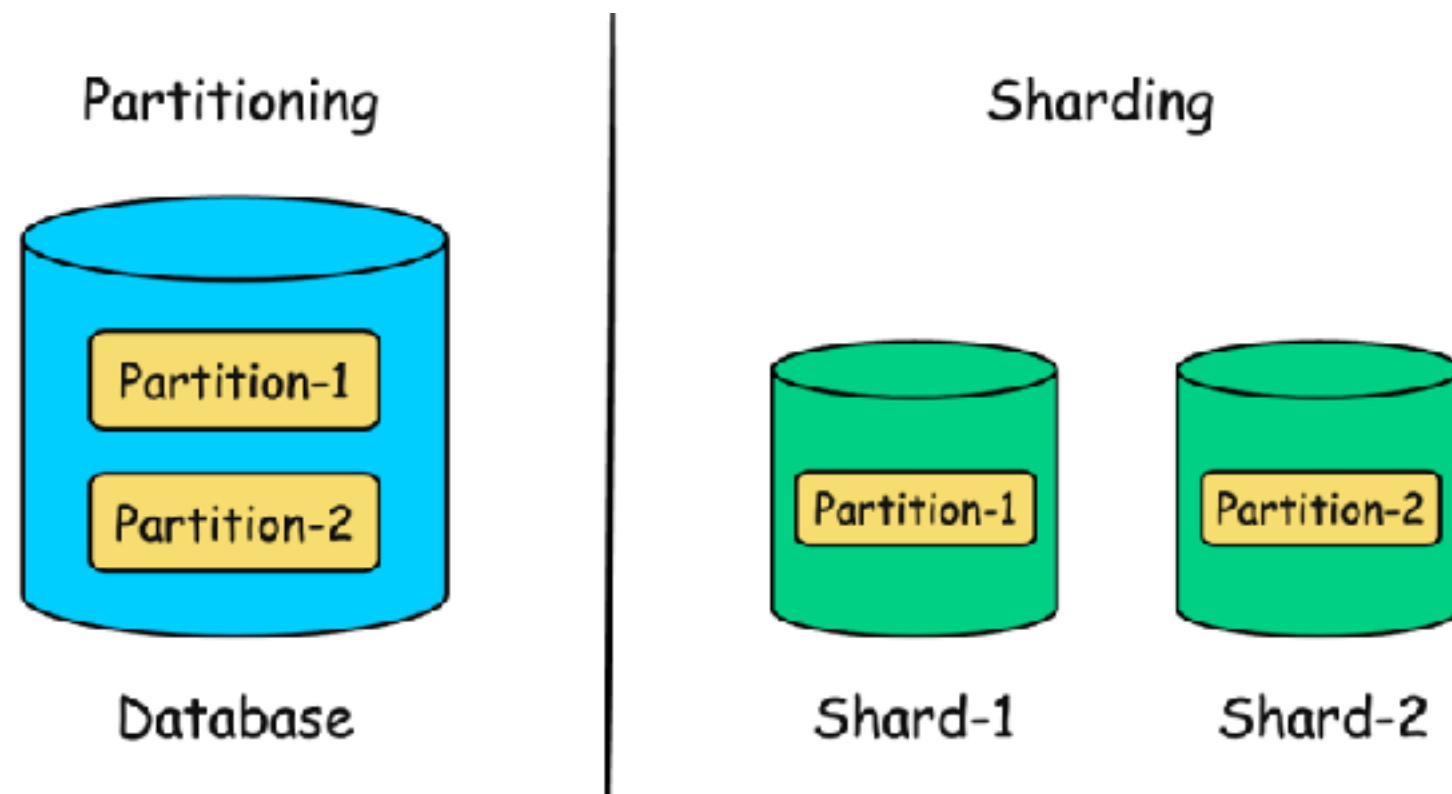
## Read-scaling solution



# Data distribution strategies

Partitioning (within a single instance)

Sharding (across multiple instances)



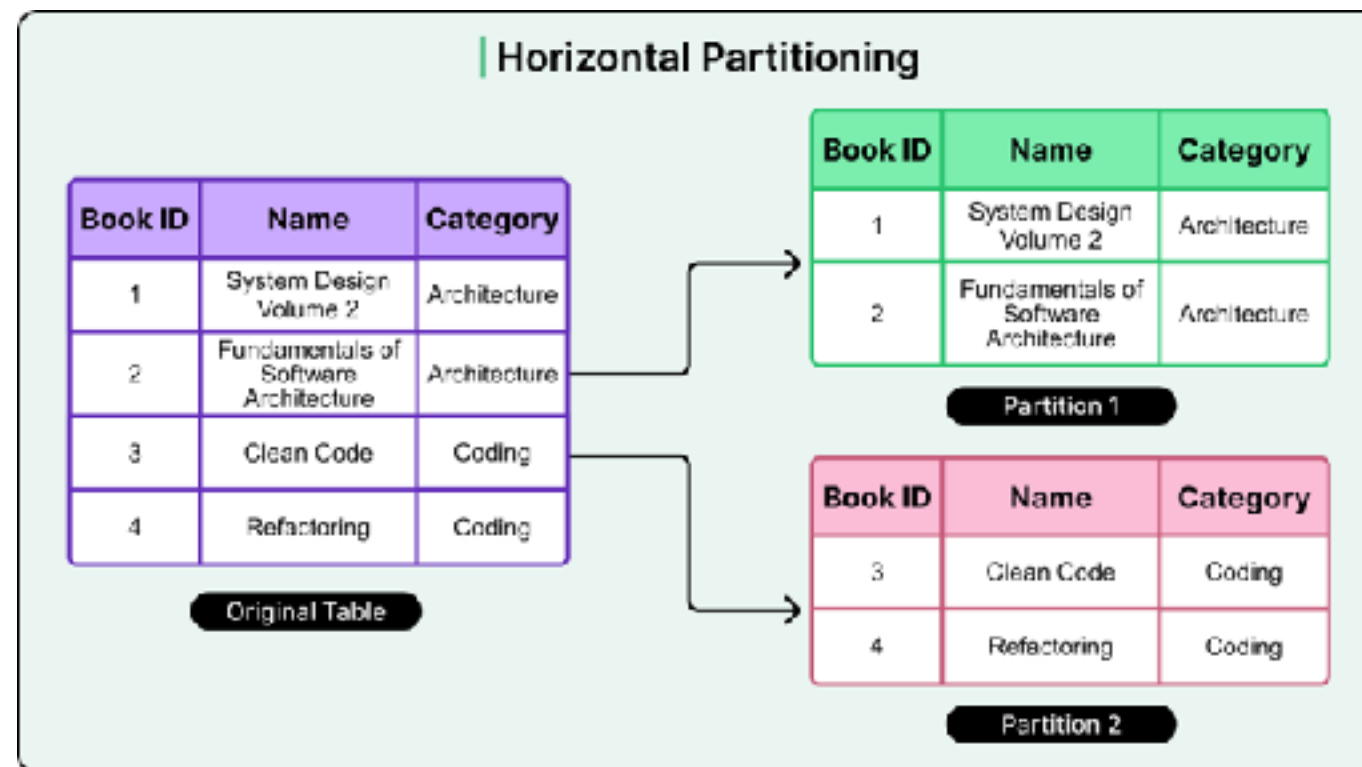
# Database sharding

Design for large dataset by split into smaller

Divided across multiple instance

Based-on horizontal partitioning

Use when to overcome single instance limitations



# Sharding techniques

Range

Hash or key-based

Directory-based

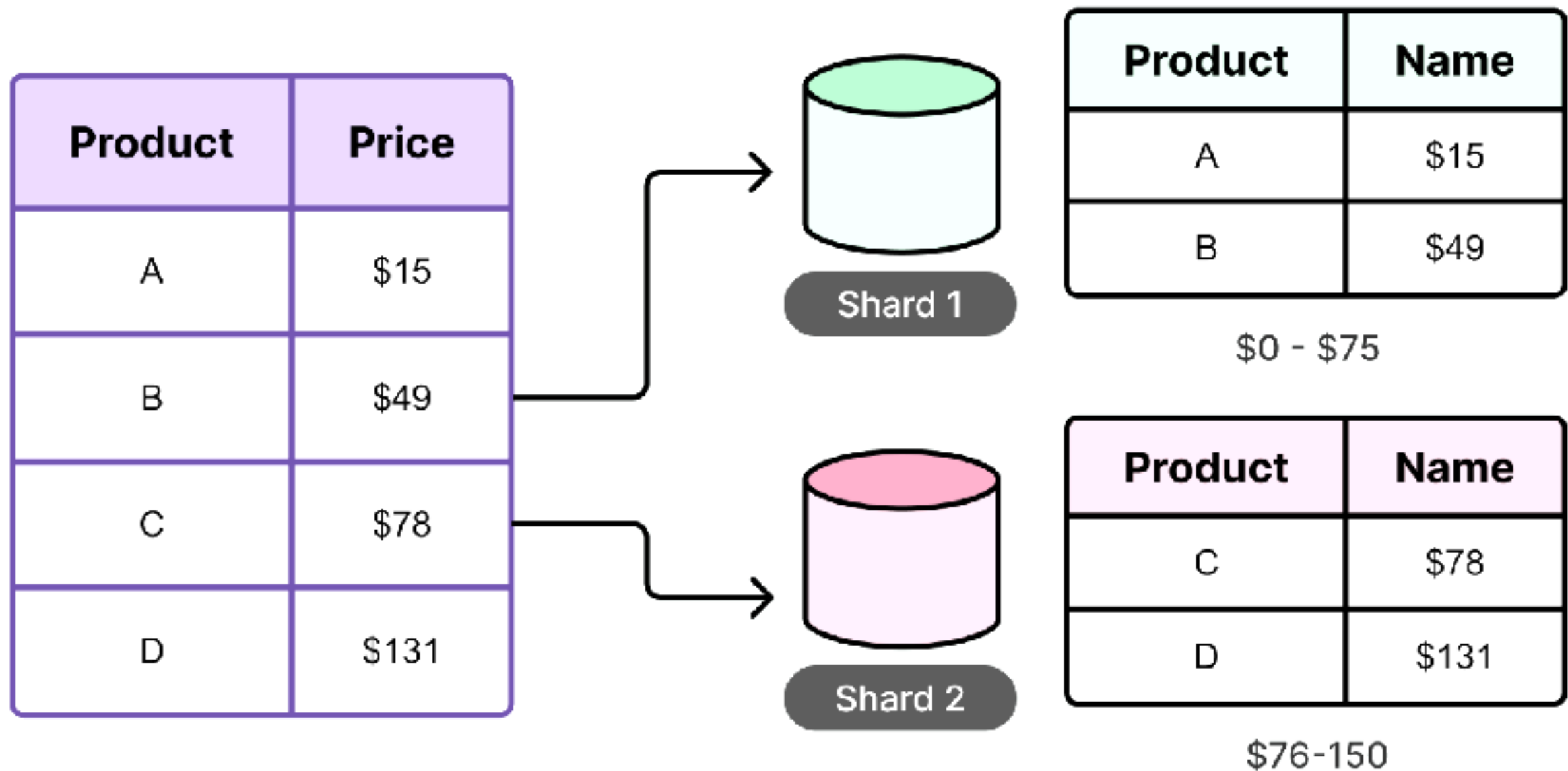
Request routing

<https://blog.bytebytego.com/p/a-guide-to-database-sharding-key>



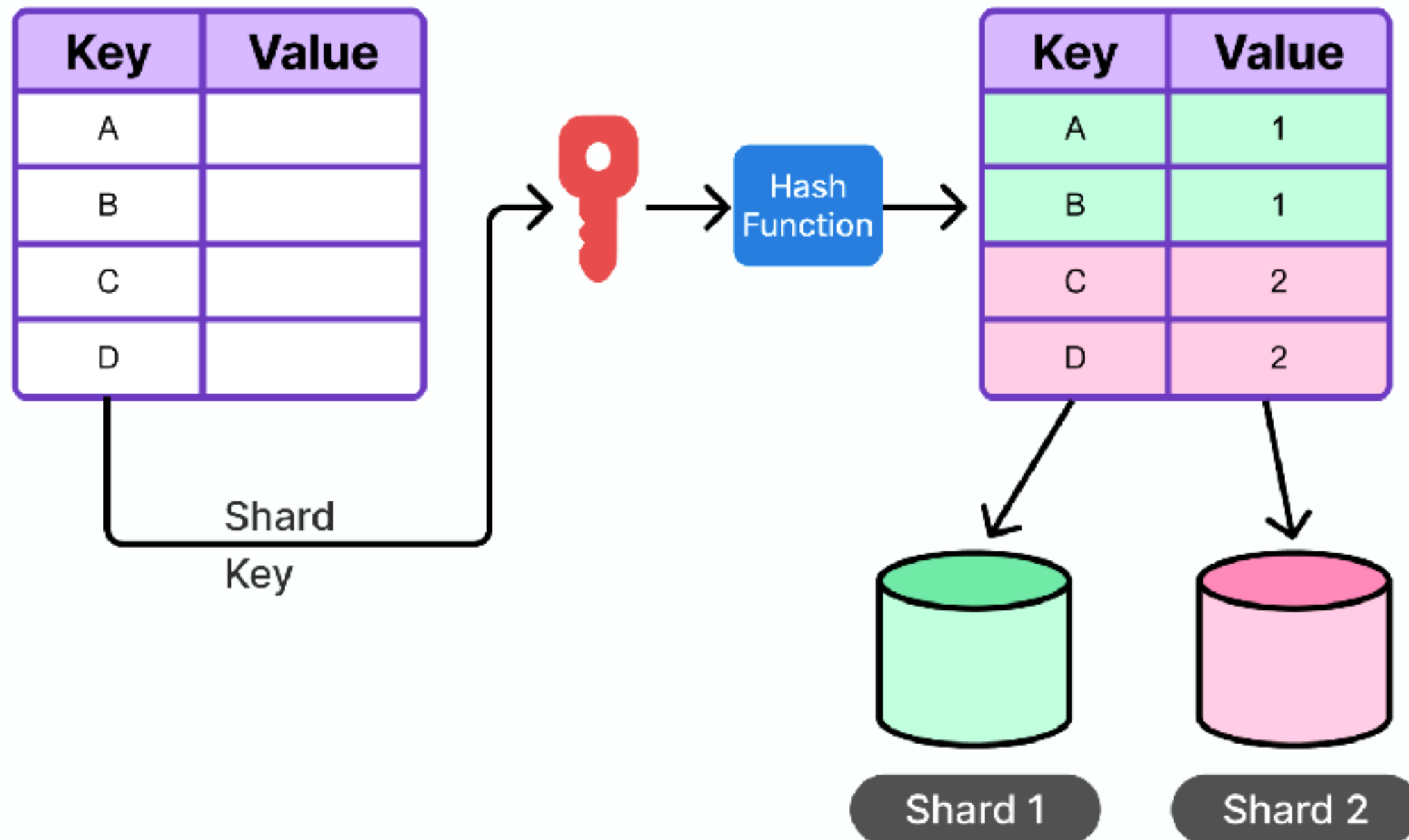


# Range-based sharding

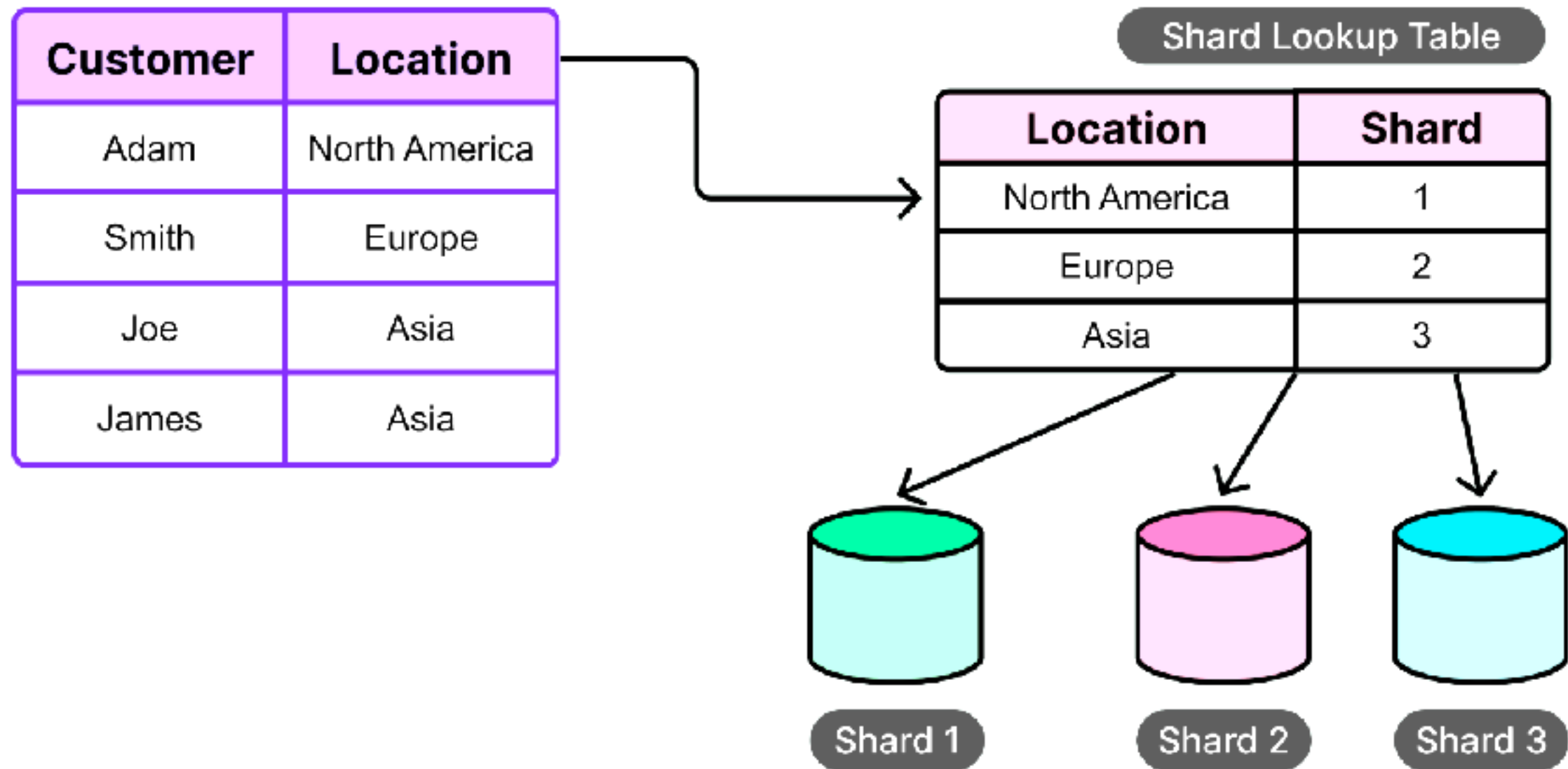




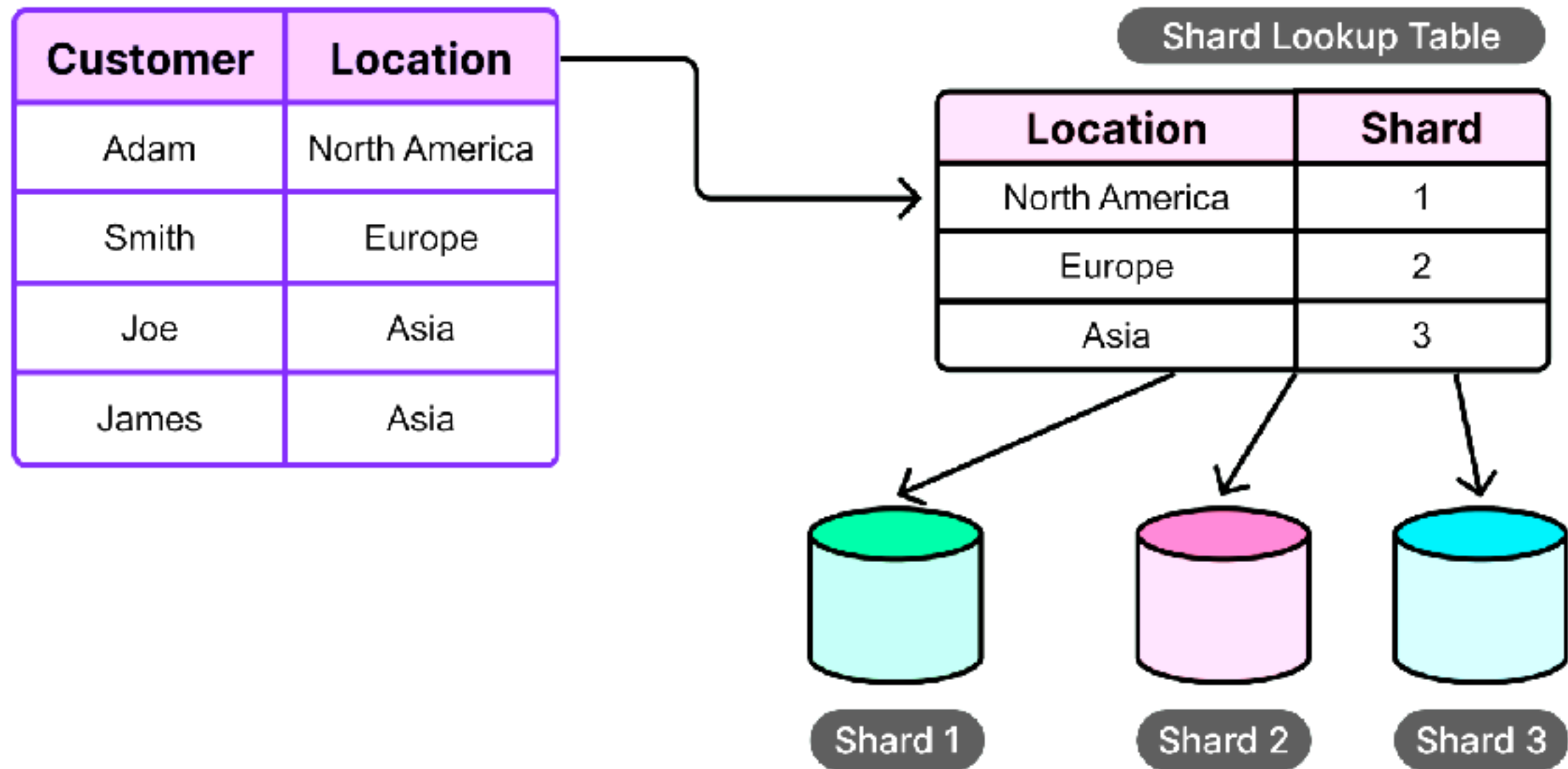
# Key-based sharding



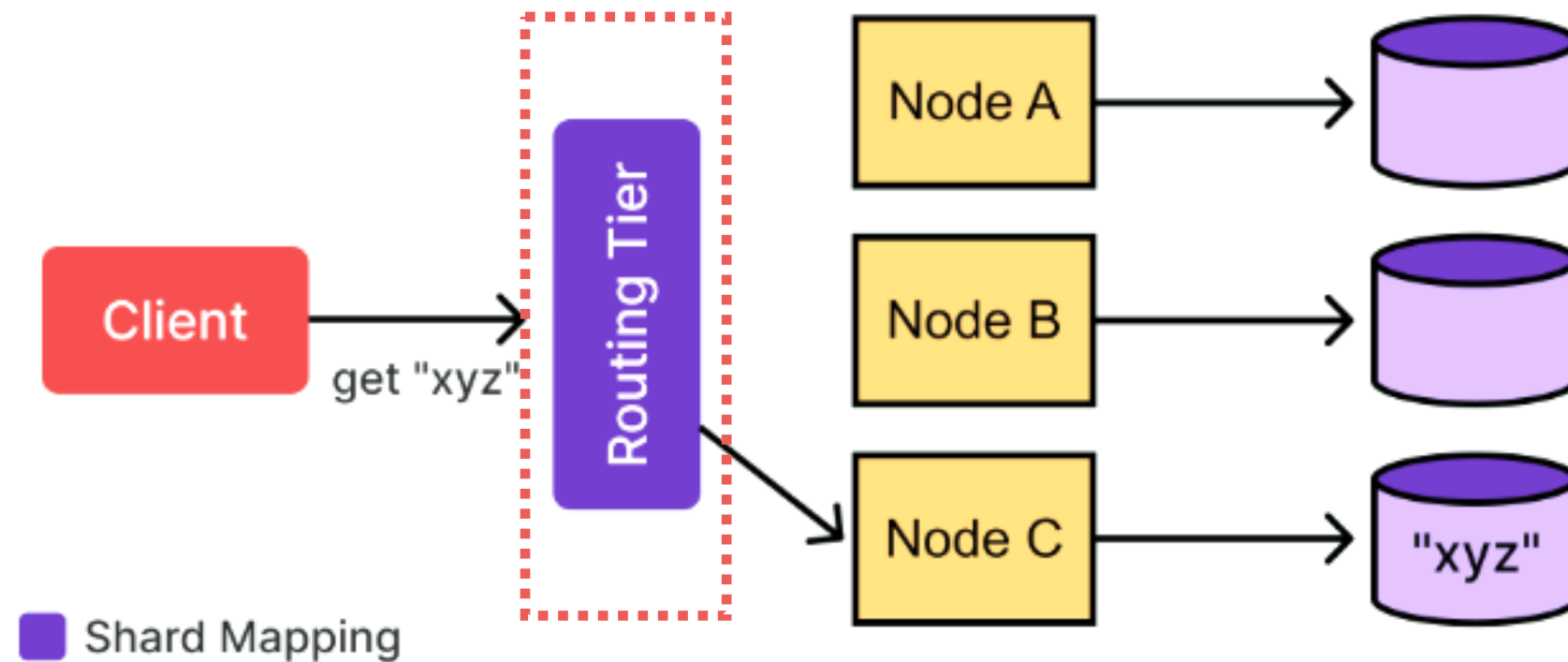
# Directory-based sharding



# Directory-based sharding



# Request routing



# Workshop with Sharding

<https://github.com/up1/workshop-database/blob/main/workshop/basic-workshop/partition.md>



# Denormalization



# Denormalization techniques

Redundant column/Pre-joining tables

Table splitting (horizontal/vertical)

Add derived columns

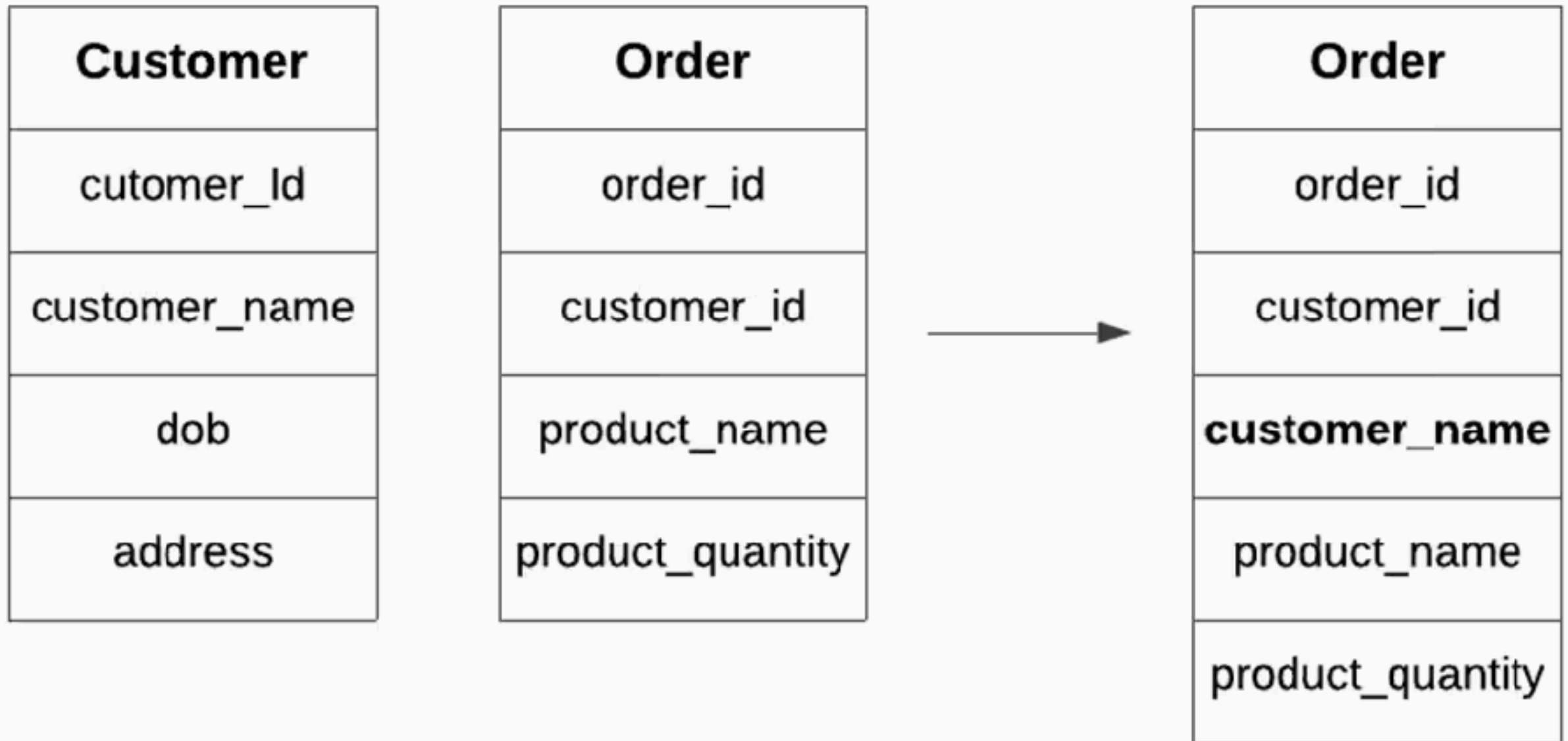
Mirrored tables

Materialized views



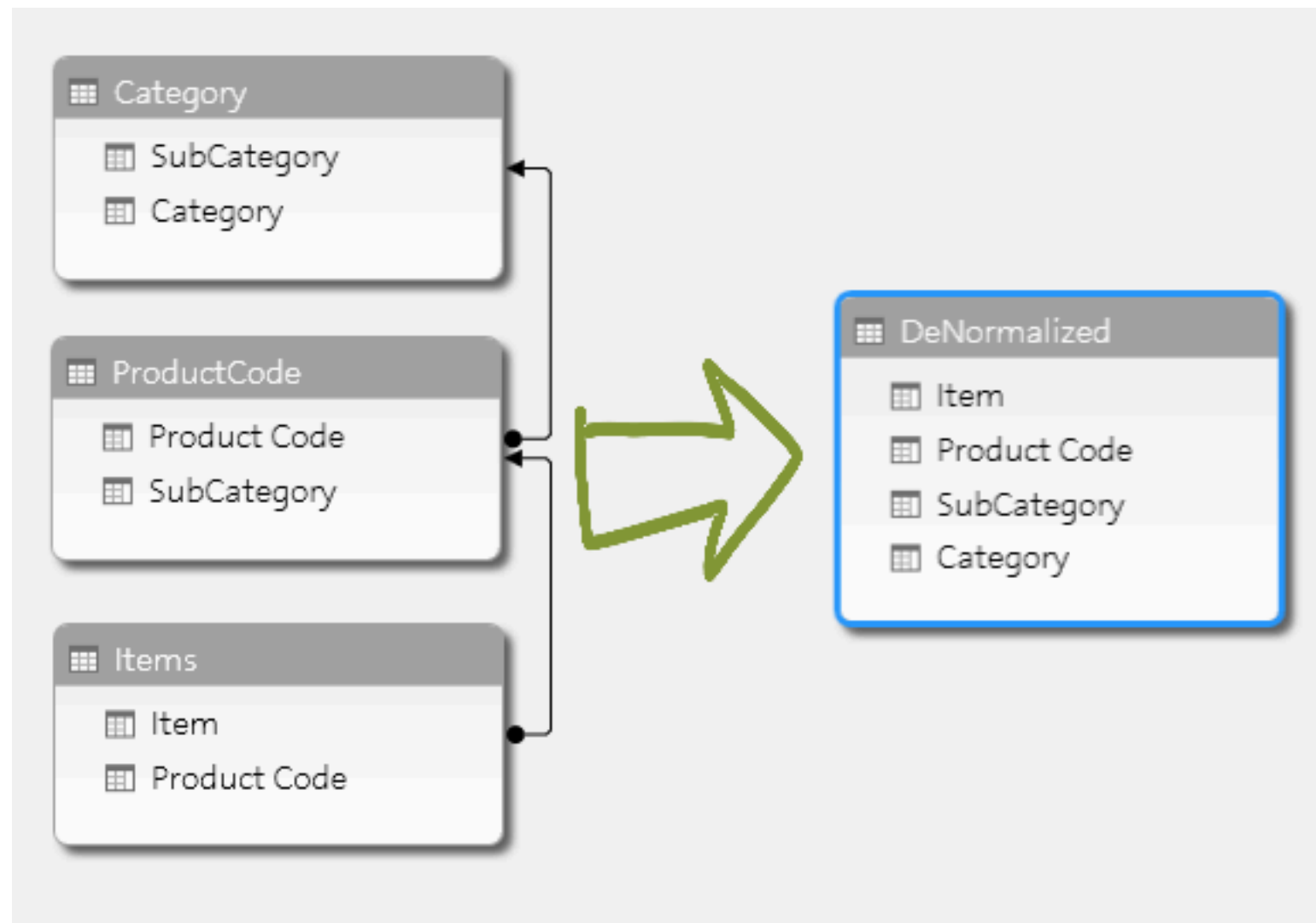


# Redundant column/Pre-joining tables





# Redundant column/Pre-joining tables



# Horizontal splitting

| Student_ID | Department_ID | Student_Name |
|------------|---------------|--------------|
| 1          | 1             | Alex         |
| 2          | 2             | Marie        |
| 3          | 1             | Sam          |
| 4          | 3             | Sara         |



Computer Science

| Student_ID | Department_ID | Student_Name |
|------------|---------------|--------------|
| 1          | 1             | Alex         |
| 3          | 1             | Sam          |

Chemistry

| Student_ID | Department_ID | Student_Name |
|------------|---------------|--------------|
| 2          | 2             | Marie        |

Botany

| Student_ID | Department_ID | Student_Name |
|------------|---------------|--------------|
| 4          | 3             | Sara         |



# Vertical splitting



# Add derived columns

| Student_ID | Name  |
|------------|-------|
| 1          | Alex  |
| 2          | Marie |

| Student_ID | Assignment_ID | Mark  |
|------------|---------------|-------|
| 1          | 1             | 20    |
| 1          | 2             | 35.50 |
| 2          | 1             | 45    |
| 2          | 2             | 45    |



| Student_ID | Name  | Total_Marks |
|------------|-------|-------------|
| 1          | Alex  | 55.5        |
| 2          | Marie | 90          |



# Pros of data denormalization

Improve query performance (UX)

Reduce complexity of data model

Improve application scalability

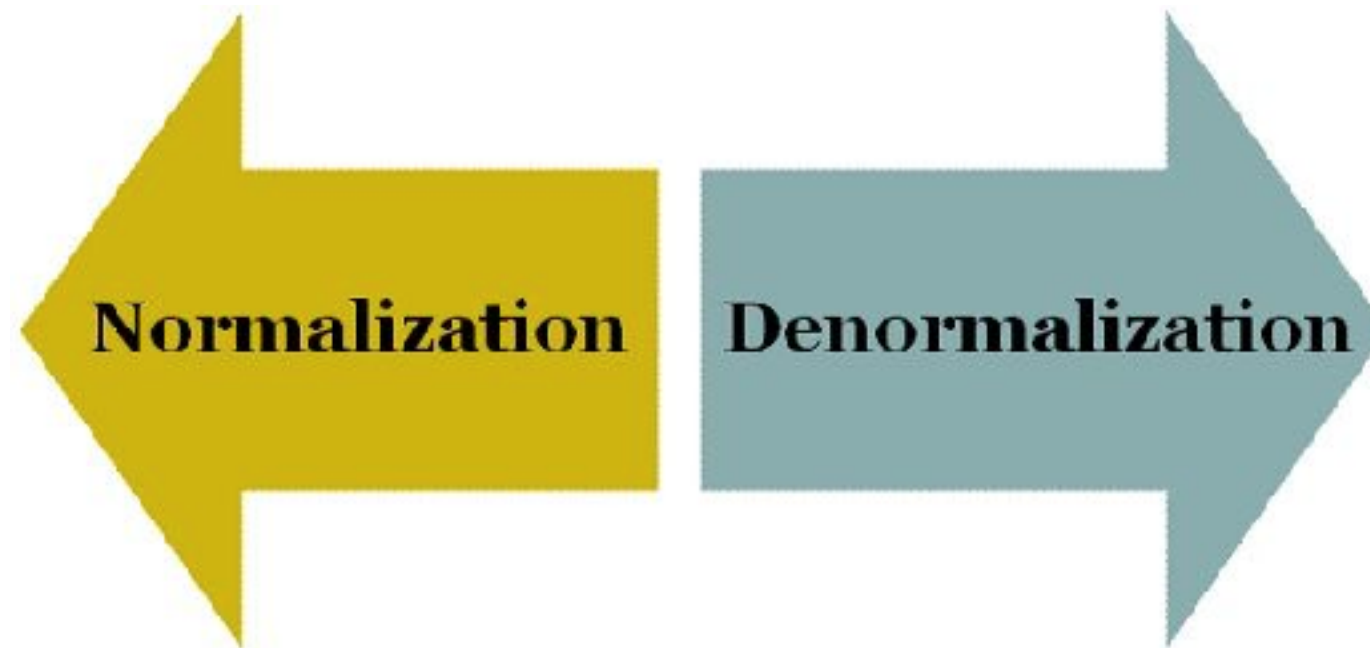
Generate data report faster



# Cons of data denormalization

- Increase data redundancy
- Inconsistency between data sets
- Require more disk space (split tables)
- More data model (hard to maintain)
- Difficult to insert and update data
- Maintenance costs







# Setup Postgres





# Database Topologies

Standalone  
Master-slave (scale-read)  
Cluster and sharding



# Essential configuration parameters



# Max connections

## Purpose

Maximum number of concurrent connections the server will accept

## Impact

Too low: connection refused

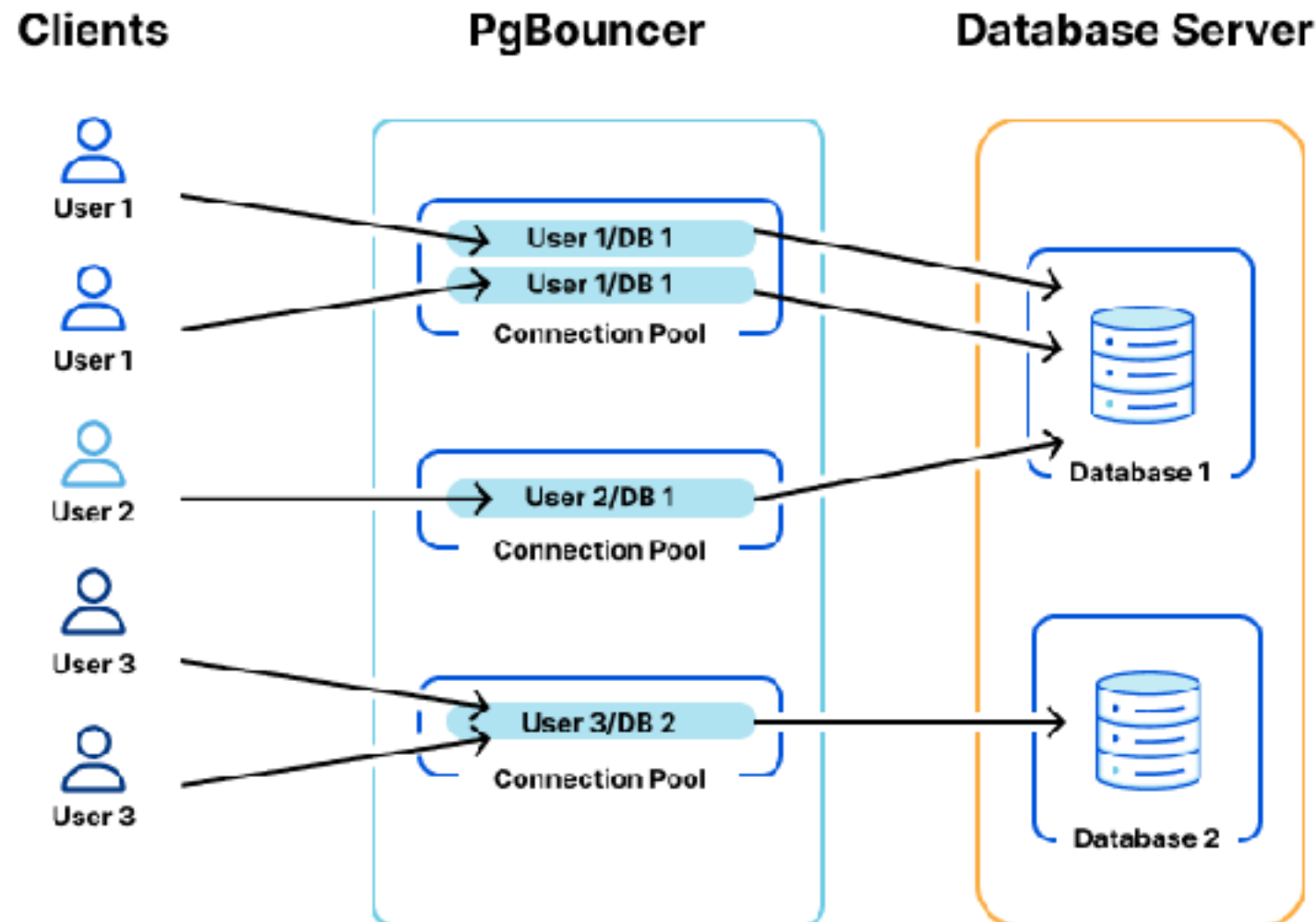
Too high: excessive memory consumption, thrashing

## Recommendation

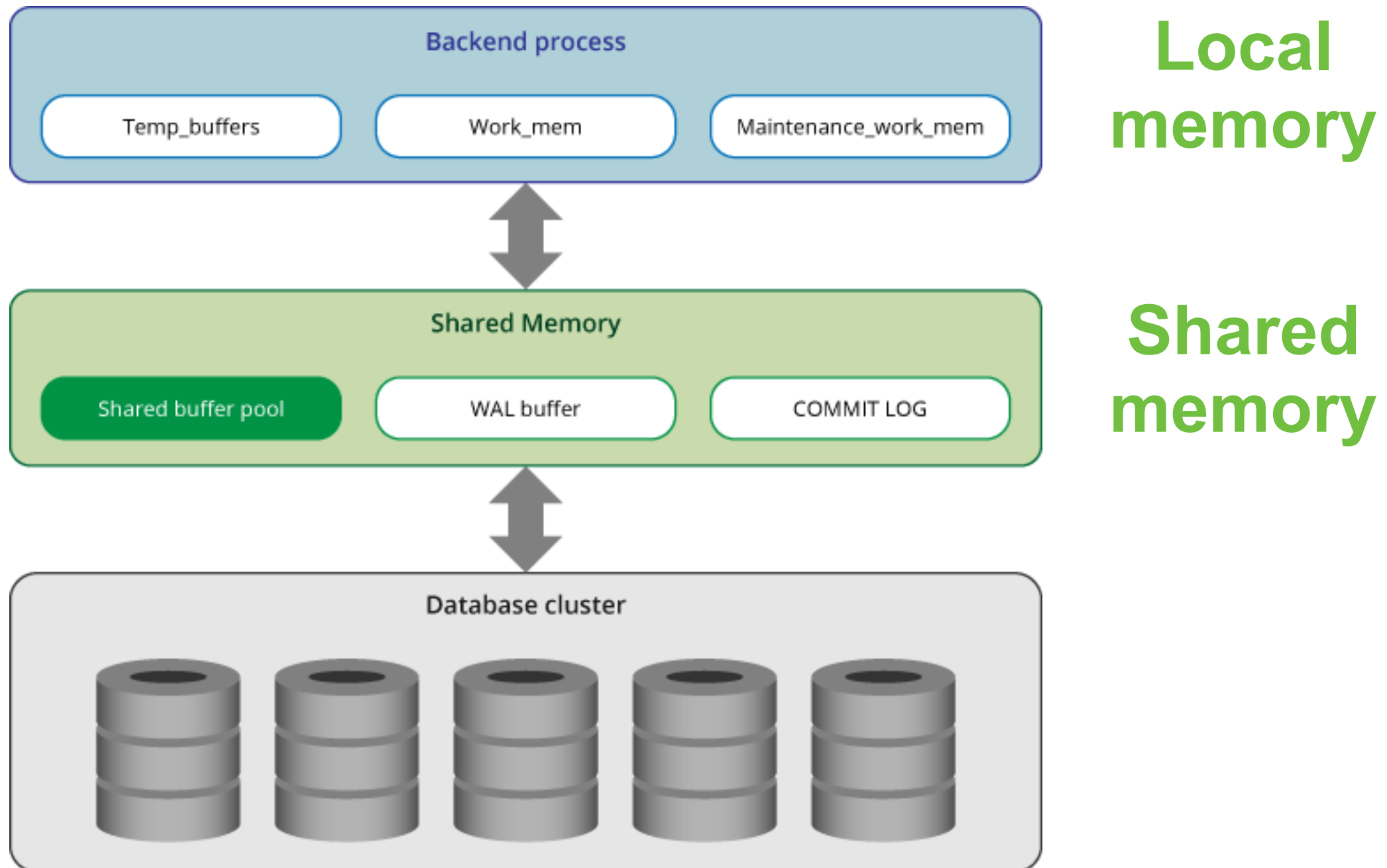
Start with a reasonable number based on application needs



# Manage connection pooling ?



# Memory architecture



# Local memory

Each backend process allocate local memory  
Query processing

work\_mem

temp\_buffer

maintenance\_work\_mem



# Shared memory

Allocate by server when it start up

shared\_buffer

wal\_buffer

Commit log (CLOG)





# Work mem (not max memory)

## Purpose

Memory used by internal sort operations and hash tables before spilling to disk

## Impact

Crucial for query performance, especially with large sorts  
Too low leads to temp file creation on disk

## Recommendation

Start with a reasonable value (e.g., 4-64MB)  
Tune based on EXPLAIN ANALYZE





# Shared buffers

## Purpose

Amount of memory dedicated to shared buffer pool for caching data pages

## Impact

Directly affects read performance by reducing disk I/O

## Recommendation

Typically 25% of total RAM, but can vary



# Effective cache size

## Purpose

Inform the query planner about the total amount of memory available for caching data (pg shared buffer + OS file cache)

## Impact

Doesn't allocate memory

Help the planner make better decision (index vs sequential)

## Recommendation

50-75% of total RAM



# **Write-Ahead Log (WAL)**

## **The foundation of Durability**



# Write-Ahead Log (WAL) ?

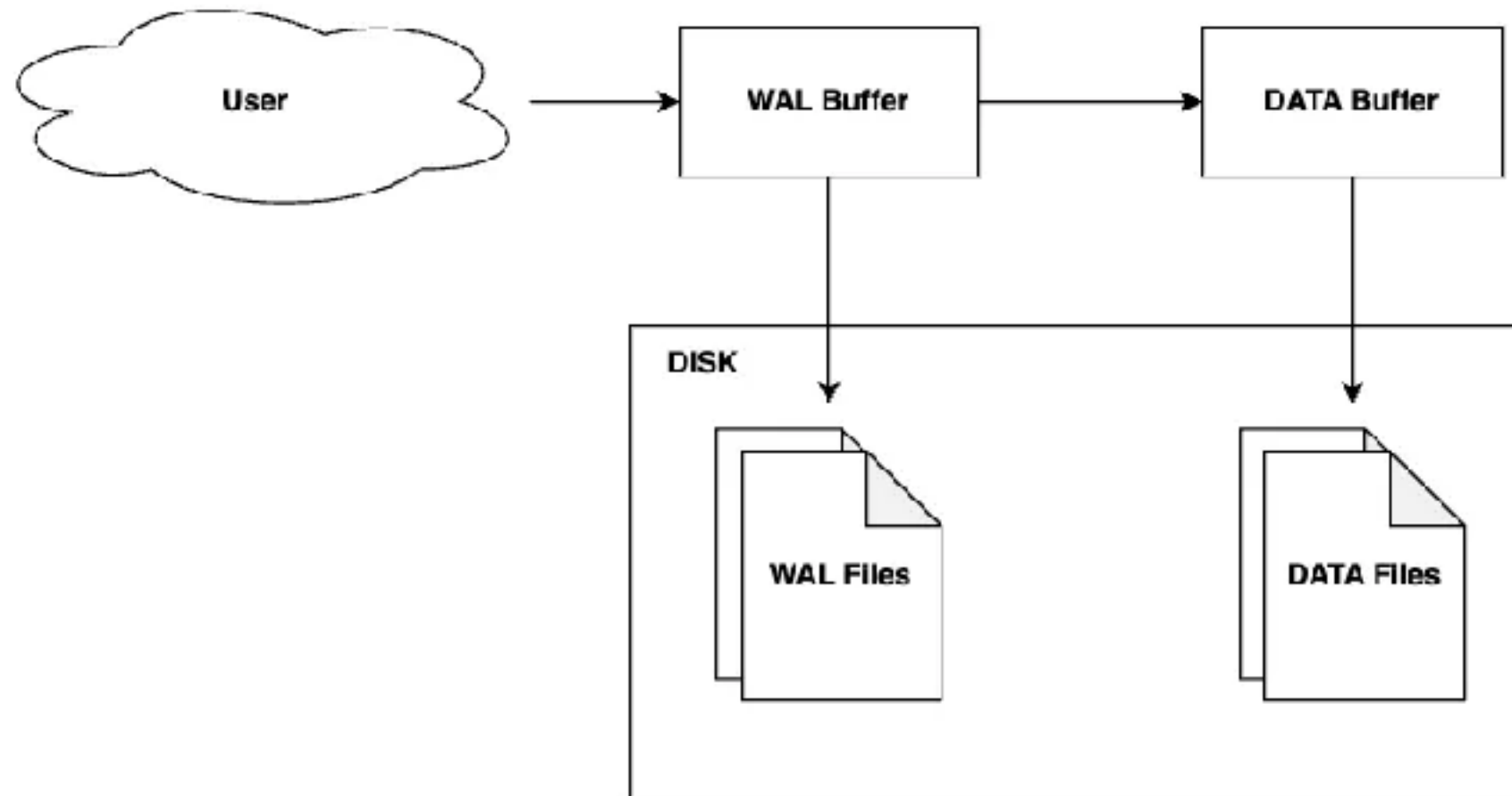
Transaction log or XLOG

Use to ensure data durability and consistency

All changes to data must be written to the WAL before applied to the actual data files on disk



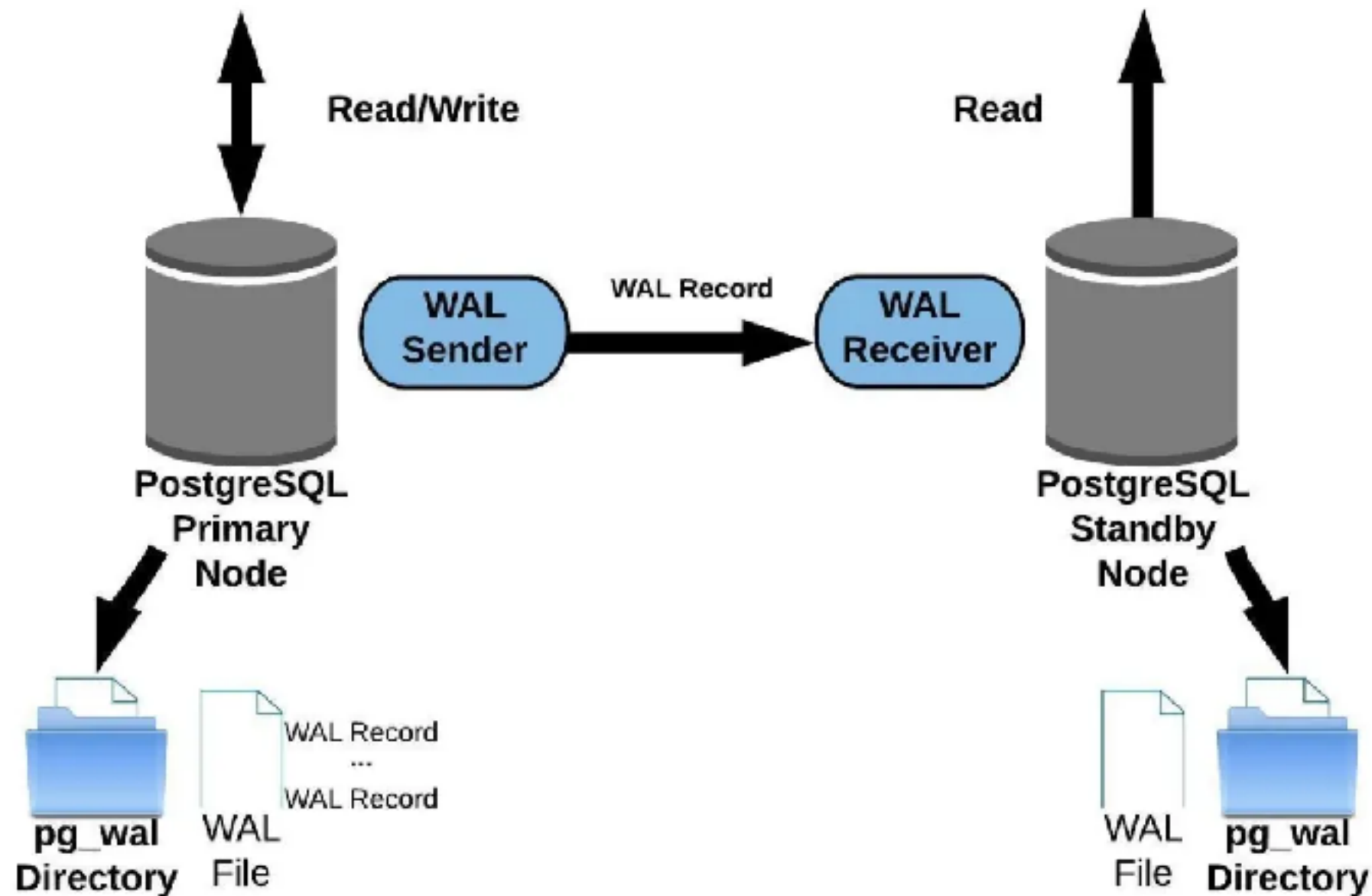
# Write-Ahead Log (WAL) ?



<https://levelup.gitconnected.com/when-databases-crash-wal-saves-the-day-write-ahead-log-wal-710e31aa9cdd>



# Async Replication



<https://levelup.gitconnected.com/when-databases-crash-wal-saves-the-day-write-ahead-log-wal-710e31aa9cdd>



# Checkpoint timeout

## Purpose

Controls how often checkpoints occur

Checkpoints flush dirty shared buffers to disk

## Impact

Too frequent: I/O spikes during checkpoints

Too infrequent: Longer recovery time, large pg\_wal directory

## Recommendation

5-10 minutes





# Max wal size

## Purpose

Controls how often checkpoints occur

Checkpoints flush dirty shared buffers to disk

## Impact

The maximum size of the WAL directory that triggers a checkpoint

## Recommendation

Based on I/O capacity and recovery time tolerance  
(e.g., 4GB-16GB+)





# High volume tuning ?

Read heavy

Write heavy

Many  
concurrent user

Complex query



# Principles for high volume tuning

Allocate more memory

Balance I/O (optimize WAL and checkpoint)

Parallelism

Monitoring (pg\_stat\_statement, EXPLAIN)

Hardware (Fast disk, RAM, CPU)

Connection pooling



# Slow Query !!



# Autovacuum

Prevent table bloat and ensure query performance



# PostgreSQL Cluster

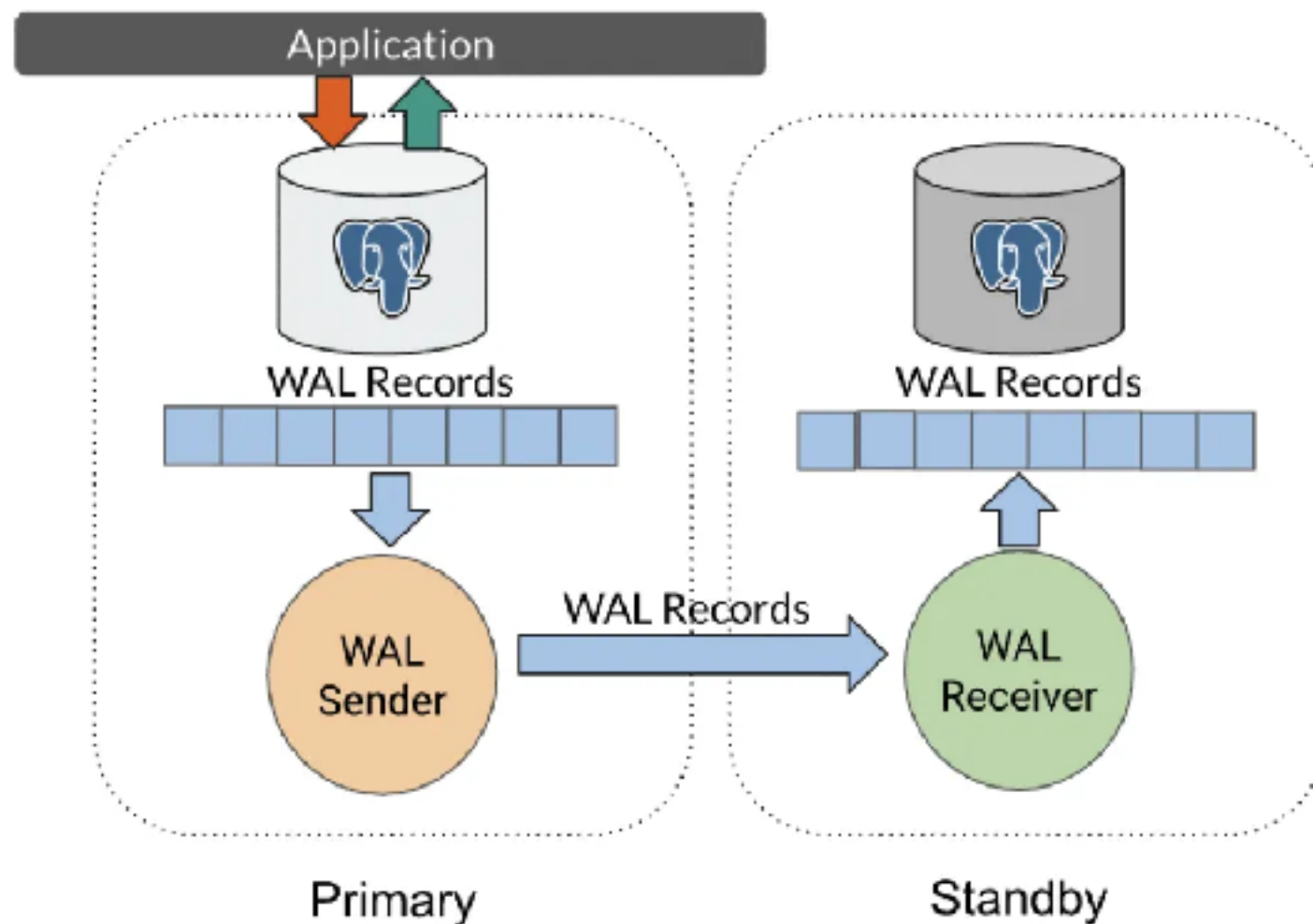


# PostgreSQL Stream Replication



# Stream Replication

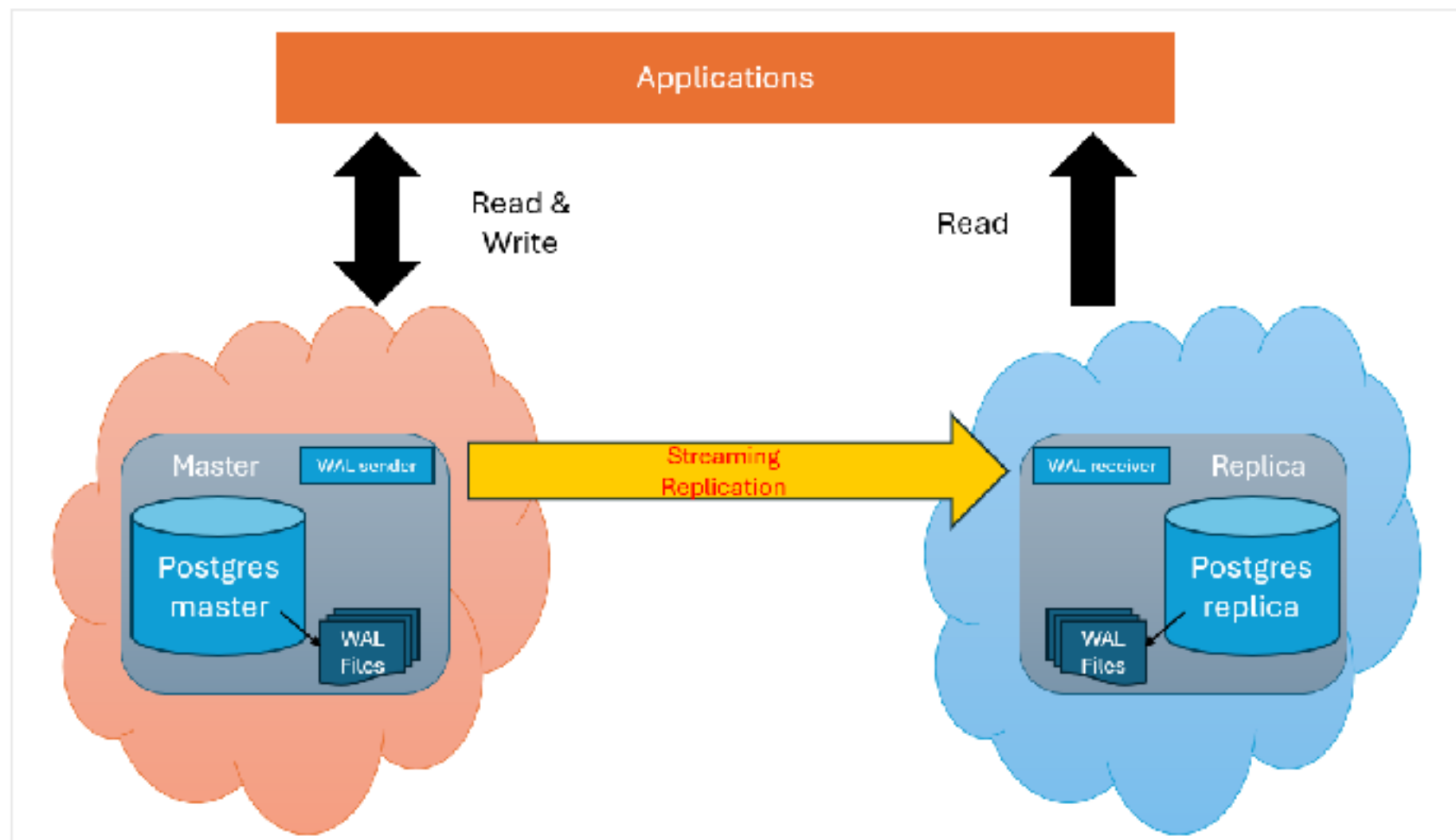
Build-in sync or **async** replication  
Simple DR site and backup cluster





# Stream Replication

## PostgreSQL stream replication (build-in)





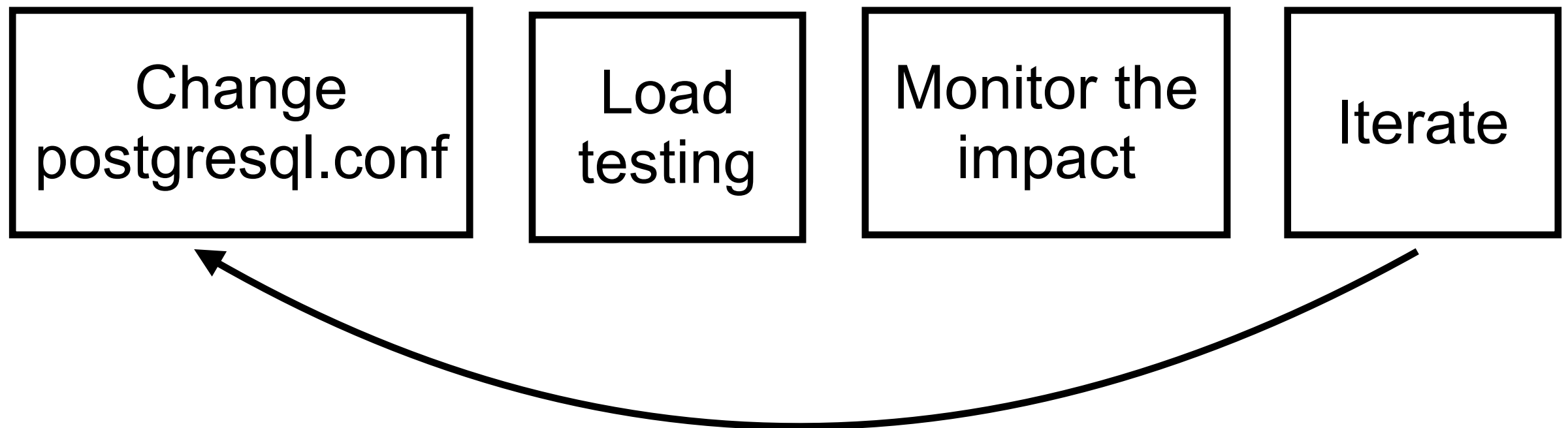
# Stream Replication configuration

Memory allocation

Checkpoints

Query planning

Send WAL data from primary to standby server



# Tuning parameters in primary

| Name                   | Location | Description                                                                      | Example value      |
|------------------------|----------|----------------------------------------------------------------------------------|--------------------|
| wal_level              | Primary  | Sets the level of information written to the WAL                                 | replica or logical |
| max_wal_senders        | Primary  | Maximum number of concurrent replication connections from standby servers        | 10                 |
| wal_keep_size          | Primary  | Minimum amount of WAL to retain in the pg_wal directory for standbys to fetch    | 4GB                |
| max_slot_wal_keep_size | Primary  | Maximum size of WAL files that replication slots are allowed to retain in pg_wal | 32GB               |



# Tuning parameters in standby

| Name                         | Location | Description                                                                                   | Example value |
|------------------------------|----------|-----------------------------------------------------------------------------------------------|---------------|
| hot_standby                  | Standby  | Allows read-only queries on the standby server while it's in recovery mode                    | on            |
| max_standby_streaming_delay  | Standby  | Maximum time to wait before canceling standby queries that conflict with incoming WAL entries | 30s           |
| hot_standby_feedback         | Standby  | Sends feedback to the primary about the oldest transaction active on the standby              | on            |
| wal_receiver_Status_interval | Standby  | Specifies the interval for the standby to report its replication status to the primary        | 10s           |



# More ..

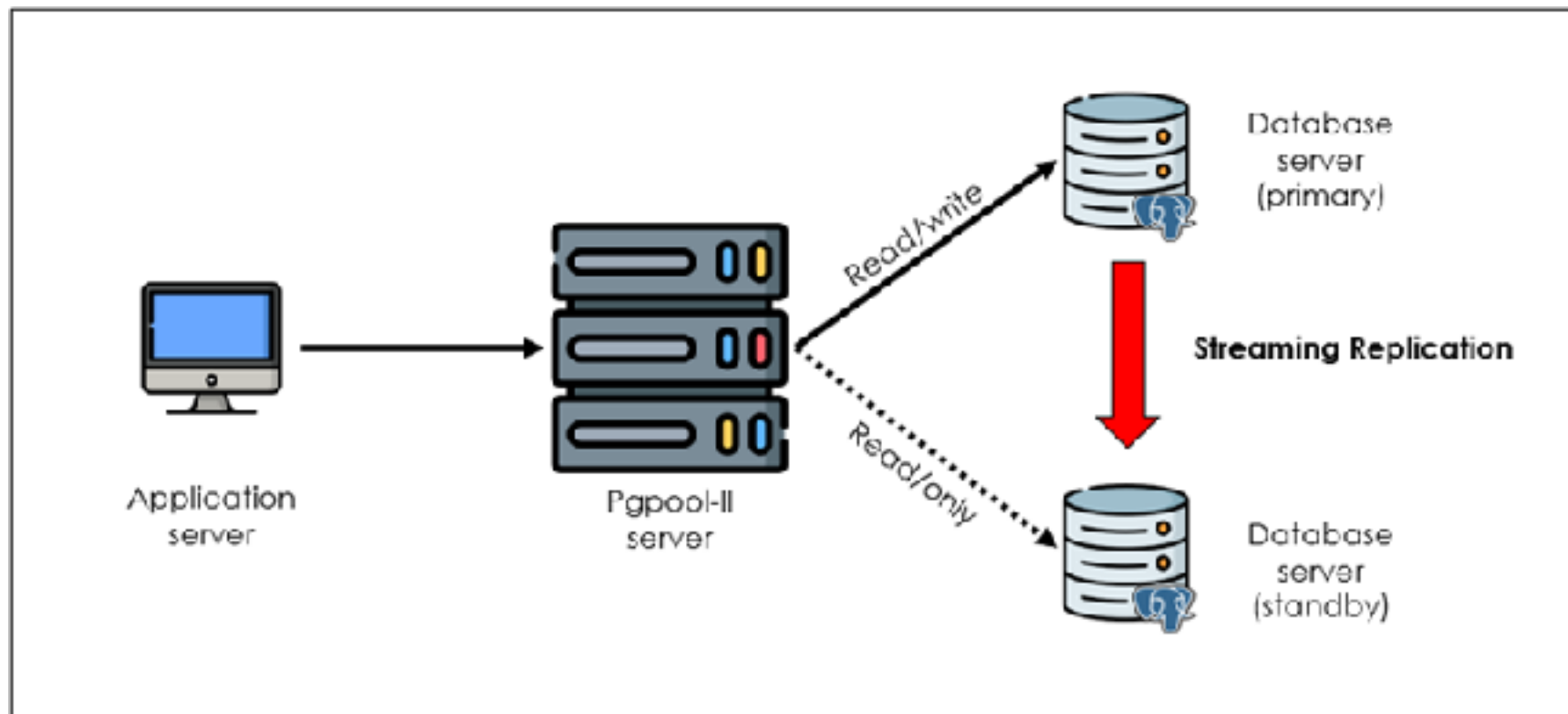


# Scale with PgPool II

Connection pooling

Load balancing

In-memory query caching

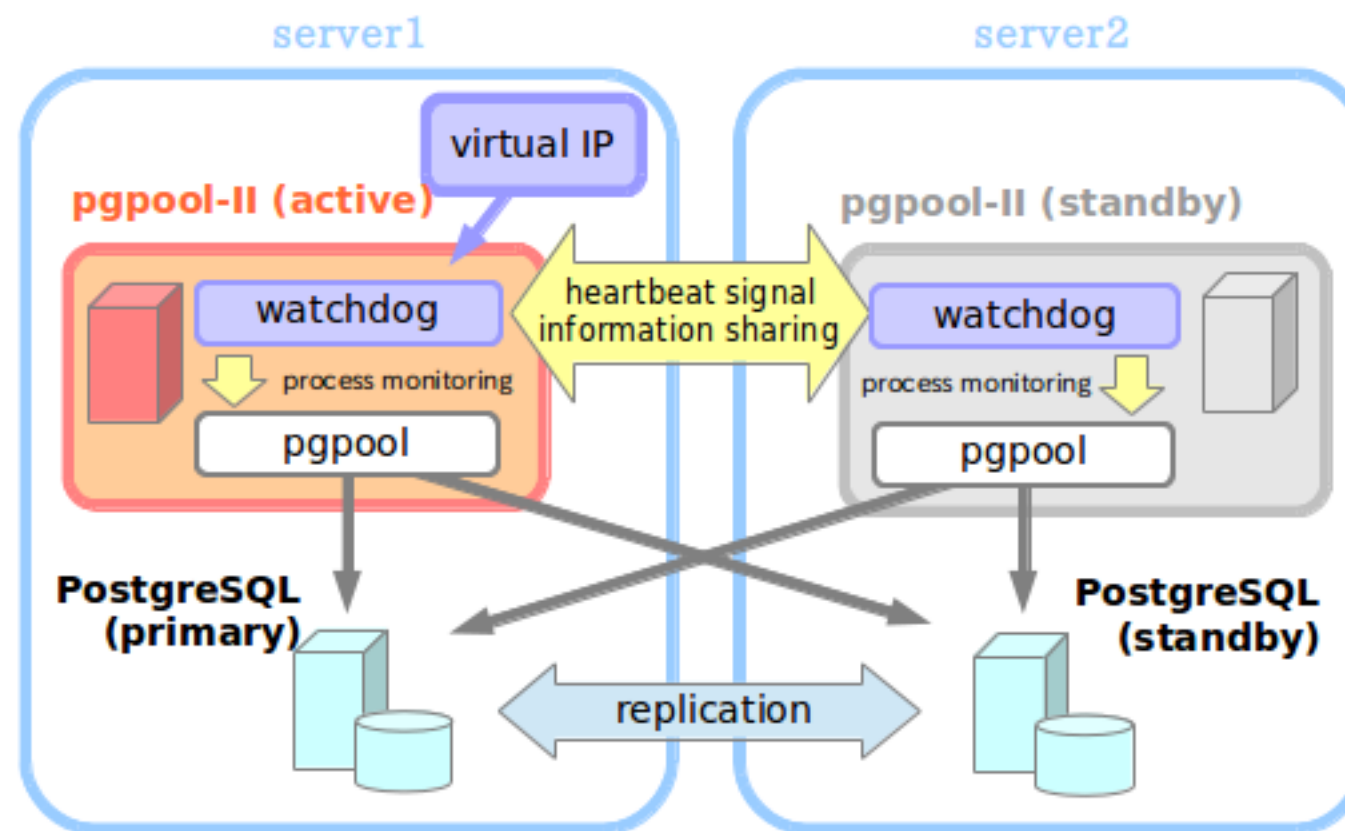


[https://www.pgpool.net/mediawiki/index.php/Main\\_Page](https://www.pgpool.net/mediawiki/index.php/Main_Page)



# Scale with PgPool II

## Master-slave mode with watchdog Failure management



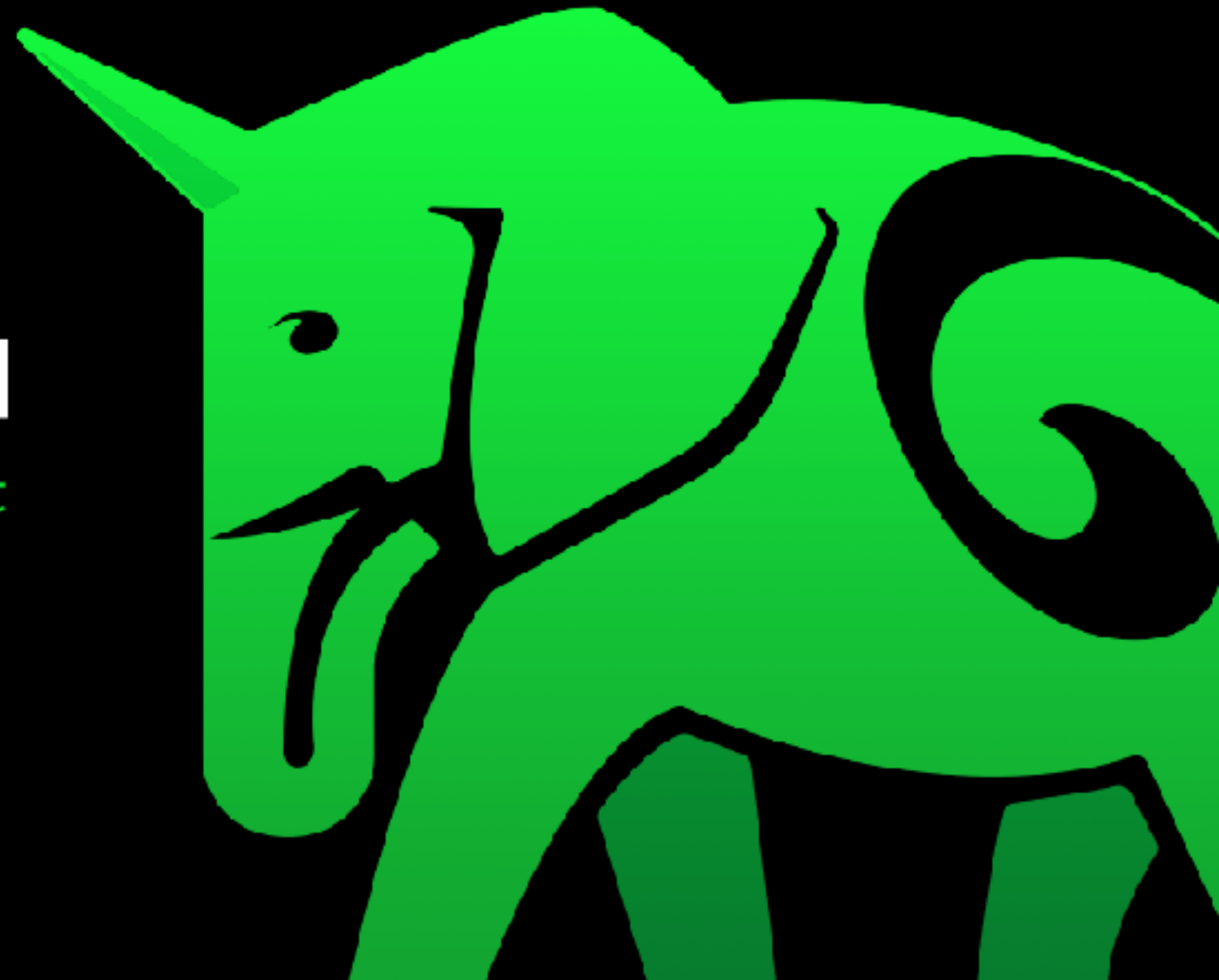
[https://www.pgpool.net/mediawiki/index.php/Main\\_Page](https://www.pgpool.net/mediawiki/index.php/Main_Page)



# Citus = Distributed PostgreSQL



[ POSTGRES AT ANY SCALE ]



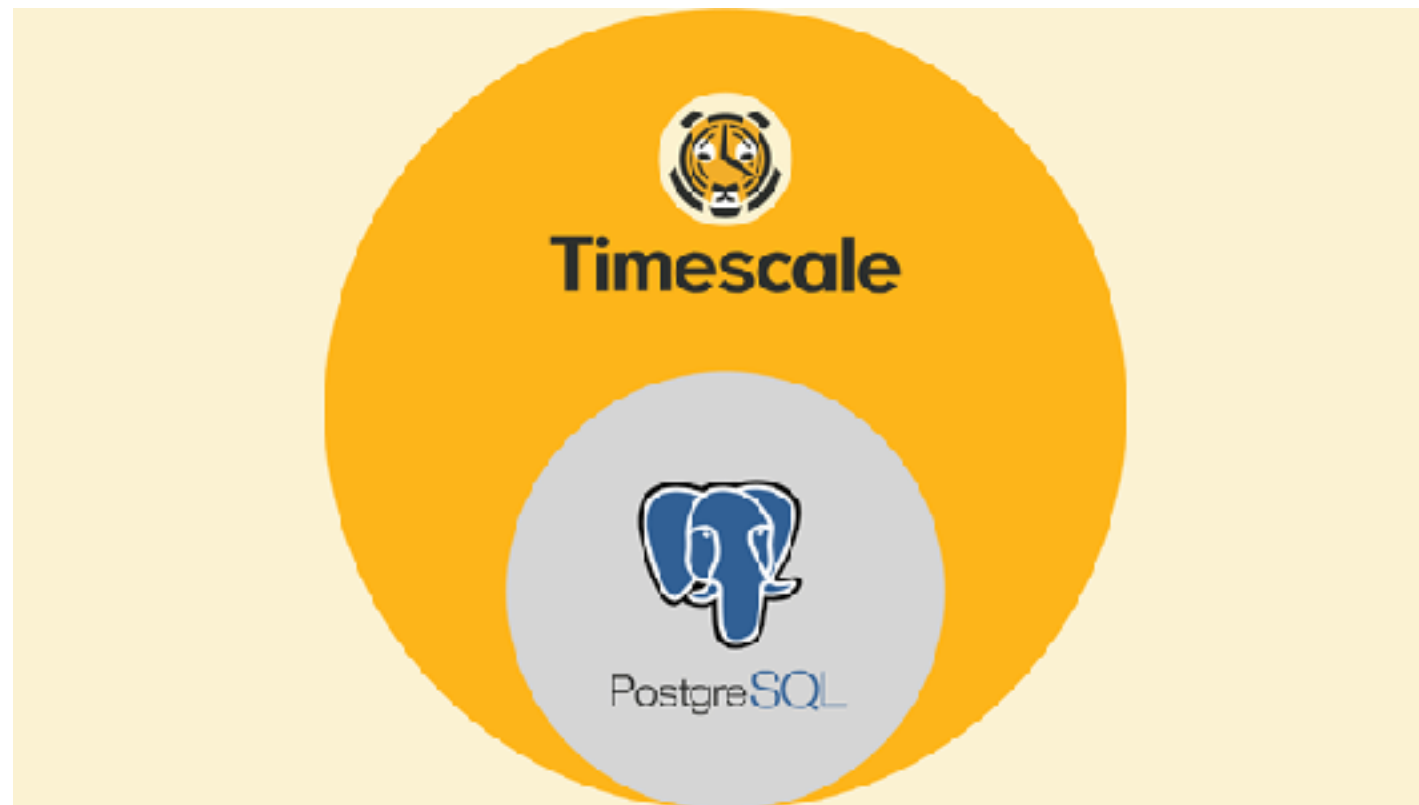
<https://github.com/citusdata/citus>





# TimeScaleDB

Time-series database for realtime analytic  
PostgreSQL extension



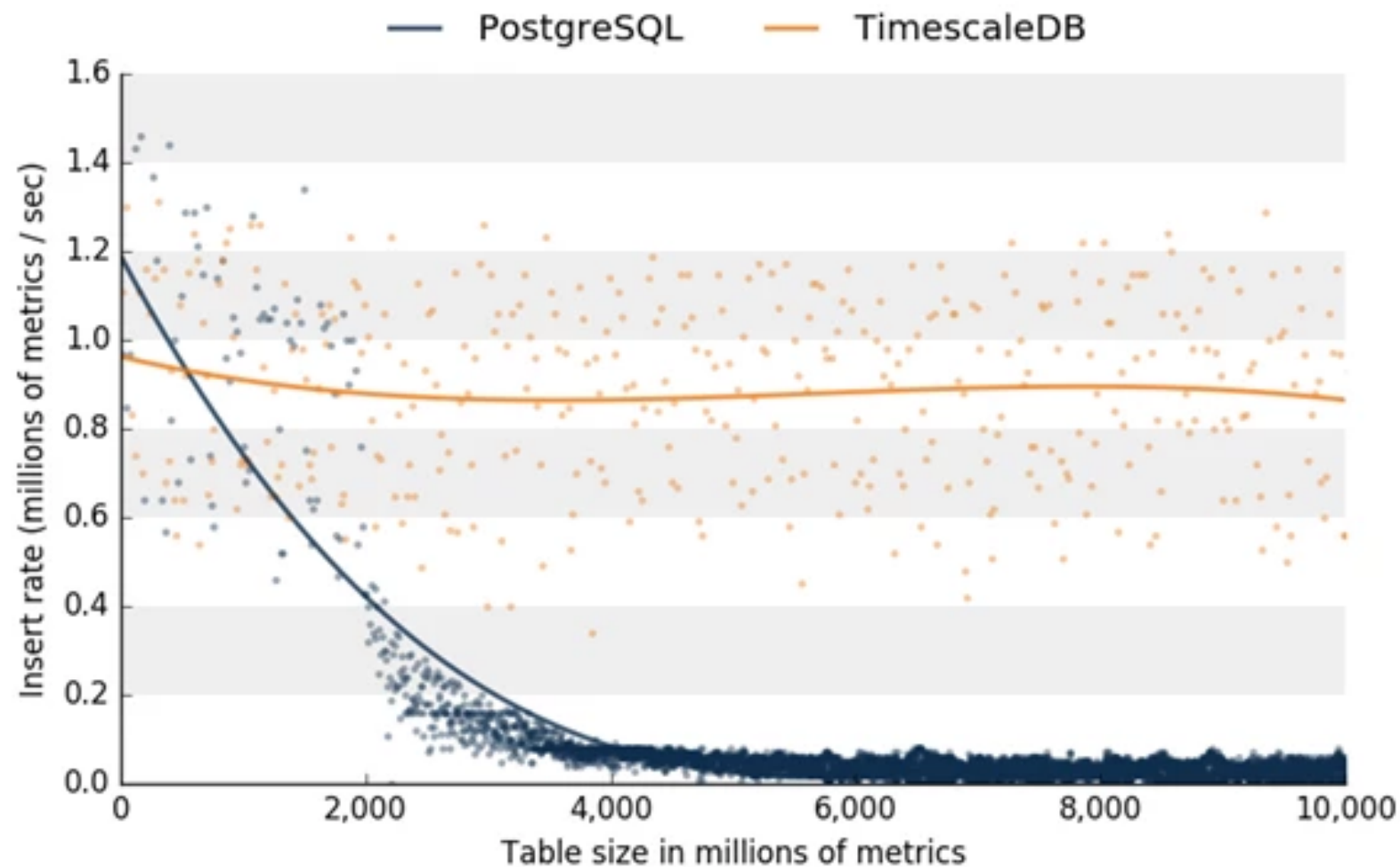
<https://github.com/timescale/timescaledb>





# TimeScaleDB

## Insert operation

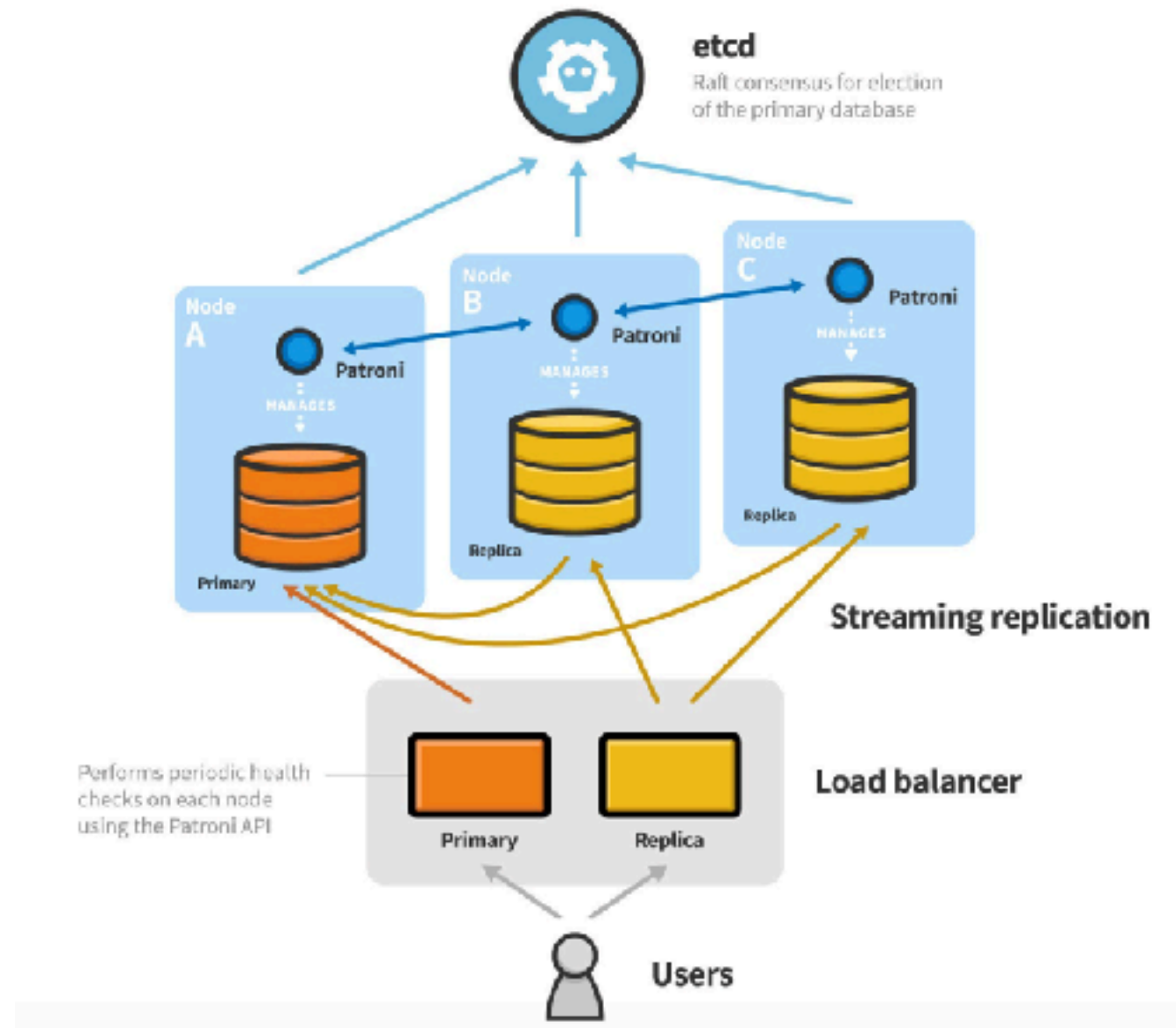


<https://azure.microsoft.com/en-us/blog/power-iot-and-time-series-workloads-with-timescaledb-for-azure-database-for-postgresql/>



# Patroni (HA for PostgreSQL)

With Etcd, Consul and Zookeeper



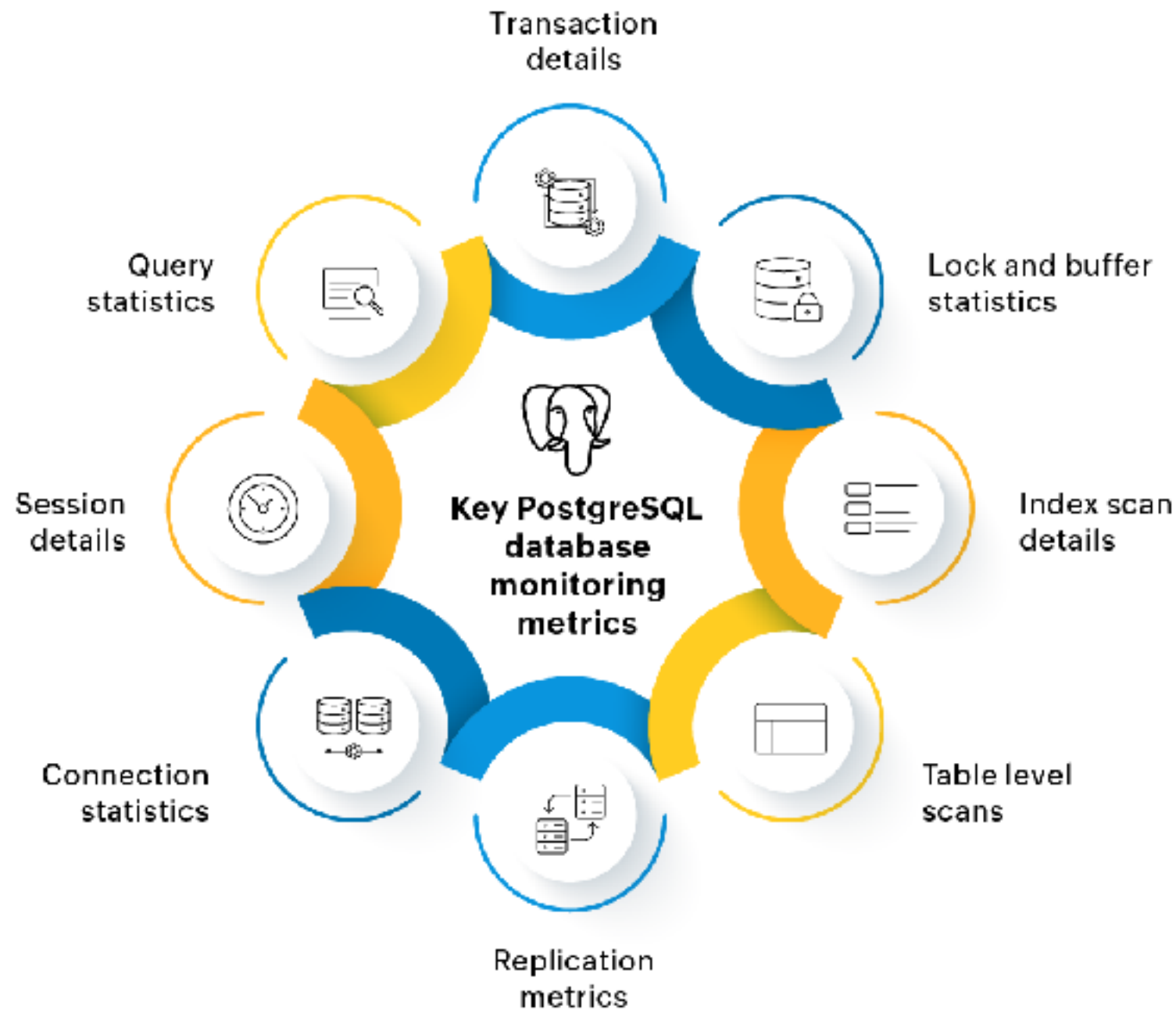
<https://github.com/patroni/patroni>



# Monitoring



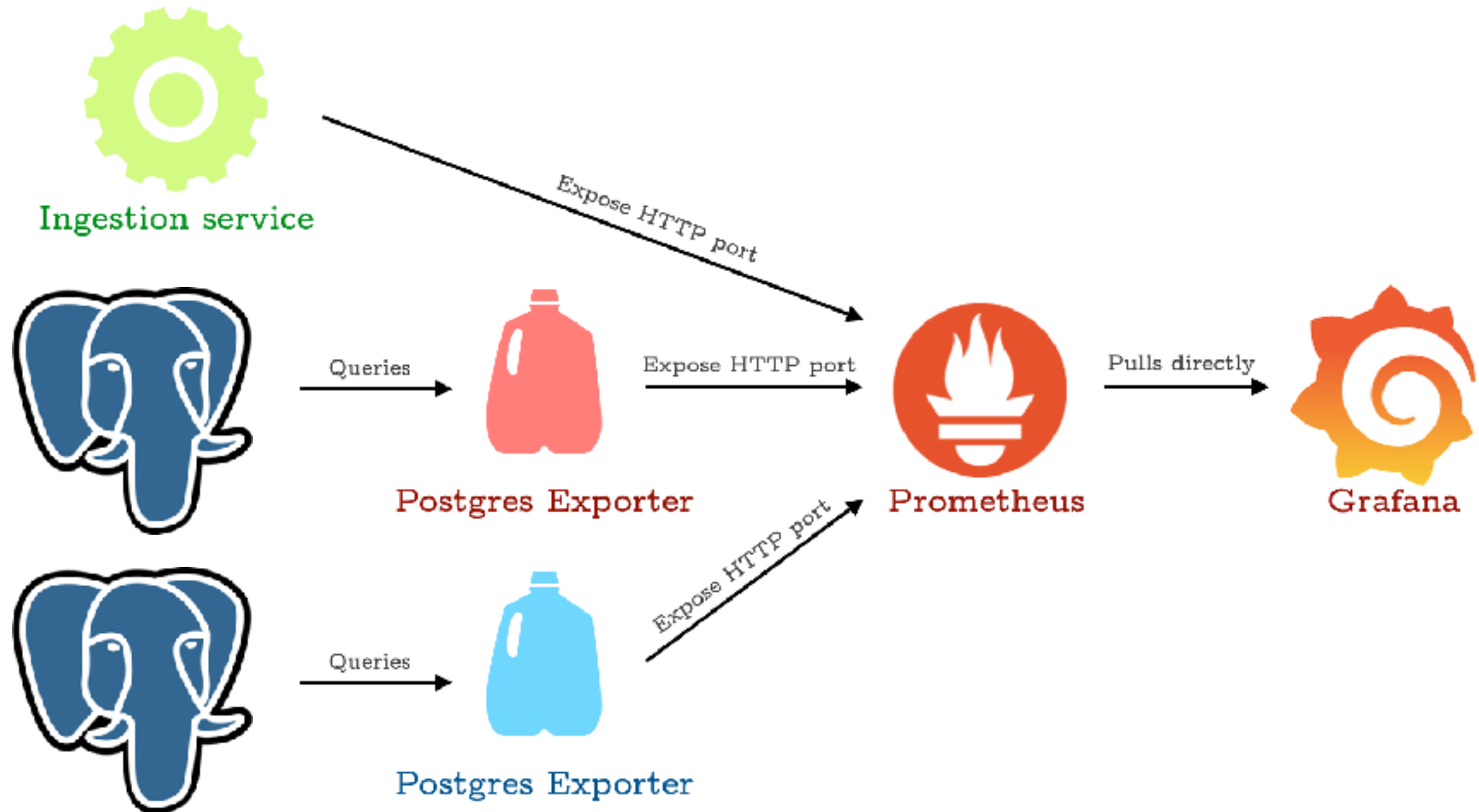
# Basic Monitoring



<https://blogs.manageengine.com/application-performance-2/appmanager/2024/04/19/postgres-performance-monitoring-best-practices-and-key-metrics.html>



# Basic Monitoring



<https://github.com/up1/workshop-database/tree/main/workshop/monitoring-testing>



# Q/A

